

Global Equity Derivatives & Strategy

Year Ahead 2026: The Bubble era

AI already big bubble or nothing to see here? Both unlikely

Some say clear bubble, others say nonsense. As we noted in Jun 2024 (see [Vol suggests AI isn't a bubble...yet](#)), a bubble may be hard to avoid as big tech leaps since 19th century led to big bubbles—and AI is government backed. Biggest IPO gains since 90s, record dip-buying, fear & vol show bubble's progression. Promise of a magical future just out of reach may usher new era of booms/busts. Key insight is bubbles' predictable behavior.

Bubble Risk Indicator says room to run, but in too deep

A hallmark of bubbles has been rising price instability near the peak. Our new BofA Bubble Risk Indicator measures instability & would've warned in 7/8 major bubbles since the 1920s. While 2025 has seen pockets of froth (Nuclear, Quantum, Asian AI), it's not yet in the US AI core. Valuations, momentum & easing policy also say more room to run. But bigger booms result in bigger busts & history suggests it's too late to avoid a bust.

Are levered ETFs or privates a risk? You need a plan

Retail & leverage are key bubble ingredients. Many fear the 500+ levered & inverse ETFs but risk is still small. Private assets are the elephant in the room with AI's \$5tn build-out, but if cracks emerge, public hedge gains can stem losses. Retail will lead the bubble (watch crypto), driven by fear-of-missing-out. Value investors need a plan before getting forced in & diversification isn't it, as in bubbles concentration + cash wins until the pop.

EU fiscal spend & sov risk; Asia AI value & cheap options

Europe outperforming US YTD shows positioning matters. Long-term tailwinds like German fiscal spend, record buybacks & pension reform favor risk assets, but beware poor execution, French sovereign stress & geopolitics. China small caps offer exciting tech innovation while China AI bubble risks looking less systemic. Korea hosts cheapest AI stocks but also bubble-like price action. An AI-heavy Nikkei can rally on stronger yen.

Equity vol supported: Big dips + rapid snapbacks normal

10%+ drawdowns & record-fast snapbacks are normal as bubbles form, and part of how vol builds. Elevated S&P P/Es, market fragility, and structural uptrend in stock vol all argue for vol support (even with correl crush). Lower vol in '26 needs narrow path of grind higher in US AI core as fundamentals reign + no irrational exuberance in periphery.

Hedges for right tail or AI crash, \$; Long vol, & dispersion

Don't be without right tail hedges in '26: NDX upvar, China calls, NKY vs ¥ hybrid. Buy dips without losing your shirt. Left tail: Hedge NVDA (AI titan), 10x equity-rates hybrid & systematic defense for multi-asset portfolios if bond pain returns. Long efficient SX5E fwd vol, cheap FX hedge for non-\$ investors, EZ Banks dispersion & China repo arb.

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Refer to important disclosures on page 68 to 70. Analyst Certification on page 67. 12906731

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Contents

2026 Outlook	3
The era of bubbles	3
What is the chance we can avoid a bubble?	4
Measuring bubble threats: BofA Bubble Risk Indicator	4
Bubble signs aplenty in 2025 even if not in AI core	7
Navigating bubbles	12
The right tail, it's all about the magic	14
The left tail, we are in too deep	16
Leverage, the kryptonite of bubbles	19
How to hedge a crack in fast-growing private assets	22
Europe: Still cheap and only a little less unloved	28
Asian AI stocks driving massive regional vol dislocations	31
Why lower US equity vol is unlikely in 2026	34
Booming derivatives income & its market impact	38
How to trade our global themes	41
Don't fight the future: A plan for the AI right tail	42
Cheapest call options on the planet includes China AI	43
How to buy-the-dip without having to catch a falling knife	44
Tail hedging the AI titan: Structured flows = opportunity	47
Hedgecraft: Systematic hedging for equity-bond portfolios	49
Hedging long-end pain with low rates vol & 10x hybrids	53
Long vol, light on carry: SX5E fwd var leads globally	54
Once bitten twice shy: Hedging FX for non-USD investors	55
Cheaply gear for EZ Banks dispersion after blistering rally	56
Nikkei up on a stronger yen for an 80% discount	57
The China repo arb – CSI 1000 calls vs ASHR calls	58
More ideas on our global radar	59
Lock in long-dated AI optionality today with convertibles	59
Cheaper German upside by >60% by fading CAC strength	59
Appendix	60
Introducing the BofA Bubble Risk Indicator	60
Summary of Trades Opened (26-Nov-25)	65
List of abbreviations	66
Research Analysts	71



2026 Outlook

The era of bubbles

Major technological leaps since the first industrial revolution in 19th century Britain and including the roaring 20s and 90s Internet boom have led to major multi-year asset bubbles, as we noted in Jun 2024 (see [Vol suggests AI isn't a bubble...yet](#)). Given the growing consensus on AI's likely profound impact, and 2025's bubble-like progression, the chance of avoiding one seems low. Unlike prior tech leaps, this one has the support of governments, as it represents a critical capability relevant to geopolitical superiority.

Thus, while not a certainty, an eventual bubble seems likely. But are we seeing signs yet? A hallmark of bubbles is prices becoming unstable as they near the peak. Our Bubble Risk Indicator does suggest froth around the edges, but not yet in the core of US tech.

With bubbles comes risk, as volatility rises into the peak and keeps rising after; and the AI boom is likely too big to avoid a bust. The biggest challenge is timing. While our 2025 outlook highlighted similar bubble risks (see [The roaring 2020s?](#)) compounded by stimulative US policy (some still forthcoming), US equities have lagged global markets in 2025, especially in USD terms. Cheaper valuations, underweight positioning and fears of "peak US exceptionalism" dominated. This could continue but is antithetical to an AI bubble peaking, as bubble-assets typically outperform until the pop. This also suggests the top is further out. Valuations in core AI not yet at extremes, a lack of obvious leverage (present in past bubbles) and a Fed easing (not hiking) argue it's still early days.

We see potential for a bubble-prone era where booms/busts are more the norm, driven by the promise of "abundance" always just out of reach and magical AI moments fueling FOMO. But bubbles have unique signatures and recognizing that can be helpful in their navigation. Often led by value-insensitive retail, they create a real dilemma for fundamental investors: chase or risk underperforming until the pop. Key is having a plan.

Our outlook on a page:

AI builds towards larger bubble: Markets became more bubble-like in 2025 & bubbles hard to avoid in big tech booms; timing the top is hard but watching for warnings is key

Leverage toxic for bubbles, but not yet: Fear is of toxic leverage in levered/inverse ETFs, but we don't see it yet; private assets perhaps a risk, but we have a hedge for it

Too late to avoid eventual bust: Even though indications suggest bigger bubble before it's over, the AI boom has come too far to avoid an eventual bust (20%+ sell-off)

Tightening financial conditions biggest risk: Kryptonite for leverage; preceded all major bubble peaks. Bond vigilantes are dormant but may not be dead: watch inflation

Vol stays supported: Risks rise as bubbles build; markets remain fragile & AI doom debate continues to drive fear, uncertainty & doubt, further inflaming instability

Europe still cheap despite outperforming US: Long-term tailwinds like German spend on Defense & infrastructure, record buybacks & pension reform favor risk assets. Poor execution, US exceptionalism, French sovereign stress & geopolitics are main risks

Asia seeing bubble-like catch-up but still offers AI upside + value: Korea/Japan seeing bubble-like moves, but value = more upside; AI bubble most room to run in China

Risks to the core view/plausible scenarios:

Grind higher as bubble takes longer: Core of US AI so big, earnings so strong, it fails to disconnect from fundamentals. Micro-bubbles boom/bust in the periphery, but too small to cause a market-wide bubble in 2026 & big upside pushed into '27 (not unlikely)

Realization AI breakthrough needed: Faith in Jensen Huang's "double exponential" token demand growth prediction stalls; delays AI replacing humans, bear market ensues



What is the chance we can avoid a bubble?

Big tech leaps result in big asset bubbles

The history of major technological leaps is one of major asset bubbles. The first industrial revolution led to numerous booms in railroad stocks in 19th century Britain, the roaring-20s in the US emanated from the invention of automobiles, radios & electronics, and the internet resulted in the late 90s tech bubble. These bubbles all built over multiple years and resulted in multi-year busts (Exhibit 1).

As we outlined in June 2024 (see [Volatility suggests AI isn't a bubble...yet](#)), bubbles form when investors coalesce around the idea that technology will create a magical future but with high uncertainty on its magnitude and timing. Two further ingredients were present in all three of the technology driven bubbles: retail participation and leverage, often via the expansion of credit.

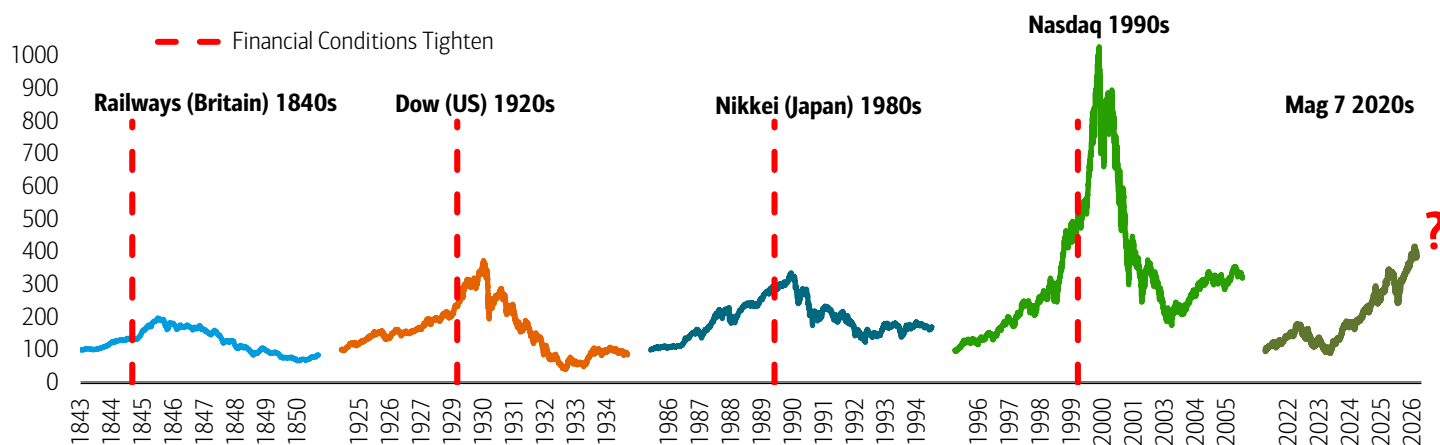
Government support makes this time different

The 1920s was further inflated by laissez-faire economic policy including tax cuts and deregulation – some of which was mimicked by the current US administration and may still impact markets. However, a major difference today is the amplification of the bubble coming from government support. Many sovereigns see integrating AI into their economy as necessary to remain competitive on the world stage. And given the perceived existential threat AI dominance presents to geopolitical power, many fear the US and China are entering a new cold war in which AI is next critical weapon.

Given this backdrop, the likelihood of avoiding a significant asset bubble in AI seems low. The question in our view is not *if* we will see a bubble, but *where* are we in the process?

Exhibit 1: Major asset bubbles in global equities in the last ~180 years have all seen financial conditions tighten before the end

Five historic equity bubbles, normalized to 100 at start, with red lines marking key financial conditions tightening events that raised leverage costs



Source: BofA Global Research. Data from 6-Jan-1843 to 24-Nov-2025. Railway data from Gareth Campbell (2013). "Deriving the railway mania." Financial History Review, 20, pp 127. Nasdaq refers to the NASDAQ-100 index; Mag 7 (BM7P) refers to the total return index of AAPL, AMZN, GOOGL, META, MSFT, NVDA, and TSLA. Each bubble shows five years before and after its peak (railways: two years). All series normalized to 100 at bubble start. Tightening of financing conditions raised leverage costs: railway companies increased share calls in 1844; brokers' loan rates were hiked in 1928; BoJ raised rates in 1989; Fed raised rates in 1999. For a broader definition of financial conditions, see this [BIS Bulletin](#). Note this chart does not contain any input from the BofA Bubble Risk Indicator.

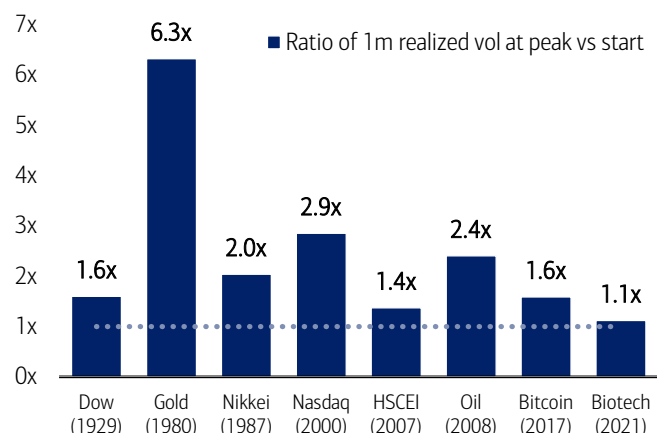
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Measuring bubble threats: BofA Bubble Risk Indicator

A hallmark of asset bubbles is prices becoming more unstable as they rise (Exhibit 2 & Exhibit 3). This is the opposite of typical behavior, as normally asset volatility falls as prices rise and rises when prices fall. When prices become more unstable on the way up, it can be due to extreme one-sided positioning due to the fear-of-missing-out (FOMO). Assets often decouple from fundamentals in part due to fundamentals themselves being very ephemeral when their future value is highly uncertain (as during technological leaps). This doesn't just occur during technological driven bubbles but is pervasive across a broader spectrum of asset bubbles, as shown in Exhibit 2 & Exhibit 3.

Exhibit 2: Looking at 8 notable asset bubbles in the last 100 years, volatility has consistently risen as the bubble developed

Rise in 1m realized vol from start to peak for historical asset bubbles

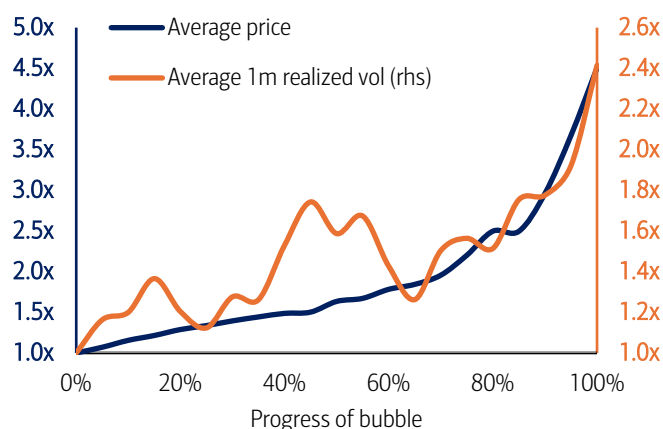


Source: BofA Global Research. Data from Oct-1926 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow, XAU for Gold, NKY for Nikkei, NDX for Nasdaq, HSCEI for HSCEI, CO1 for Oil, XBTUSD for Bitcoin, SPSIBI for Biotech.

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Exhibit 3: Asset bubbles have historically shown the signature of rising volatility with prices

Average price & 1m realized vol in historical bubbles from start to peak



Source: BofA Global Research. Data from Oct-1926 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow, XAU for Gold, NKY for Nikkei, NDX for Nasdaq, HSCEI for HSCEI, CO1 for Oil, XBTUSD for Bitcoin, SPSIBI for Biotech.

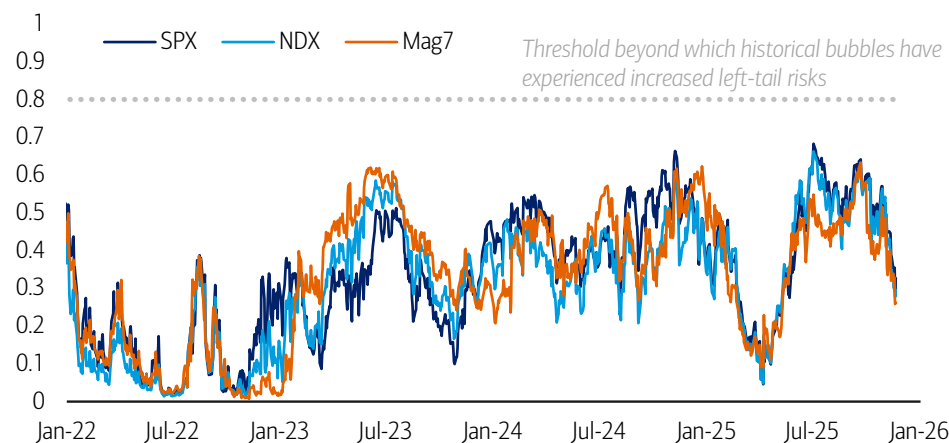
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Informing bubbles from the first 4 moments

The overly exuberant sentiment and lopsided positioning backdrop during an asset bubble often bleeds into price action in the form of instability. Rallies get larger and faster, but with volatile swings and rapidly bought drawdowns. To quantify the degree of this bubble-like instability, the **BofA Bubble Risk Indicator ("BRI")** takes inspiration from the way the first 4 moments describe a statistical distribution: returns, vol, momentum and fragility (see Appendix for a description of the indicator's construction, how it's behaved in a number of historical bubbles and what it's designed to capture). Indeed, high returns coupled with high vol, high momentum, and elevated fragility have characterized the exuberant price action that most asset bubbles have shown over the past century, especially at their frothiest points. Recall that fragility represents the degree of "local tail events" an asset displays, as we first introduced in our 2016 outlook (see [Fragility is the new volatility](#)).

Exhibit 4: While pockets of froth have appeared in places in 2025, the BofA Bubble Risk Indicator suggests that we are yet to see bubble-like instability in broad US equities or core tech

BofA Bubble Risk Indicator on SPX, NDX and the Magnificent 7 stocks



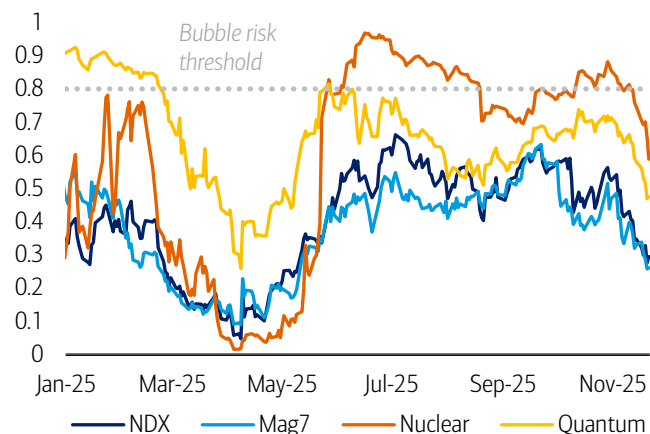
Source: BofA Global Research. Data from 1-Jan-22 to 21-Nov-25. Mag7 / Magnificent 7 refers to Apple, Amazon, Alphabet, Meta, Microsoft, Nvidia & Tesla; BM7P Index used for certain Mag7 data. See the More ideas on our global radar for further details on the BofA Bubble Risk Indicator. The BRI is intended to be an indicative metric only and may not be used for reference purposes or as a measure of performance for any financial instrument or contract, or otherwise relied upon by third parties for any other purpose, without the prior written consent of BofA Global Research. The BRI was not created to act as a benchmark.

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Focusing on today, the Bubble Risk Indicator suggests that we are yet to see typical bubble-like instability in broader US equity markets and including the core of US tech (Mag 7), which tends to be more acute when the indicator exceeds 0.8 (Exhibit 4). To be clear, the BRI is not saying a bubble is not possible from these levels, but rather that broad US markets are not experiencing the “typical” instability we have seen near the peak of past asset bubbles in the last 100 yrs. Some markets, however, such as US nuclear-related and quantum computing stocks along with Asian equity indices like Nikkei & Kospi have been experiencing bubble-like instability (see Exhibit 5 & Exhibit 6)

Exhibit 5: In contrast to broader US equities, pockets such as nuclear-related and quantum computing stocks have shown bubble-like price action and froth in 2025

BofA Bubble Risk Indicator on NDX, Magnificent 7, Nuclear and Quantum stocks

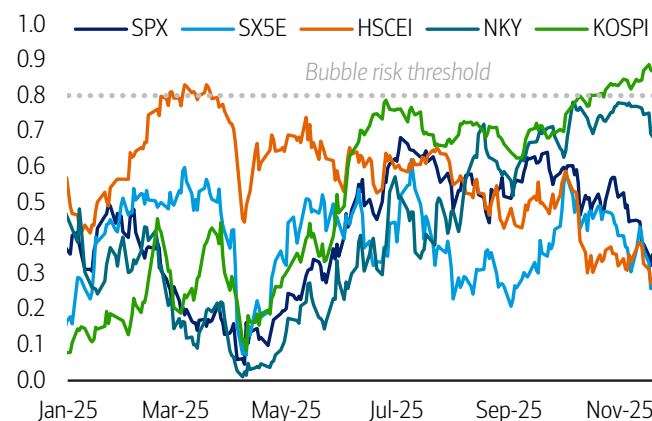


Source: BofA Global Research. Data from 1-Jan-25 to 21-Nov-25. Mag7 / Magnificent 7 refers to Apple, Amazon, Alphabet, Meta, Microsoft, Nvidia & Tesla. Underlying indices: NDX (NDX Index), Mag7 (BM7P Index), Nuclear (MVNLR Index), Quantum (BQTUM).

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Exhibit 6: Looking at global equity indices, the Nikkei and Kospi have seen far more bubble-like price action in 2025 than equity markets in the US, Europe and China

BofA Bubble Risk Indicator on global equity indices



Source: BofA Global Research. Data from 1-Jan-25 to 21-Nov-25.

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Valuation and late-stage momentum also say more room to run

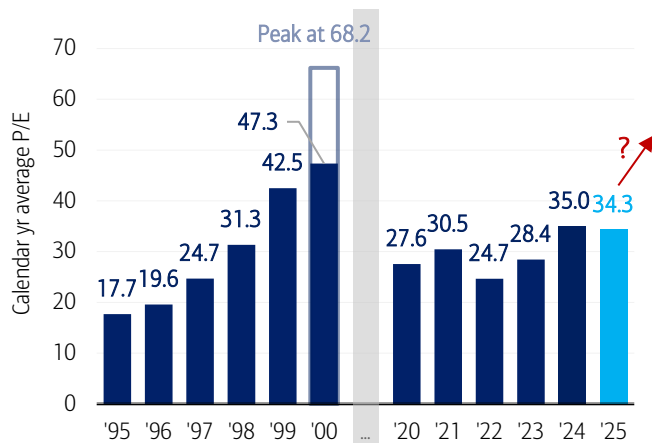
While US tech valuations have risen materially since 2022, they remain below the levels that were seen during the frothiest points of the late '90s dotcom bubble (Exhibit 7). More broadly, the dotcom bubble saw a clear decoupling of stock prices from fundamentals, which we are yet to see in the broader market (Exhibit 8). From this perspective, valuations (and hence prices) may still have room to expand.

As another indication there may be room to run, US equities/core AI are yet to see the rapid acceleration in momentum typical near bubble peaks. Indeed, seven out of eight asset bubbles within our universe saw nearly their highest upside momentum at their peaks (Exhibit 9), with the subset of equity bubbles (not including Gold, Oil and Bitcoin which are not tethered to “fundamentals”) typically seeing peak prices ~20% above their 3m moving averages (vs approx. -2% in the Nasdaq today; Exhibit 10).



Exhibit 7: While tech valuations have risen materially since 2022, they remain below those seen during the latter stages of the dotcom bubble, suggesting there may be further room for expansion

S&P 500 tech sector P/E during dotcom bubble ('95-'00) & the AI era ('22-)

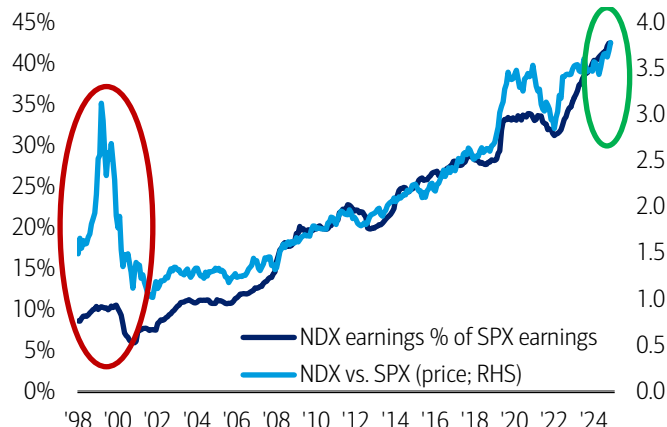


Source: BofA Global Research, Bloomberg. Data from 1-Jan-95 to 21-Nov-25. P/E ratio based on NTM expected EPS, and is calculated as a calendar year average.

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Exhibit 8: During the dotcom bubble, stock prices clearly decoupled from fundamentals, a phenomenon yet to be seen recently

Nasdaq 100 vs S&P 500 – earnings & performance

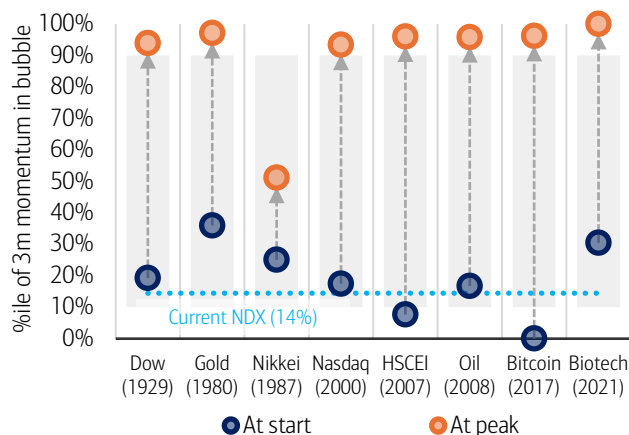


Source: BofA US Equity & Quant Strategy, FactSet. Data from Jan-98 to Oct-25.

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Exhibit 9: Most asset bubbles over the past century have seen close to their highest momentum at their peaks, with 7 out of 8 asset bubbles in our universe seeing their 3m momentum in the top decile

%ile rank of 3m momentum at the start and peak of historical asset bubbles

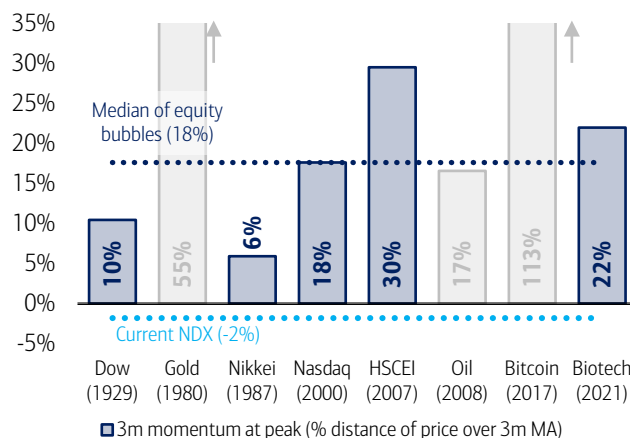


Source: BofA Global Research. Data from Oct-1926 to Feb-2021. NDX as of 21-Nov-25. 3m momentum = % distance of price above/below 3m moving average. Underlying Bloomberg tickers: INDU for Dow, XAU for Gold, NIKY for Nikkei, NDX for Nasdaq, HSCEI for HSCEI, CO1 for Oil, XBTUSD for Bitcoin, SPSIBI for Biotech.

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Exhibit 10: Equity market bubbles over the past century have typically seen peak prices be ~20% above their 3m moving average, which could be used as a rule of thumb when worried of a blow-off top

3m momentum at the peak of historical asset bubbles



Source: BofA Global Research. Data from Jan-1929 to Feb-2021. NDX as of 21-Nov-25. 3m momentum = % distance of price above/below 3m moving average. Underlying Bloomberg tickers: INDU for Dow, XAU for Gold, NIKY for Nikkei, NDX for Nasdaq, HSCEI for HSCEI, CO1 for Oil, XBTUSD for Bitcoin, SPSIBI for Biotech.

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Bubble signs aplenty in 2025 even if not in AI core

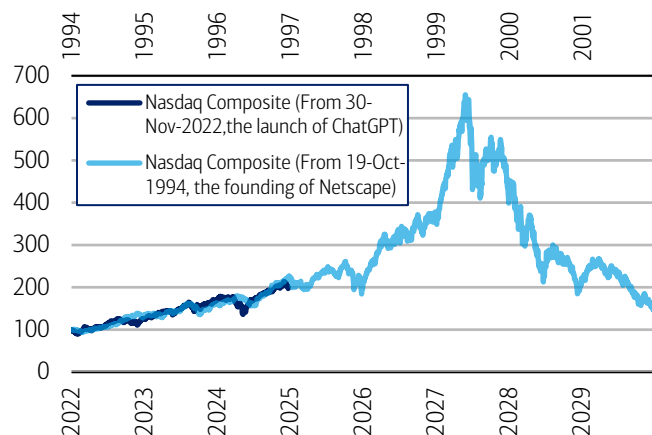
Like the growing political polarization, we see a similar divide in the bubble debate. Some argue AI is clearly a bubble about to pop (or maybe it has), while others say no bubble at all. The nuance in part hinges on where you look, and we agree it's not obvious it's in the core of US AI stocks. However, we still see evidence of US markets moving towards a more bubble-like state in 2025. Why?

- One simple observation (see Exhibit 11) is that the rise of the Nasdaq since the launch of ChatGPT in Nov-22 is following a similar trajectory (so far) to the period following Netscape's launch in Oct 94 (maybe just a coincidence?)

- Fears of a bubble by some sentiment measures are the highest since the GFC (see Exhibit 12). An AI bubble is the #1 risk expressed in our latest BofA Fund Manager survey (see Exhibit 8 in the [Global Fund Manager Survey](#)). Perception of risk (even if irrational) can result in assets behaving as if the risk is real.
- Our bubble indicator has broadly continued its uptrend in AI-exposed assets since ChatGPT's launch in Nov-22. It has also shown signs of froth in sub-sectors such as nuclear and quantum computing (see Exhibit 5).
- US tech IPOs in 2025 have seen some of their best first-week performance since the late 90s tech bubble (Exhibit 13).
- Markets are snapping back from drawdowns at a pace only commonly seen during the 90s tech bubble (Exhibit 32) and buy-the-dip strength is near 100yr extremes (Exhibit 14), typical of when FOMO dominates.
- Vol in the US has reset higher in '25 (even if marginally, Exhibit 74), despite markets rising (vol up/spot up), with the VIX floor increasing from 12 to 15.
- Importantly, it was profitable to carry long vol positions in 2025, whether via VIX futures paired with long S&P (which continues to reverse prior years of underperformance, as seen in Exhibit 15), or via the NDX upvar structure we recommended last year (see our [2025 Outlook](#)).
- Dispersion, a classic hallmark of bubbles, has continued to rise. This is true both for equal-weighted returns dispersion (Exhibit 16) as well as cap-weighted vol dispersion, which has been higher only 11% of the time since 1995, primarily in the dotcom bubble (Exhibit 17).

Exhibit 11: Nasdaq's post-ChatGPT rally is tracking a path similar to its climb after Netscape's 1994 launch. Is history rhyming or coincidence? The parallel invites questions about tech cycles

Index performance normalized to 100 at the start of each period; if the pattern holds, the current peak would arrive around April 2028.

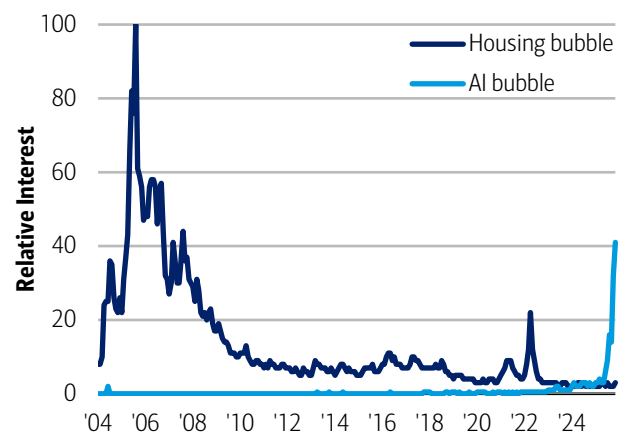


Source: BofA Global Research. Daily data from 19-Oct-1994 through 9-Oct-2002 and 30-Nov-2022 through 21-Nov-2025. Comparison uses Nasdaq Composite, not Nasdaq-100. If Nasdaq-100 were used, the 1990s internet surge would be outpacing today's AI-driven move. While Nasdaq-100 dominated in the '90s, its performance since ChatGPT's launch has been broadly in line with the Composite.

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Exhibit 12: AI bubble fears have risen to their highest so far in 2025 but remain far below the level recorded during the housing bubble leading to the GFC. Fear can become self-fulfilling

Google Trends data comparing US search interest in "Housing bubble" and "AI bubble" from 2004 to 2025 as a bubble sentiment indicator



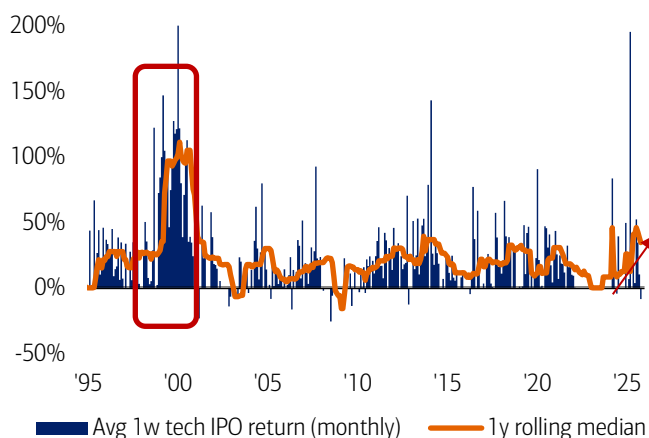
Source: BofA Global Research, Google Trends. Monthly data from Jan-04 to Nov-25. "Numbers represent search interest relative to the highest point on the chart for the given region and time. A value of 100 is the peak popularity for the term. A value of 50 means that the term is half as popular. A score of 0 means there was not enough data for this term" (Google).

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Exhibit 13: Recent tech IPOs have seen some of their strongest performance since the '90s dotcom boom, but on a sustained basis remain far from the frenzy of that era

Average US tech/communications IPO 1st week performance (monthly)

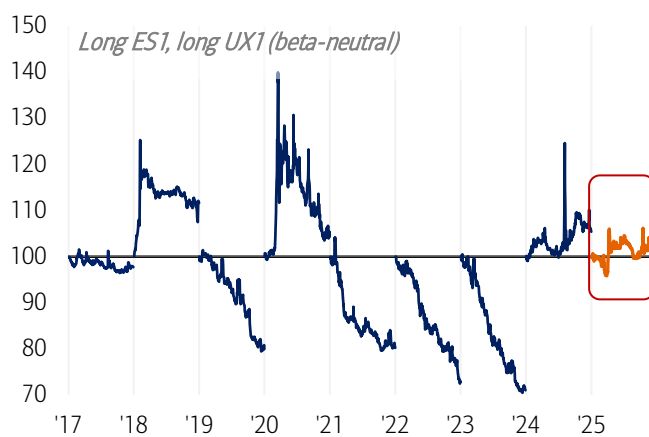


Source: BofA Global Research, Bloomberg. Data from Jan-95 to Oct-25. Min. size threshold of \$50m. Tech IPOs include those with GICS classification of Information Technology and Communication Services.

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Exhibit 15: Hedging long equities with long VIX futures would have been profitable in 2025 as it was in 2024, and would have continued to reverse prior years of underperformance

Hypothetical performance of long ES1, long UX1 with beta-neutral sizing

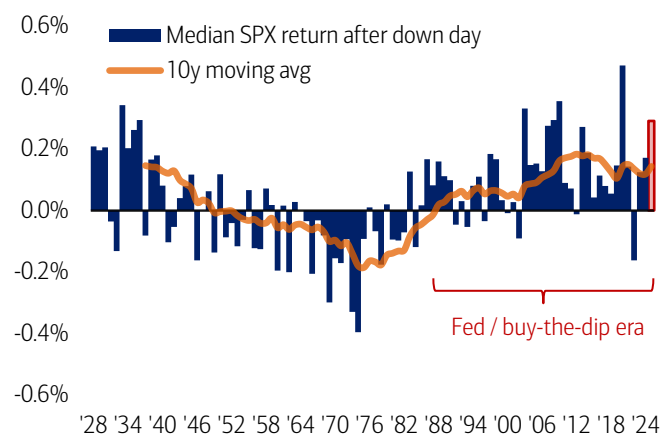


Source: BofA Global Research. Data from 1-Jan-2017 to 21-Nov-25. Futures are rolled 1 business day before respective expiries. Beta neutral sizing uses trailing 1y realized beta. This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein

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Exhibit 14: Dip-buying became increasingly attractive following Greenspan's introduction of the Fed put in 1987 and in 2025 has been among its strongest in 100yrs

SPX median 1-day return after down days (by calendar year)

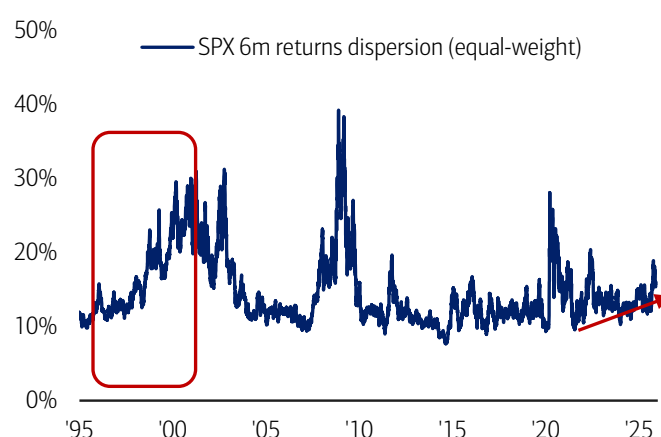


Source: BofA Global Research. Data from Jan-1928 to 21-Nov-25. 2025 bar is YTD.

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Exhibit 16: Returns dispersion in the S&P, which rose significantly in the late '90s as the dotcom bubble expanded, rose in 2025 as well (even on an equal-weighted basis)

SPX equal-weighted 6m returns dispersion



Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25.

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Exhibit 17: S&P vol dispersion, which also rose significantly during the dotcom bubble, has continued its trend higher in 2025 and now sits at levels higher than those seen in 2008

S&P Top 50 1y realized vol dispersion



Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25.

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Size, the real risk this time is different

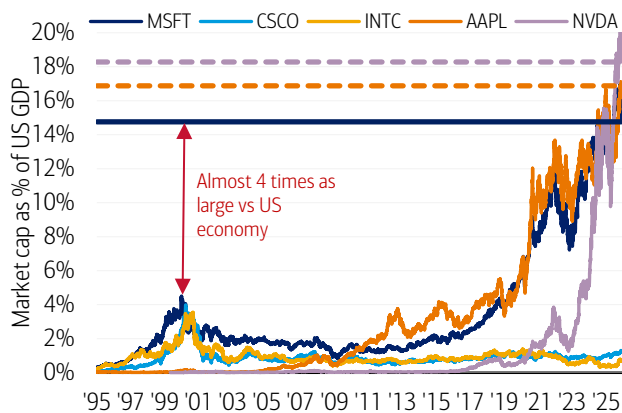
While our base-case is that a larger AI bubble will develop before it's over, as we first noted in Jun 2024, it may require the market to broaden into a wider and smaller set of AI winners. Large-cap US tech may struggle to generate the instability common in an asset bubble simply due to its sheer size and profitability.

Markets are more concentrated than any time in history, and currently NVDA is worth more than any country in Europe. With earnings forecasts continuing to outpace its share price, it may be difficult for NVDA to decouple from fundamentals in the way the largest stocks in the 90s tech bubble did. For example, if NVDA were to re-rate to Cisco's peak P/E in 2000 (~200), it would be an \$20.8tn company.

The risk would be that a negative surprise in forward earnings expectations cause it to sell-off without ever generating the instability typical of stocks in bubbles.

Exhibit 19: Compared to the dotcom bubble, today's tech megacaps are ~4 times as large with respect to the entire US economy, which may be a hurdle for the pace at which an AI bubble forms

Market cap as % of US GDP for largest 3 tech stocks in tech bubble & now

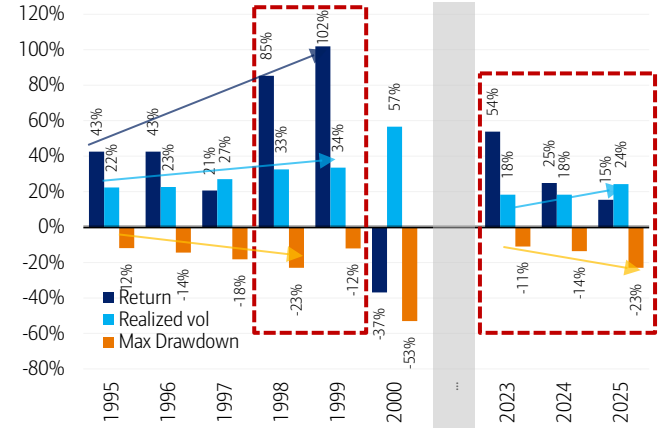


Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25.

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Exhibit 18: The dotcom bubble saw the Nasdaq 11x, with rising vol & drawdown sizes over the 5 year span. The AI era has been relatively muted so far, but vol and drawdowns have risen in 2025

Nasdaq 100 returns, realized vol & max drawdown for the dotcom bubble ('95-'00) & the AI wave ('22-')

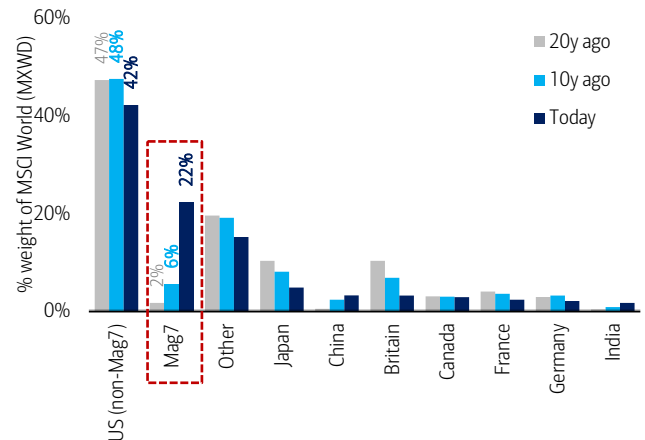


Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25. Calendar year metrics.

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Exhibit 20: The US is now ~65% of global equity market cap, up from 50% 20yrs ago, with nearly all gains being driven by large-cap tech...the next 10 years could simply be an extension of this supercycle

Geographical breakdown of MSCI World Index (MXWD) constituents



Source: BofA Global Research. Data from Oct-2005 to Oct-2025. Mag7 / Magnificent 7 refers to Apple, Amazon, Alphabet, Meta, Microsoft, Nvidia & Tesla.

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What about the risk AI doesn't live up to the hype?

This is the risk we are already in a bubble. Many project a multi-fold increase in demand for AI in the next several years. Jensen Huang, CEO of NVDA, believes we could see \$3-\$4trn/year in AI spend by 2030, and \$5trn/year long-run given its potential to amplify the productivity of the global knowledge economy. In addition, McKinsey ([see report](#)) projects ~\$7trn of cumulative spending by 2030 on data centers.

However, machines making steady progress towards fully replacing humans is a hot debate in the AI world and key questions remain:

- To what extent does the projected “double exponential” in token demand (from exponential AI user growth on top of the exponential growth/user as AI becomes more capable at reasoning) depend on progress towards AGI (human-level intelligence)? Most believe token demand is coming even if AGI is still far out.
- Can LLMs (Large Language Models) even get us to AGI, or will new technological breakthroughs be required? While many think new methods are required, they believe we will get to AGI, though the timing is highly uncertain.

The risk is that those currently experimenting with AI are too hopeful on its potential to augment productivity, and the reality is it takes more time to progress, resulting in disillusionment and a material market pull-back.

Again, the risk simply comes back to timing as the implications of achieving AGI or even perhaps ASI (super-human intelligence) are more profound than any technological achievement in human history – and until we achieve it, many will argue it's just out of grasp. The extreme potential of AI combined with high timing uncertainty will likely leave markets susceptible to bubble-type booms/busts for the foreseeable future.

Navigating bubbles

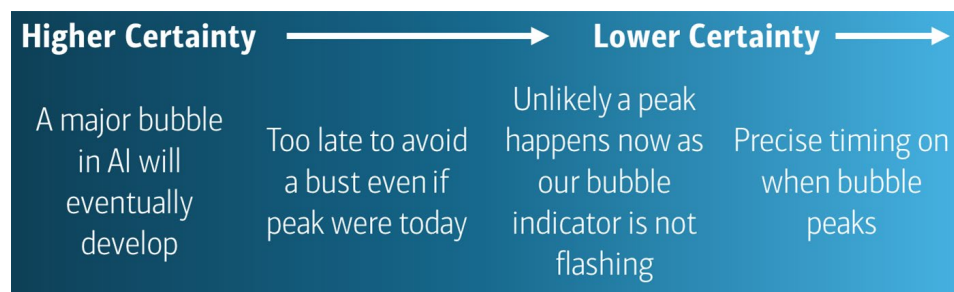
Timing becomes the biggest risk

As noted, it's easier to forecast that an eventual bubble is likely and whether markets are experiencing "bubble-like" behavior than it is to time the top. Bubbles often persist, not because you don't know you are in one, but rather due to the inability to time the end (as Chuck Prince infamously noted in July 2007).

Framing the level of conviction around what's more and less certain is key to developing a plan for bubble navigation. The fact AI will result in a larger bubble seems most probable. Next, given today's momentum, history suggests it's too late to avoid a bust (see, Too late to avoid a bust). The fact the core of US tech has yet to breach the top quintile of our bubble indicator since GPTs' launch suggesting more room to run would be next, and the least knowable is the precise timing of the top.

Figure 1: It's more certain a bubble develops than being able to time the top

Our assessments of which events are more certain vs. less certain



Source: BofA Global Research

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Diversify at your own risk, concentration + cash wins in bubbles

A common reaction to today's record concentration in US equities and uncomfortable valuations is to seek safety in diversification. However, while this may feel safe, the history of asset bubbles suggests it's sub-optimal and risky.

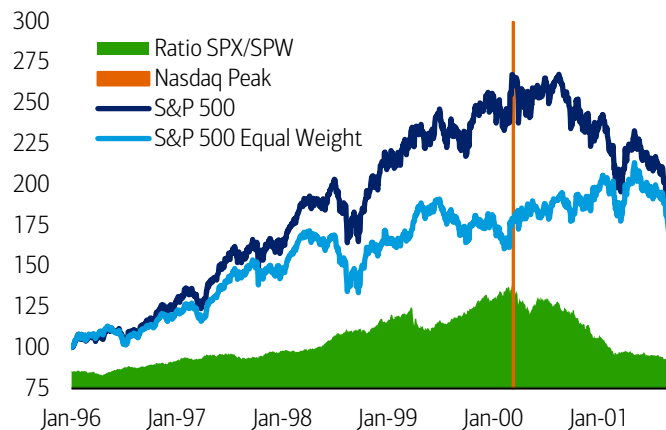
Assets at the forefront of the bubble tend to outperform until the peak. Hence diversifying away from the leaders is tantamount to calling the top (the riskiest proposition in our view). Staying long the concentration but diversifying with cash or adding a derivatives overlay would have been superior in the late 90s tech bubble (Exhibit 22).

Going back to the roaring-20s, the history of major equity asset bubbles also shows the regions at the center of the bubble tend to outperform global markets (Exhibit 23 - Exhibit 25). This is why the idea of "peak US exceptionalism" that took off as a theme in 2025 is antithetical to an eventual AI-bubble developing further.



Exhibit 21: The late 90s warned us diversifying too soon was costly as the S&P 500's concentrated cap-weighted exposure beat equal-weight approaches until the Nasdaq peak

Performance of S&P 500 (cap-weighted) vs. S&P 500 Equal Weight highlighting the Nasdaq peak in April 2000

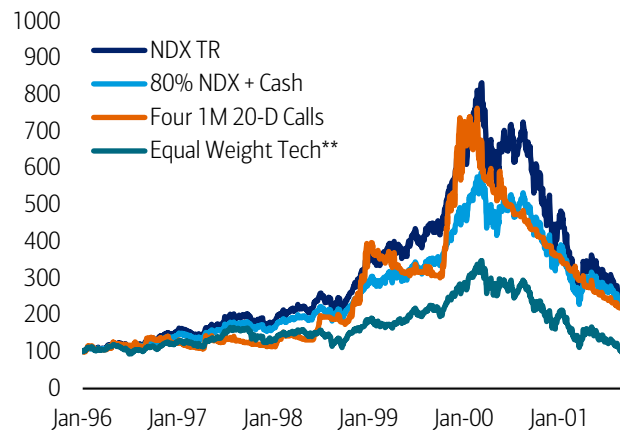


Source: BofA Global Research. Daily data from 19-Jan-1996 through 21-Sep-2001.

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Exhibit 22: During the '90s bubble, using cash or call options was a more effective way to de-risk Nasdaq exposure ahead of the peak than shifting to equal-weight, which outperformed after the crash

Performance of Nasdaq-100 strategies during the tech bubble, comparing pure NDX, NDX with cash or monthly NDX calls, and equal-weight tech

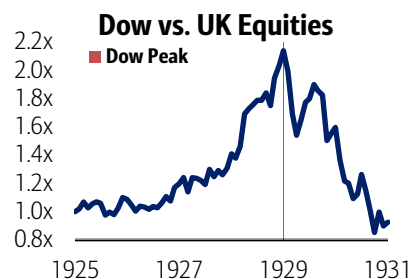


Source: BofA Global Research. Daily data from 19-Jan-1996 through 21-Sep-2001. **Tech includes names in the Information Technology sector under the Global Industry Classification Standard, along with companies in other sectors that are primarily internet/tech-driven (e.g., Amazon). A cap-weighted benchmark of our Tech names closely tracks the Nasdaq during the 90s. We equal-weight our Tech names due to the unavailability in that period of both the Nasdaq equal-weight index and Nasdaq membership data needed to estimate equal-weight exposure.

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Exhibit 23: History shows bubbles reward concentration...

US led the 1920s bubble—UK lagged

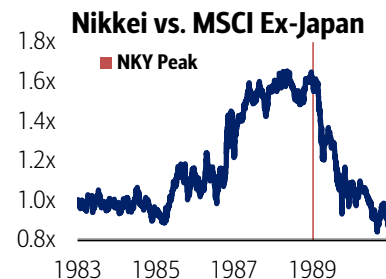


Source: BofA Global Research. Monthly data from Aug-1925 to Aug-1931. We compare to UK equities as a global non-US benchmark at that time.

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Exhibit 24: ... leaders dominate global benchmarks until the top...

Japan dominated the 1980s boom

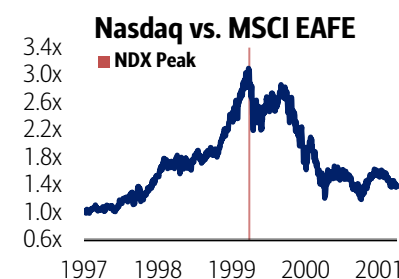


Source: BofA Global Research. Daily data from 30-Dec-1983 to 29-Dec-1991. MSCI ACWI Ex-Japan data starts on 31-Dec-1987. Prior data is proxied with an allocation to the S&P 500 and FTSE 100 that tracks MSCI ACWI Ex-Japan through Dec-91.

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Exhibit 25: ...then the unwind erases excess gains.

US tech beat global peers until bubble peak



Source: BofA Global Research. Daily data from 31-Dec-1997 to 27-Mar-2002.

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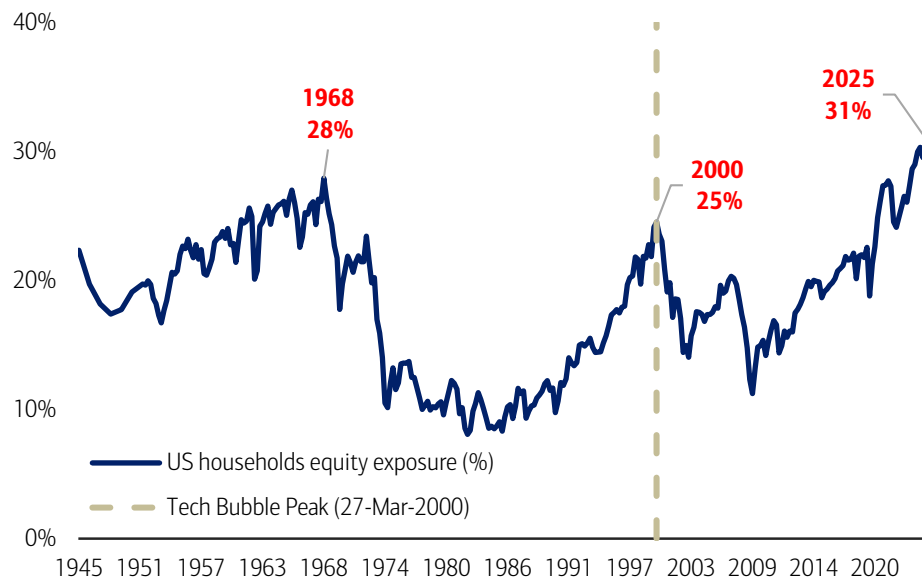
The right tail, it's all about the magic

Retail is the forcing function

Value-insensitive investors with long, indefinite investment horizons (mainly retail) will likely lead the asset bubble. Their competitive fear of missing out (FOMO) on owning a slice of the magical new future is the basis for a bubble. As retail leads, they will painfully pressure value-sensitive investors (professionals) with much shorter time-horizons to chase. In terms of US stocks, this will also likely force in those that abandoned US assets due to fears over peak US exceptionalism. No doubt the AI bubble would be further progressed if this “de-risking” from the US had not occurred in 2025.

Exhibit 26: US household equity holdings are at an all-time high

Ratio of US household holdings of equities to total financial assets



Source: BofA Global Investment Strategy, Haver, Federal Reserve.

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Crypto could be a leading indicator for retail in AI

Equity and crypto markets were once distinct pools of capital and risk, but the line between the two is becoming increasingly ambiguous with price action in crypto now often mirroring traditional financial (TradFi) markets. However, in some instances crypto appears to move in advance of public equities. For example, weakness in bitcoin preceded the decline in Nasdaq in Jan & Nov (Exhibit 27). There is also empirical evidence that when bitcoin materially underperforms Nasdaq then Nasdaq is biased to have negative returns, and the reverse if bitcoin outperforms Nasdaq (Exhibit 28).

We may rationalize crypto as being a leading indicator to equities due to:

- Crypto markets have lower barriers to entry relative to TradFi allowing a wider array of investors to participate, potentially allowing more information to be captured in decentralized markets and sometimes in advance of equities.
- Retail is likely to lead in the inflation of an asset bubble, as we have argued, which could move crypto markets first or more quickly than stocks due to the higher retail ownership & lack of fundamental values to anchor prices in crypto vs public equities.
- Crypto may be among the first markets to be impacted by investors reaching for liquidity to cover margin by virtue of being a 24/7/365 “cash machine”; for example, in the 05Aug24 fragility shock, which primarily occurred outside US market hours, bitcoin plunged in connection with the Yen carry trade unwind.



Exhibit 27: In some recent instances bitcoin has acted as a leading indicator to equities, for example, BTC sold off before Nasdaq in January and November

Performance of Nasdaq-100 and Bitcoin over the last 1yr

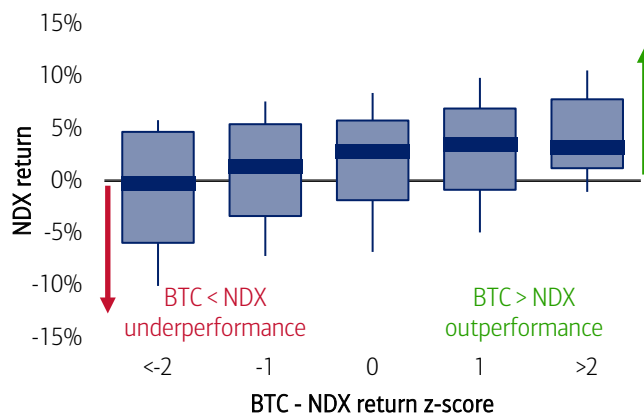


Source: BofA Global Research, Bloomberg. Data from Nov-24 to 21-Nov-2025.

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Exhibit 28: Empirically BTC underperforming NDX in z-score terms generally results in NDX forward returns becoming biased more negative and the reverse if BTC outperformed NDX

Distribution of forward NDX 1m returns conditional on the spread between the BTC and NDX 1m return z-score*



Source: BofA Global Research, Bloomberg. Data from Jan-2020 to 21-Nov-2025. *For BTC and NDX the EWMA z-score of 1m returns is computed independently, then the difference is taken. Under/out-performance of BTC vs NDX is measured in z-score terms not raw returns. The box charts represent the 10/90th (whiskers), 25/75th (box edges), and 50th (middle bar) percentiles.

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Watch out for magical moments

The latest AI boom kicked off in November 2022 with the launch of ChatGPT, a clear magical moment. The risk is that we get more of these. Many have argued that Agentic AI is the next big wave of innovation. Progress seems faster in non-consumer facing applications so far, but an AI agent booking a holiday or planning a party could be another ChatGPT moment that fuels optimism.

A ChatGPT moment is also coming in physical AI (the robots are coming) and most likely will first play out in the expansion of self-driving car offerings with Google's Waymo and Tesla at the forefront.

Exhibit 29: Further ChatGPT moments in new AI modalities risk fueling further FOMO

Potential "magical moments" in AI

Watch out for magical moments



Source: BofA Global Research

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The left tail, we are in too deep

A thing to fear is fear itself

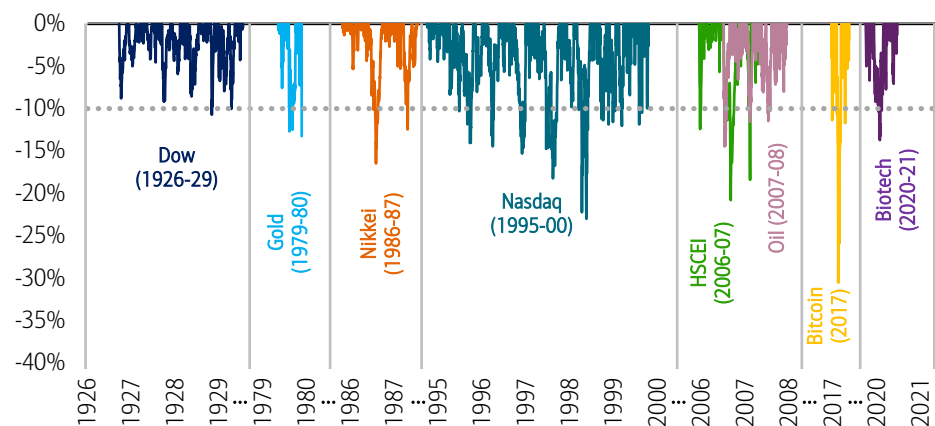
Many factors including our bubble indicator, valuations, a lack of sharp acceleration in returns, and easing financial conditions (not tightening), suggest we may be far from a bubble top; nonetheless bubble fears are real.

The latest example is concerns over rising debt-loads among those financing the AI build-out, circular financing, and chip depreciation extensions. All of this derives from the high future uncertainty over how the benefits of today's massive AI investments will pay out. It's this uncertainty that causes larger pullbacks as bubbles build (but also faster snapbacks) and ultimately is how vol rises at the same time as price.

As Exhibit 30 illustrates, each of the 8 major asset bubbles we have studied since 1900 has witnessed 10%+ drawdowns in the lead-up, and longer bubbles have seen multiple. Also notable is the tendency for these drawdowns to get larger as the bubble progresses, as happened in the late '90s for example, which is natural as valuations and volatility both become more extended as momentum accelerates.

Exhibit 30: Within the universe of 8 asset bubbles over the past century, every bubble has seen a correction of at least 10% over the course of the bubble build (and prior to the ultimate bust)

Max drawdowns during historical asset bubbles



Source: BofA Global Research. Data from Jan-1929 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow (Oct-1926 to Sep-1929), XAU for Gold (Jul-1979 to Jan-1980), NIKY for Nikkei (Jan-1986 to Oct-1987), NDX for Nasdaq (Jan-1995 to Mar-2000), HSCEI for HSCEI (Jun-2006 to Oct-2007), CO1 for Oil (Jan-2007 to Jul-2008), XBTUSD for Bitcoin (Jul-2017 to Dec-2017), SPSIBI for Biotech (May-2020 to Feb-2021).

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Faster snapbacks also a feature of bubbles

As Exhibit 31 illustrates, snapbacks from pre-peak drawdowns also tend to be faster. Among the 10 fastest historical snapbacks from 10%+ drawdowns, at least one of them were during the asset bubble periods (but prior to the peaks) we noted (blue shaded cells). For the two “tech-driven” bubbles, the Dow in the ‘20s and Nasdaq in the late ‘90s, the fastest snapbacks in history were during the bubble, and for the late 90s internet bubble, all 10 of the fastest snapbacks were during the bubble. If looking at the fastest snapbacks from 15%+ drawdowns, the fastest three in Nasdaq’s history were in the late ‘90s bubble and the 5th fastest was this summer following “Liberation Day” (Exhibit 32).



Exhibit 31: Bubbles have historically seen pre-peak drawdowns snap back with rapid speed, with the reversions often being amongst the fastest in these assets' histories

Top 10 drawdowns (10%+) with fastest reversions for 8 assets in which bubbles occurred (days to new ATH in parentheses); blue cells indicate drawdown during asset bubble period

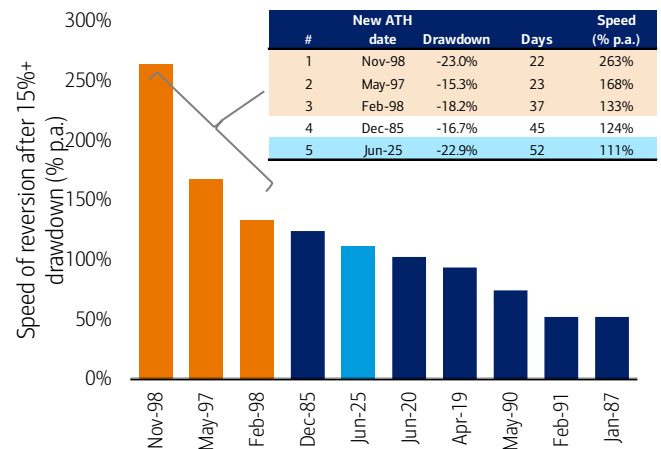
#	Dow (1929)	Gold (1980)	Nikkei (1987)	Nasdaq (2000)	HSCEI (2007)	Oil (2008)	Bitcoin (2017)	Biotech (2021)
1	-11% (13)	-20% (44)	-12% (26)	-10% (4)	-18% (6)	-20% (6)	-25% (2)	-16% (6)
2	-11% (16)	-13% (40)	-16% (47)	-12% (5)	-21% (46)	-13% (5)	-32% (3)	-12% (9)
3	-23% (39)	-15% (59)	-15% (43)	-12% (8)	-89% (2055)	-10% (6)	-23% (3)	-36% (39)
4	-10% (21)	-13% (66)	-26% (86)	-12% (11)		-15% (20)	-19% (3)	-13% (20)
5	-12% (32)	-29% (217)	-21% (88)	-23% (22)		-29% (39)	-19% (3)	-14% (24)
6	-19% (59)	-44% (385)	-21% (98)	-15% (23)		-11% (16)	-11% (2)	-14% (41)
7	-10% (40)	-21% (306)	-13% (70)	-12% (18)		-11% (19)	-20% (4)	-13% (40)
8	-12% (48)	-45% (1195)	-24% (261)	-14% (22)		-28% (56)	-38% (10)	-29% (120)
9	-37% (166)	-70% (2180)	-37% (852)	-11% (19)		-11% (22)	-14% (6)	-49% (489)
10	-17% (87)		-82% (3659)	-10% (18)		-28% (61)	-24% (12)	-79% (2606)

Source: BofA Global Research. Data from Jan-1929 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow, XAU for Gold, NKY for Nikkei, NDX for Nasdaq, HSCEI for HSCEI, CO1 for Oil, XBTUSD for Bitcoin, SPSIBI for Biotech.

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Exhibit 32: The Nasdaq's most recent rally back to all-time highs (after Liberation Day) was its 5th fastest reversion from a 15%+ drawdown, with the fastest being those in the late '90s dotcom bubble

Fastest Nasdaq 100 rallies to new all-time highs after a 15%+ drawdown



Source: BofA Global Research. Data from 4-Feb-85 to 21-Nov-25. 'Speed of reversion' is calculated as an annualized ratio of drawdown to the number of days taken to reach new all-time highs.

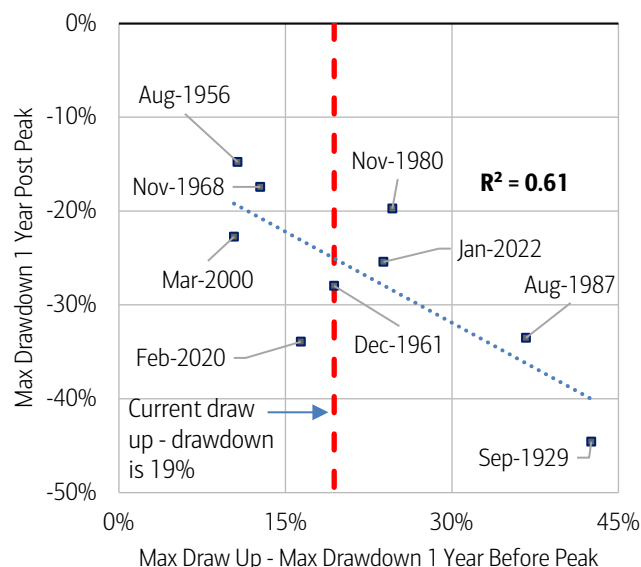
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Too late to avoid a bust

As we noted in our 2025 outlook (see [The roaring 2020s?](#)), history shows that larger booms result in larger busts. In fact, the Nasdaq100 drawdown we witnessed in 2025 was nearly exactly in-line with the historical relationship we published last year (see Feb-2025 dot on Exhibit 34). But as the vertical red lines show for both the S&P and NDX, given the draw-ups we have seen since liberation day lows, it suggests that if we had a meaningful drawdown from here, it would generate an ~ 25% drawdown in the S&P and 30% drawdown in NDX. This could be caused by exogenous factors (the most obvious risk we believe would be financial conditions tightening, for example due to a surprise in inflation) but it could also result from an unknown-unknown. The bottom line is that markets become more vulnerable to pullbacks the more bubbles build and the larger the boom the larger the bust.

Exhibit 33: Bigger the boom, the bigger the bust; relationship has held generally well in the last 100yrs of S&P history

Largest draw ups – drawdowns in US equities leading into peaks vs max drawdowns 1 year post the peak.

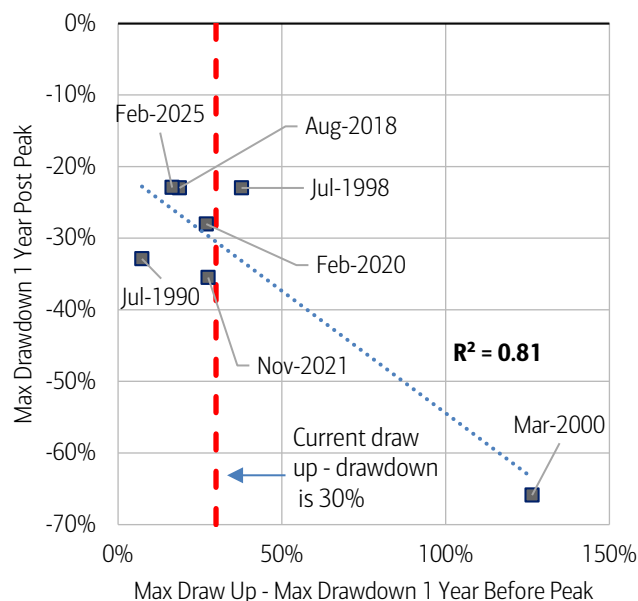


Source: BofA Global Research. Data from 1-Jan-1921 to 25-Nov-2025. Prior to the S&P 500 data beginning on 30-Dec-27 we use Dow Jones Industrial Average returns. Peaks are defined as the highs before a drawdown greater than 20% and peak dates are noted in the labels. On the x-axis is the difference between the max draw up and max drawdown in the year prior to reaching the peak. A draw up is the return from the low in the year prior to the peak while a drawdown is the return from the high in the one year before the peak to the low. On the y-axis is the max drawdown in the year following the peak, defined as the return from the peak to the low over the next year.

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Exhibit 34: While the 2000s tech bubble wasn't the largest boom in S&P, it was in Nasdaq; it was also the biggest bust

Largest draw ups -drawdowns in the NASDAQ-100 leading into peaks vs max drawdowns 1 year post the peak



Source: BofA Global Research. Data from 4-Feb-1985 to 25-Nov-2025. Peaks are defined as the highs before a drawdown greater than 20% and peak dates are noted in the labels. On the x-axis is the difference between the max draw up and max drawdown in the year prior to reaching the peak. A draw up is the return from the low in the year prior to the peak while a drawdown is the return from the high in the one year before the peak to the low. On the y-axis is the max drawdown in the year following the peak, defined as the return from the peak to the low over the next year.

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Mid-terms means Trump put strike rising, but what if they lose?

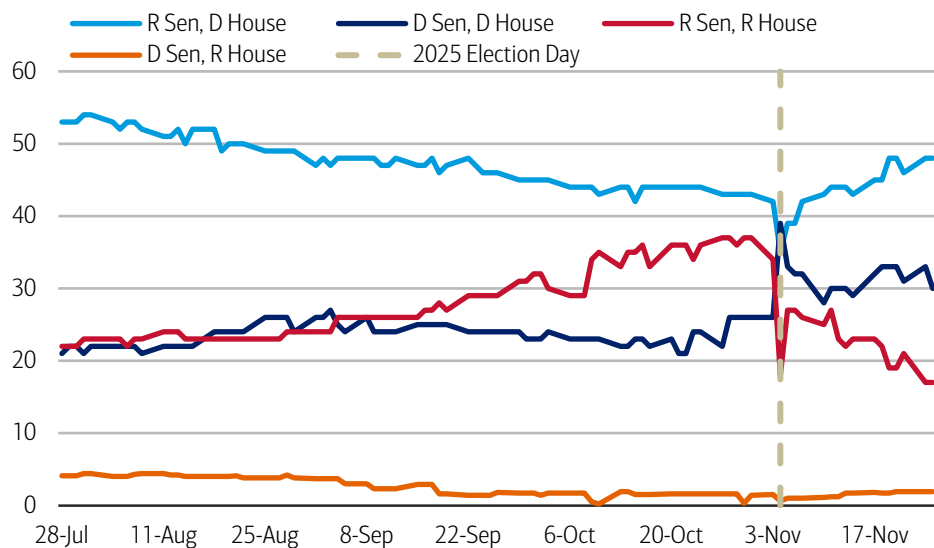
An observation we have repeated throughout this year is that policy volatility is a feature of the US administration not a bug. However, the administration's ability to take risk is clearly diminishing as the Nov-26 US mid-term elections approach (their biggest tail risk is losing the mid-terms) and suggests the strike of the "Trump put/Bessent put" is rising as a result. This supports the right tail.

However, if Republicans lose unified control in Washington (Polymarket calls for about 80% probability currently), it will likely throw sand in the gears of any stimulative administration policy and result in political morass or possibly worse. While it's possible it doesn't derail AI momentum, particularly if it was becoming late-stage bubble like (note bubbles have tremendous power to absorb macro risk in their final stages, see [The roaring 2020s?](#)), it does represent a risk of a momentum slowdown or even potentially triggering a significant pullback.



Exhibit 35: According to Polymarket there is an 80% chance Republicans lose unified government

Polymarket 2026 US midterm balance of power odds



Source: BofA Global Research, Bloomberg. Data from 28-Jul-25 to 25-Nov-25. Bloomberg tickers: R Sen, D House: PLYMC2A6, D Sen, D House: PLYMC2A4, R Sen, R House: PLYMC2A7, & D Sen, R House: PLYMC2A5.

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Leverage, the kryptonite of bubbles**Rarely the same, but it shows up every time**

During the largest of the railroad stock bubbles in Britain in the 1840s, it was common practice for investors to make only a small initial payment for their shares with a commitment to produce additional funding as the company called for it. Hence there was embedded leverage (10 x 1 not uncommon) in what Gareth Campbell in his paper, “Deriving the railway mania” describes as derivative-like positions.¹

In the US during the 1920s, 10 x 1 leverage was also commonly provided by banks to retail investors to purchase shares on margin. In early 1928, prior to the crash in October 1929, the Fed attempted to reign in speculation by tightening policy, which they continued despite push-back from some banks until the crash in 1929.

During the internet boom in the late ‘90s, leverage stemmed from corporate capital structures of the telecom companies which overbuilt fiber infrastructure and overspent on the belief that there would be continued funding rounds. Following IPOs, investors used margin and options to trade companies that lacked proven business models.

While the bubble leading up to the GFC in 2008 is not one we discuss here (as it did not express bubble-like characteristics within public markets in the runup), it was a golden era for hedge fund capital and embedded leverage growth (see Exhibit 9 in our [2024 Year Ahead](#)). And there was also significant leverage embedded into banks and holders of structured credit products.

But where is it this time? Levered & inverse ETFs, private assets?

Today, many fear that the toxic leverage is embedded in levered and inverse ETFs or perhaps in private assets which have seen rapid growth in recent years. While the number of ETF products has exploded in recent years, it does not tell the whole story.

¹ Gareth Campbell (2013). “Deriving the railway mania.” *Financial History Review*, 20, pp 127

Levered and inverse ETFs not yet a tail risk for equities

The popularity of levered & inverse funds is rising but are the risks growing too?

Leveraged and inverse (L&I) funds are by no means a new phenomenon and easily pre-date the global financial crisis (Exhibit 36). But these products have undergone a renaissance with a deluge of new L&I ETF listings including at the individual stock level. The growing AUM and “short gamma” nature of L&I funds (which requires them to buy in a rally & sell in a decline) has caused some investors to raise concerns that L&I products could cause stocks to become more fragile and ultimately may be a lurking tail risk waiting for the right catalyst to drive a severe broad based equity market decline.

There is precedent for L&I products to cause a “tail wags the dog” style event, for example as famously occurred in the VIX ecosystem on 05-Feb-2018 when 1m VIX futures doubled and the large inverse VIX ETP complex violently unwound. However, the US stock market has far deeper liquidity than the VIX ecosystem and despite a steady stream of new products, the size of US L&I ETFs relative to all US ETFs is quite small and has seen little growth (Exhibit 37). Moreover, the bulk of L&I assets are on highly liquid underliers, such as the S&P 500, Nasdaq 100, NVDA, and TSLA setting a high bar for L&I funds to have broad market impact (Exhibit 38).

Deep equity market liquidity is a high hurdle for the L&I rebalance to overcome

L&I market impact is a function of two main parameters: (i) the return in the underlying, which determines the L&I rebalance size, and (ii) market liquidity which we assess via volume, though we grant volume isn’t always a perfect reflection of likely price impact. By intentionally considering scenarios with extreme market drawdowns and constrained liquidity we can infer if indeed L&I funds are a lurking tail risk for the US stock market.

- Adjusting the rebalance of L&I products by the underlier’s S&P beta implies ~\$12bn of S&P notional to sell from all L&I funds per 1% drop in the S&P (Exhibit 39)
- Comparing the total L&I rebal. across all funds only to S&P/Nasdaq volumes ensures we understate market liquidity (by excluding stock volumes) and overstate impact
- Over the last decade a >5% 1-day SPX drawdown has always resulted in S&P + Nasdaq volumes of 1.75x the norm and generally far higher (Exhibit 40)

Combining these observations, in a repeat of 1987’s 1-day 20% SPX decline the total L&I rebalance across all underliers (stock, index, and thematic funds) could account for ~16% of volumes (1.75x volume multiple) if trading activity is historically low – an unlikely outcome for such an extreme decline. If, more reasonably, trading is elevated then L&I funds could be ~7% of volume (3.75x volume multiple), see Exhibit 41.

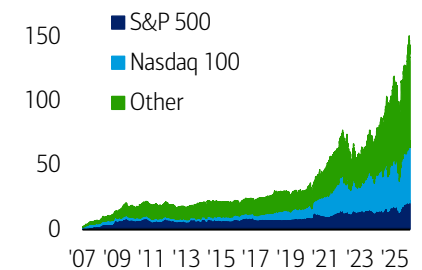
In a selloff even if lev-long investors “capitulate” L&I are unlikely to be a tail risk

To steelman the argument that L&I funds don’t pose a threat to broad market equities, we can further assume that in a 20% 1d SPX decline all lev-long fund investors capitulate and fully liquidate their positions due to poor performance (an assumption not made previously), counter to the usual buy-the-dip behavior; see Exhibit 42. In the extreme case of only 1.75x volumes, the combined L&I rebalance and lev-long liquidation flows could account for ~23% of trading activity though <11% may be more realistic as volumes would likely be quite elevated in practice (Exhibit 43). Clearly 23% of volumes would have noticeable market impact, however, even in this overly aggressive scenario L&I funds would still undoubtedly be (i) the side show to the catalyst which instigated a 20% SPX decline & (ii) unlikely to cause a runaway spiral of price impact leading to more rebalance flows (and yet more price impact) as occurred with the inverse VIX ETP unwind in 2018.

In a more realistic redux of the ‘87 crash: (i) volumes would likely set all-time record highs, (ii) L&I funds wouldn’t be forced to only use S&P / Nasdaq to hedge thematic & stock products as we have assumed, (iii) market wide circuit breakers, a post ‘87 invention, may help reduce the severity of the equity decline, & (iv) the well-trained buy-the-dip response could lead to historic inflows to lev-long products as investors position for Fed policy put intervention – all of which would greatly reduce the L&I equity selling pressure & impact.

Exhibit 36: US levered and inverse (L&I) funds now have ~\$130bn of AUM

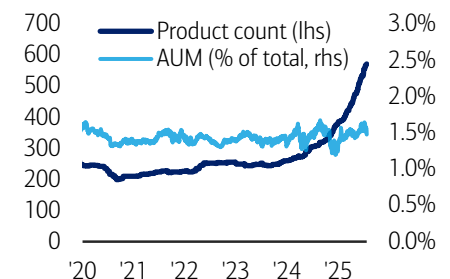
US L&I assets under management (\$bn)



Source: BofA Global Research. Data from Jan-07 to 17-Nov-25. The universe primarily includes US based L&I ETFs and ETNs with daily leverage resets. ETFs/ETNs on US underliers with non-US tickers are not included as they are a small fraction of the US domiciled AUM and have little impact on the analysis.
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Exhibit 37: Despite >300 new L&I ETFs since 2024, the L&I complex has not grown relative to the broader ETF market in AUM terms

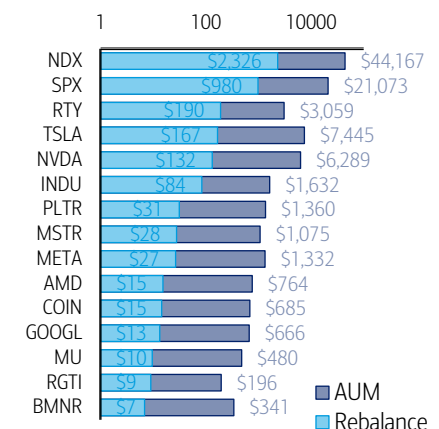
US domiciled L&I ETF (i) product count and (ii) AUM vs total US ETF AUM



Source: BofA Global Research, EPFR. Data: 2020 to 13-Nov-25.
BofA GLOBAL RESEARCH

Exhibit 38: The bulk of L&I assets are on deeply liquid underliers but L&I funds do exist on some small / lower quality stocks

L&I AUM and notional rebalance per 1% move for indices & stocks (\$mn, log scale)

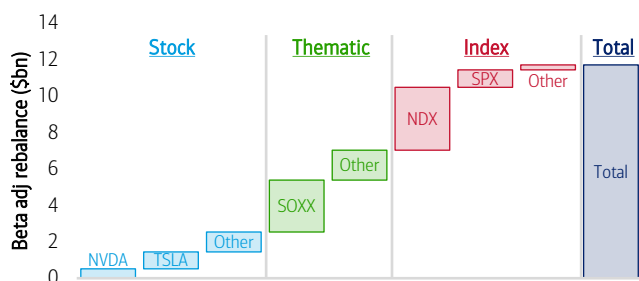


Source: BofA Global Research. Data as of 17-Nov-25. Top 15 products with largest L&I rebalance. Thematic L&I omitted.
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Exhibit 39: Aggregating the beta adjusted L&I rebalance across all funds implies ~\$12bn of S&P notional to sell per 1% S&P decline

L&I beta adjusted notional rebalance per 1% move in the S&P*

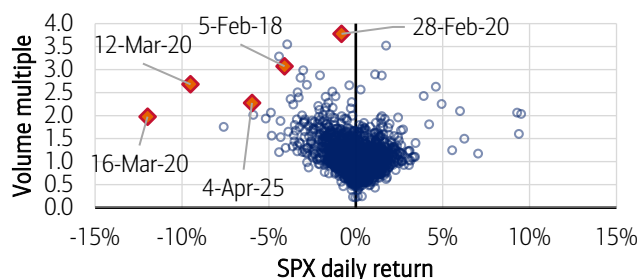


Source: BofA Global Research. Data through 17-Nov-25. *The adjusted rebalance translates the rebalance per 1% move in the L&I underlying into an equivalent amount of S&P notional to trade per 1% move in the S&P. The "Thematic" category captures products on sectors & themes (e.g., semis).

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Exhibit 40: Since 2015, the S&P declining by >5% in 1-day has always resulted in volumes rising by at least 75% (and often far more)

SPX 1d return vs S&P + Nasdaq volume multiple (= 1d volume / 1yr avg)*



Source: BofA Global Research. Data from 01-Jan-15 to 17-Nov-25. *S&P volume includes ES + SPY while Nasdaq includes NQ + QQQ. Nasdaq volumes are beta adjusted prior to combining with S&P.

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Exhibit 41: In a repeat of the 1987 crash the L&I rebalance could account for 7-16% of volumes. While in a repeat of March 2020, L&I rebalance flows may be 4-10% of trading activity

Theoretical total equity L&I mechanical rebalance as a percentage of S&P + Nasdaq volume depending on: (i) the 1-day decline in the S&P and (ii) the multiple of a typical full day's volume traded

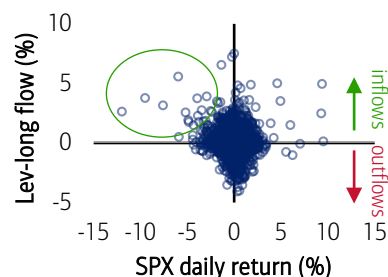
S&P 1-day decline	Multiple of a typical days volume														
	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00		
20%	27.8	22.2	18.5	15.9	13.9	12.4	11.1	10.1	9.3	8.6	7.9	7.4	7.0	1987 Crash (-20%)	
18%	25.5	20.4	17.0	14.6	12.8	11.4	10.2	9.3	8.5	7.9	7.3	6.8	6.4	March 2020 (-12%)	
16%	22.7	18.2	15.1	13.0	11.4	10.1	9.1	8.3	7.6	7.0	6.5	6.1	5.7	Liberation Day (-6%)	
14%	19.9	15.9	13.3	11.4	9.9	8.8	8.0	7.2	6.6	6.1	5.7	5.3	5.0		
12%	17.1	13.6	11.4	9.7	8.5	7.6	6.8	6.2	5.7	5.2	4.9	4.5	4.3	Volume "lower bound" (1.75x volume mult.)	
10%	14.2	11.4	9.5	8.1	7.1	6.3	5.7	5.2	4.7	4.4	4.1	3.8	3.6		
8%	11.4	9.1	7.6	6.5	5.7	5.1	4.5	4.1	3.8	3.5	3.2	3.0	2.8	Feb 2020 Volume (3.75x volume mult.)	
6%	8.5	6.8	5.7	4.9	4.3	3.8	3.4	3.1	2.8	2.6	2.4	2.3	2.1		
4%	5.7	4.5	3.8	3.2	2.8	2.5	2.3	2.1	1.9	1.7	1.6	1.5	1.4		
2%	2.8	2.3	1.9	1.6	1.4	1.3	1.1	1.0	0.9	0.9	0.8	0.8	0.7		
0%	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Source: BofA Global Research. Data through 17-Nov-25. A volume multiple of 1.75 is denoted as a "lower bound" since, per Exhibit 40, over the last decade a 5% S&P decline has always resulted in a minimum 75% increase in volumes. Figures here assume 0 investor outflows.

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Exhibit 42: The learned response to buy-the-dip in equity markets is clearly present in levered long funds which typically see inflows during large S&P declines

SPX 1d return vs the lev-long product flow rate

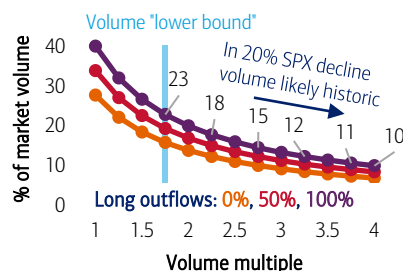


Source: BofA Global Research. Data from Jan-15 to 17-Nov-25. The "flow rate" aggregates change in shares outstanding across all levered-long products. For more details on this approach see our [GFI 21-Oct-25](#) report on VIX ETPs.

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Exhibit 43: In a repeat of the '87 crash the L&I rebalance + a full unwind of levered longs could be ~23% of volumes if liquidity is poor but <11% may be more realistic

L&I rebal. as % of vol. assuming: (i) a 20% 1d SPX decline & (ii) lev-long ETFs have 0-100% outflows

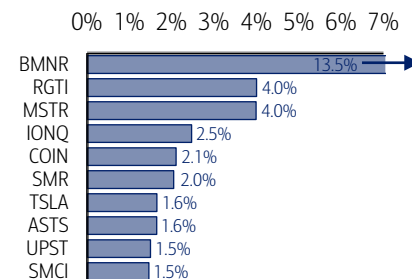


Source: BofA Global Research. Data through 17-Nov-25. Volume is the aggregate S&P + Nasdaq volume. 100% outflows from lev-longs results in a complete unwind of all assets in those products. Volume "lower bound" corresponds to a volume multiple of 1.75x as per Exhibit 40.

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Exhibit 44: L&I funds may be more prone to affect stocks which have a mismatch between L&I size, volatility, and liquidity

Top 10 stocks with the largest L&I standardized rebalance*



Source: BofA Global Research. Data through 17-Nov-25. *The standardized rebalance = $A \times B / C$, where (A) = the rebalance per 1% move, (B) = the underlying EWMA daily realized vol, (C) the underlier's typical notional volume. BMNR screens high partly from extreme realized vol. Figures above don't include impact of index/thematic L&I funds on stocks.

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What about private markets?

Private markets have grown rapidly, becoming an increasingly important part of global capital formation. (See our Thematic Investing team's [Private Companies Primer](#)). This expansion is supported by innovations in liquidity solutions and financing structures that enable scale, credit expansion and balance sheet recycling, which in turn allow markets to meet rising demand for financing. As these mechanisms proliferate, leverage is gradually building across the ecosystem, creating interconnections that could amplify future stresses. The anticipated surge in AI-related investment, estimated at \$5.2 trillion by 2030, is likely to accelerate these dynamics, reinforcing private markets' role in fueling growth while also highlighting growing fragility and the importance of monitoring systemic risk. We address these vulnerabilities by constructing private asset hedges using liquid public instruments that can help mitigate private market exposures.

How to hedge a crack in fast-growing private assets

Private assets: Big and growing, led by credit

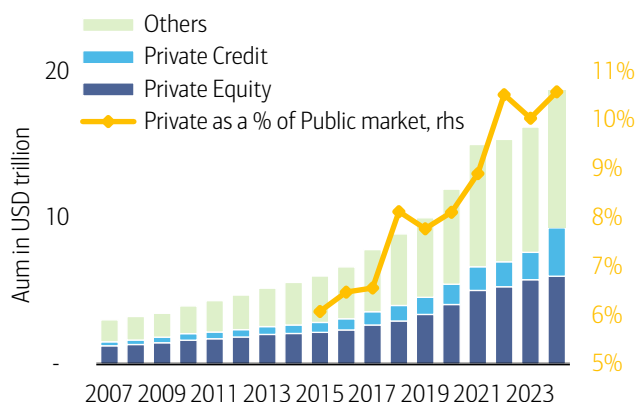
Global private market assets of ~\$18trn represent ~10% of the public equivalent, a meaningful share of global capital formation that has doubled in the last 10y (Exhibit 45). Private credit has emerged as the market's fastest-growing segment (up 75% YoY to \$3trn, Exhibit 46). Now, estimates² suggest private assets could grow to as much as \$24trn by 2029, propelled by a perceived 'illiquidity premium' and regulatory constraints that limit traditional bank lending, thus creating space for private lenders to step in.

Private exposure in the public domain: Banks and insurers

Banks have found it more capital-efficient to lend indirectly via private credit vehicles rather than on their own balance sheets, reinforcing private market growth. Indeed, US bank lending to non-bank financial intermediaries (NBFIs) has surged 160% in the past 5y to \$1.7trn (10% of total bank lending, Exhibit 49). 47% of this is directed to private equity and private credit funds. Institutional investors (pensions, insurers, sovereign wealth) have also allocated meaningfully to these markets, embedding them deeper into the financial ecosystem (Exhibit 47, Exhibit 48). This is a structural shift, as what began as an alternative source of capital now operates as a core component of global financing, amplifying its systemic relevance and risk transmission mechanisms.

Exhibit 45: Not only have private markets doubled in size in the last 6y, their size as a share of public assets has doubled since 2015

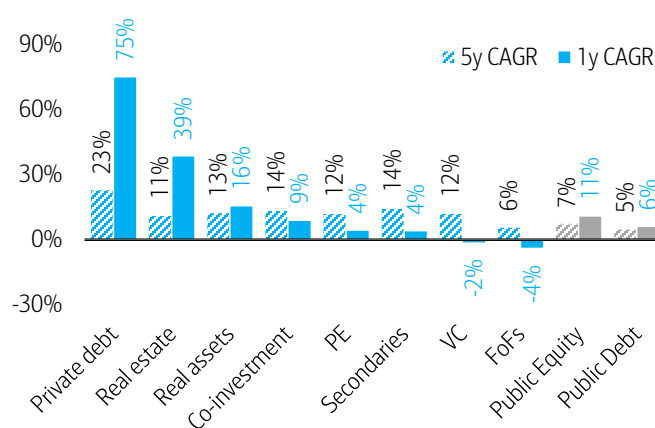
Global private markets AuM in USD trillions, and as a % of the public market



Source: BofA Global Research, PitchBook Data, Inc, www.pitchbook.com, Bloomberg. Data from Jan-07 to Dec-24. Others: VC, Real Estate, Real Assets, Secondaries, fund-of-funds, co-investment
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Exhibit 46: Private credit (PC) has seen the largest yoy growth. This outpacing vs PE growth suggests growing leverage in private assets

1y ('22 to '24) and 5y ('19 to '24) growth in global private, public market AuM



Source: BofA Global Research, PitchBook Data, Inc, www.pitchbook.com, Bloomberg. Data from Dec-19 to Dec-24

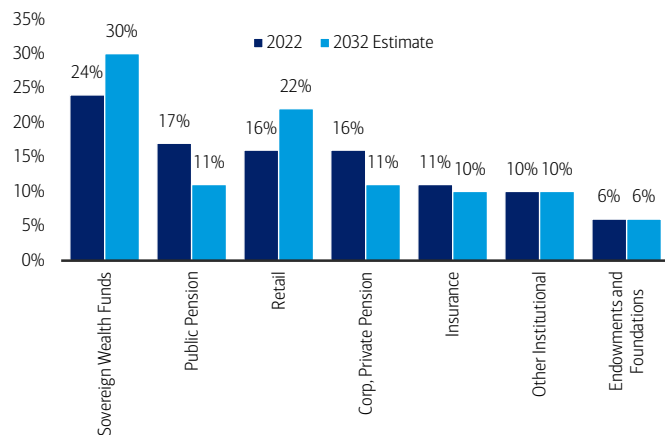
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² See "2029 Private Market Horizons" by Pitchbook Data, Inc.



Exhibit 47: Private markets are now held by institutional investors who were primarily involved in public markets in the past

Global alternatives AUM by investor (2022 and 2032 projections)

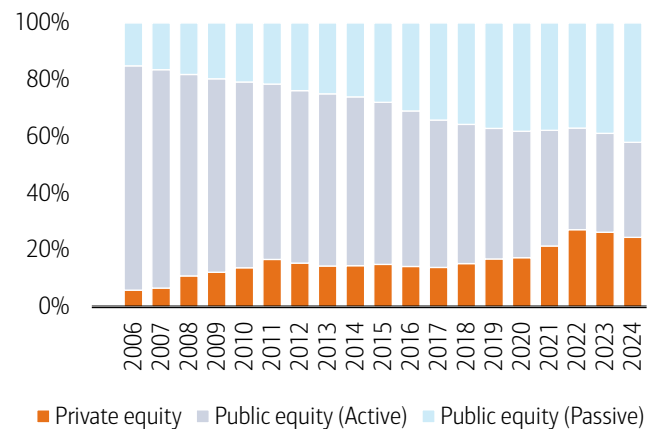


Source: BofA Global Research, Bain. Based on charts from our European Other Financials colleagues' report 'Feedback on Private Credit and Private Wealth meetings', published on 13 February 2025

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Exhibit 48: US Pensions are seeing growing allocations to Private equity, at the cost of Active public equities

Pension allocation as a % of total equity, 2006-2024



Source: BofA US Equity & Quant Strategy, FactSet. Data from Jan-06 to Dec-24. Pensions & Investments Research Center (as of 9/2024), Strategic Insights, SimFund, BofA US Equity & US Quant Strategy. Based on US pension funds within the top 1000 that have available asset allocations. Total equity includes public equity (global, international, domestic) and private equity.

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Hidden risks in non-transparent channels: Liquidity, Leverage and Opacity

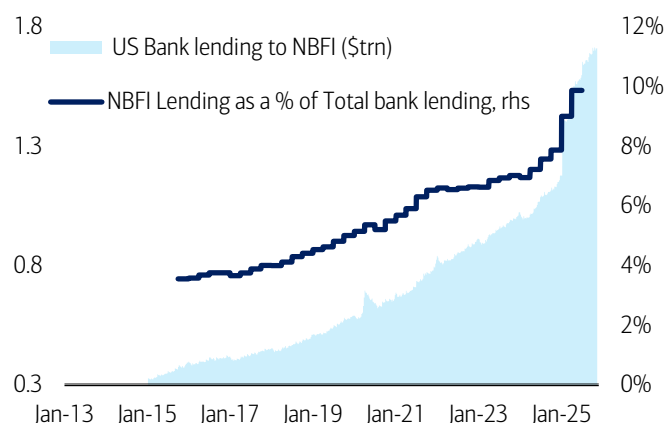
Beneath the headline growth, vulnerabilities are emerging. Liquidity is pressured as legacy vintages face slow exits and secondary transactions clear at discounts (McKinsey³). This leaves limited partners (LPs) with smaller distributions and tighter cash flows, creating fundraising headwinds. To bridge these gaps, managers increasingly rely on liquidity-smoothing mechanisms (continuation funds, NAV financing, payment-in-kind or PIK features, covenant-lite lending), that provide short-term relief but effectively borrow liquidity from the future, thus increasing the risk of a correlated liquidity shock.

At the same time, off-balance-sheet structures and leveraged vehicles such as Private Credit CLOs have grown in popularity (Exhibit 50, see our Securitized finance research team's latest [CLO weekly](#)). While these CLOs typically hold senior secured assets and maintain contained leverage, their use to expand credit via balance sheet recycling heightens the market's sensitivity to downgrades and defaults. This migration of capital away from regulated banking channels into opaque structures also raises credit quality concerns. Taken together, these dynamics concentrate risk in opaque channels, raising correlation and mark-to-market fragility risk should fundamentals deteriorate.

³ See McKinsey's 2025 Private Market reports, "Braced for shifting weather"

Exhibit 49: US Banks' loans to NBFIs has been on the rise, increasing its share within total lending. c47%* of NDFI loans go to PE/PC funds

US Bank loans to NBFI, as a proportion of total bank lending

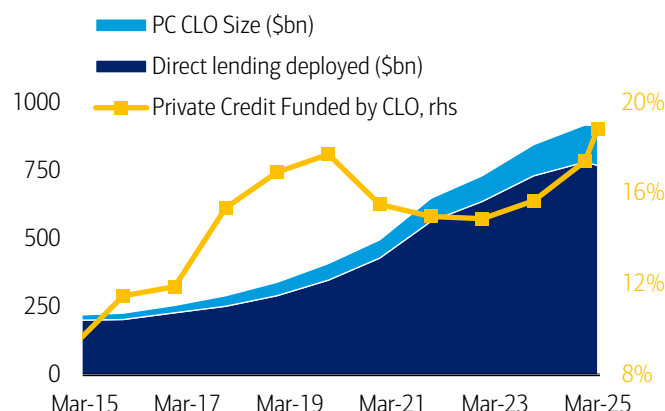


Source: BofA Global Research, Bloomberg, *Moody's Ratings, Federal Reserve H.8 release.

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Exhibit 50: Private Credit CLOs are gaining ground vs direct lending, thus expanding credit by allowing PC balance sheet recycling

Size of deployed Private Credit CLOs, Direct Lending and their ratio



Source: BofA Global Research, LCD, LSEG Data & analytics, KBRA DLD. Based on charts from our Securitized Product Strategy team's "CLO Weekly", dated 16-Nov-25

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The AI surge: A private asset growth engine but also a fragility amplifier

The global AI infrastructure build-out is set to demand unprecedented levels of financing (McKinsey⁴ estimates a spend of \$5.2 trillion on datacentre processing of AI loads by 2030). Public markets cannot meet demand of this scale alone (U.S. corporate bond issuance in 2025 has been \$2.5 trillion) and Oracle's recent rapid CDS widening may deter companies from adding significant debt to balance sheets. Private credit, SPVs, and leverage will likely play a critical role here. If private credit were to participate in even just a third of anticipated AI financing need, it would lead to ~50% growth on this theme alone. Recent AI mega-deals with hyperscalers building out data centres financed by private credit and SPV-financing have already set the tone for the coming years.

However, this private-debt-fuelled buildout likely heightens risks too. Leverage is rising, while technology obsolescence/faster depreciation is a real threat. A mitigating factor could be sovereign support as governments may view compute capacity and AI infrastructure as national security matters. Subsidies and credit guarantees could cushion systemic risk even in volatile markets. Net net, however, AI is likely a double-edged sword. While it can defer existing cracks by redirecting flows to high-demand sectors, it raises systemic leverage and fragility by way of contagion from a potential AI bust. The AI outlook lifts or sinks all boats, even private ones.

Potential triggers that could kickstart an unravelling of private markets

Several catalysts could trigger today's latent vulnerabilities in private assets into active dislocations (see Exhibit 51):

- **AI valuation reset:** Aggressive assumptions on AI-driven growth could face reality checks if adoption slows or technology cycles shorten, leading to repricing of AI-linked private assets and associated contagion across investment portfolios.
- **Credit cycle shock:** Downgrades can overwhelm CCC buckets in PC CLOs, tripping reinvestment limits and tightening liquidity. Public credit repricing compresses equity cushions and raises refinancing costs, as widening spreads in liquid markets impact valuations for private assets too.

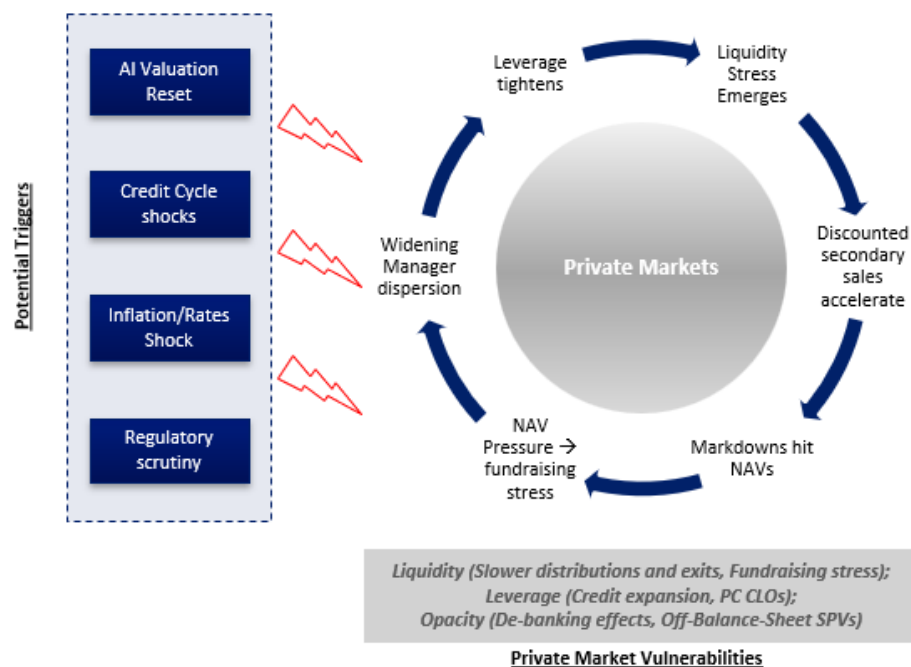
⁴ See McKinsey Quarterly, Apr 28, 2025, "The cost of compute"



- **Rates/inflation shock:** Rates or inflation surprises can hit private credit from both the borrowing and lending angles, accelerating covenant breaches & liquidity stress.
- **Regulatory scrutiny:** Heightened scrutiny of fund asset quality, ratings and leverage could slow capital flows materially. Increased audit requirements and stress testing may exacerbate manager dispersion and impact NAVs.

Exhibit 51: Potential triggers that could expose cracks in private markets

Private market vulnerabilities and potential triggers that may kickstart a negative spiral



Source: BofA Global Research

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The Convergence: Today, Public and Private markets co-determine outcomes

The divide between public and private markets is eroding in our view, creating feedback loops that can amplify systemic risk. Aside asset profitability, private IRRs depend on tradable macro variables like inflation, rates, credit spreads and FX. Public equities are increasingly reflecting themes driving private assets (e.g. infrastructure, AI, energy transition). Public financials (banks, insurers, pension funds) act as a risk-conduit from one market to the other; Banks, through their increased lending to NBFIs and institutional investors through allocations to private assets. Insurers, in particular, have growing allocations to private credit, including PC CLOs, entrenching them alongside private market players in the risks of these markets. Moreover, both public and private markets are exposed to the 'denominator effect'; for instance, public market pain can add pressure to reduce private investments to maintain allocation targets, while news of leverage excesses or defaults in the private arena can sour sentiment in public markets.

The convergence of private and public market risk creates opportunities for **hedging private risk with liquid, public market hedges**. Crucially, stress in public assets closely linked to private markets may ultimately impact private valuations, while potential hedge gains will give investors the flexibility to add risk at better terms.

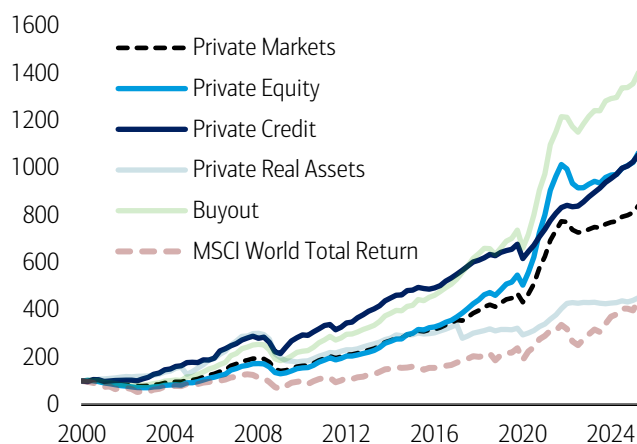
Crafting a public hedge for private assets: Liquidity & proximity to shared risks

We look for liquid, optionable public underliers that are related to the private risks we've identified and offer scalability in terms of hedge size:

- **SPX:** Geared to AI risk; has arguably the most liquid options market on the planet.
- **XLF:** US Fins, exposed to growing bank lending to NBFIs & insurance PC allocations.
- **MSCI EM:** Liquid exposure to the illiquidity risk premium and exposure to USD vol.
- **HYG:** Exposure to US high yield credit, the public face of corporate debt market risk
- **TLT:** Exposure to 20y+ US Treasuries in case of another bearish shock in rates.

Exhibit 52: Notwithstanding infrequent observations, benchmarks for private assets have performed well.

Hamilton Lane private market benchmark indices, quarterly data

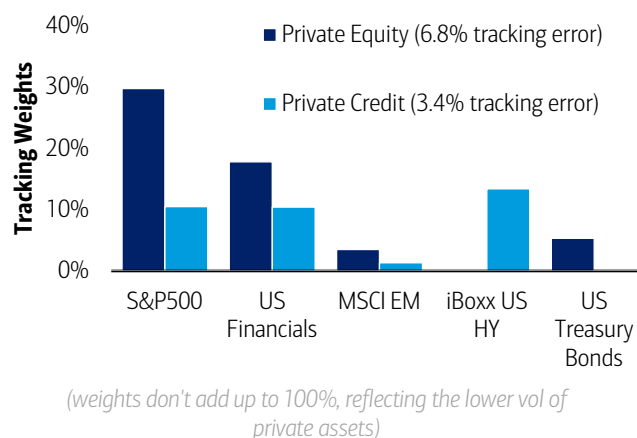


Source: BofA Global Research, Bloomberg, Hamilton Lane. Data: Mar-00 to Jun-25.

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Exhibit 53: A combination of liquid public market underliers can be used to proxy and hence hedge illiquid private markets.

Proxy assets with tracking weights for PE & PC (tracking err. in parentheses)



Source: BofA Global Research, Bloomberg. Data: 01-Oct-16 to 01-Jul-25. Solving for weights that minimise tracking error using quarterly returns in the last 10y to respective Hamilton Lane indices.

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Note: While some may consider more directed hedges like options on single stock proxies of private credit risk like BDCs (Business Development Companies) or put options on levered loan ETFs, these are unlikely to be scalable enough to be of practical value for many institutional investors. Moreover, BDCs are already showing significant signs of strain: The S&P BDC index (SPBDCUP) is down 17% YTD while US levered loans (proxied by SPBDALB) are down less than 1%.

Public hedges for private assets

Private credit: Long Dec26 80% put on basket of HYG (38%), SPX (29%), XLF (29%) & EEM (4%) for 1.46% (27% cheaper vs avg of individual legs), or for **0.52%** of notional invested in Private credit, indicatively. For reference, this premium outlay is <10% of the 550bps of excess spread typically generated by private credit investments.

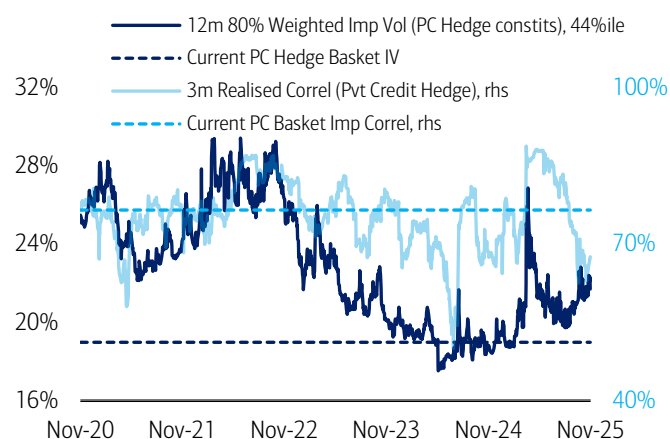


Private equity: Long a basket of Dec26 90-70 put spreads on individual underlyings: SPX (53%), XLF (32%), TLT (9%) & EEM (6%) for 2.66% (weighted max payout ratio of put spreads is 7.7x), spending **1.5%** of notional invested in Private equity on hedge premium, indicatively. Given the higher volatility of the PE hedge constituents (Exhibit 55) and significant decorrelation potential, we prefer buying put spreads on individual constituents rather than on the basket.

Risk: Limited to premium paid.

Exhibit 54: Vol on the Private Credit hedge basket is low vs the vol of its constituents, cheapening the price of basket puts

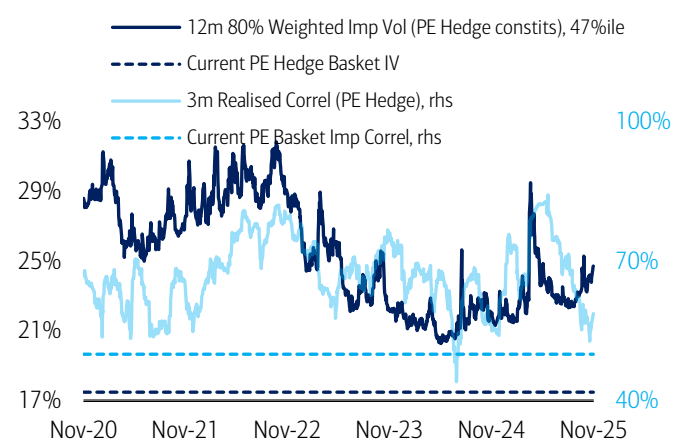
Private Credit Hedge: Weighted component IV, Basket IV, Basket Correls



Source: BofA Global Research, Bloomberg. Data from 20-Nov-20 till 19-Nov-25. Weighted Implied Vol is the weighted average (scaled to 100% weights) vol of the PC Hedge basket constituents.
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Exhibit 55: The Private Equity hedge basket constituents are more volatile, making a put spread on the constituents an attractive hedge

Private Equity Hedge: Weighted component IV, Basket IV, Basket Correls



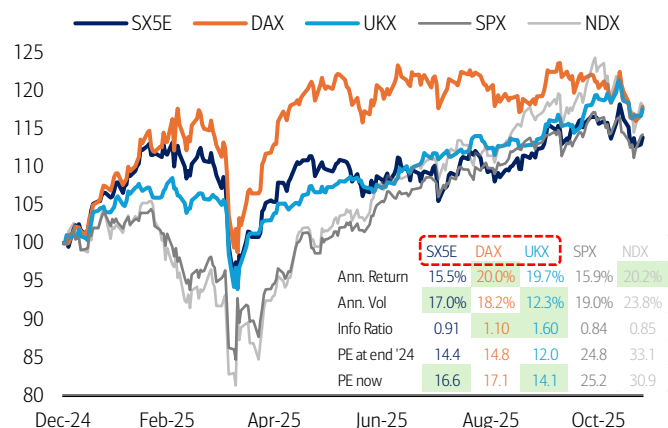
Source: BofA Global Research, Bloomberg. Data from 20-Nov-20 till 19-Nov-25. Weighted Implied Vol is the weighted average (scaled to 100% weights) vol of the PE Hedge basket constituents.
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Europe: Still cheap and only a little less unloved

A year ago, we called for upside hedges in EU equities as the region was cheaply valued yet unloved by investors against the backdrop of a new and more purposeful 2nd Trump presidency and a European leadership vacuum. Exhibit 56 shows just how important positioning (or lack thereof) can be for asset returns, with the DAX and EU equities outperforming the US in H125. Where do we stand today? EU equities are still cheap vs the US, despite rerating higher. And Exhibit 57 shows that while there've been EU inflows YTD, this is <10% of global equity inflows (EU is >15% of global equities) and only about 20% of cumulative regional outflows since the Russia-Ukraine war in 2022.

Exhibit 56: While US stocks rebounded harder after 'Liberation day', EU equities are top of the pile YTD & cheaper despite rerating higher

YTD perf. of major EU & US equity indices, their volatility & valuations

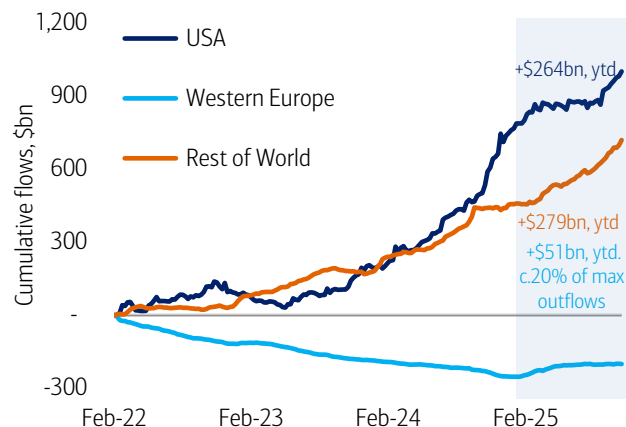


Source: BofA Global Research. Data from 31-Dec-24 to 25-Nov-25. Green shaded cells in table highlight the two best indices on each metric (cheaper = better for valuations).

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Exhibit 57: While EU equities have seen inflows YTD, it is less than 10% of global equity inflows, and only 20% of EU outflows since 2022*

Cumulative inflows (\$ bn) into Western European funds



Source: EPFR, BofA Global Research, Bloomberg. Data: 23-Feb-22 (*when the Ukraine war began) until 5-Nov-22. Fund flows are cumulative, from retail and institutional investors in/out of developed market equity ETFs and Mutual funds, domiciled in Western Europe.

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Europe has structural tailwinds (for a change), but execution will be key

Germany's change of (fiscal) heart came as a surprise to most in Q125, and for good reason. Federal approval for spending on defence (Exhibit 58) and on infrastructure (Exhibit 59) are not only substantial (2.4% and 0.9% of 2025 GDP, respectively), but also expected to grow by >35% and nearly 60%, respectively in 2026. As always with large endeavours, execution will be key and there are already implementation concerns and signs of stress in Chancellor Merz's coalition. However, this remains an important tailwind for Germany in the coming years.

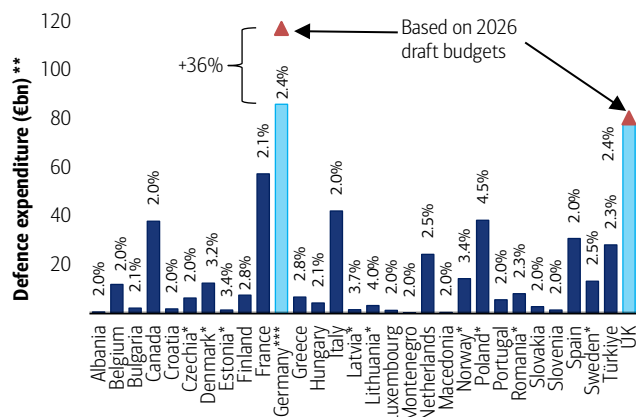
Pension reform in Holland & Finland suggest more risk-taking and hedging flow

Dutch pensions, the largest in Europe with €1.6trn in assets (Exhibit 60), are undergoing structural reform starting 1-Jan-26. Much focus has been on the rates impact of this change given reduced duration hedging needs in the new Defined Contribution (DC) system. Our rates team notes that the asset mix post-transition will likely involve more risk-taking (see [here](#)), driven by younger members with higher risk appetites and fewer capital charge constraints on holding riskier assets. In our view, the lead up to transitions could also see demand for equity hedging as it is relatively cheap for pensions to protect high funding ratios (see our [GEVI](#)). After the transition, we could see more equity exposure in the allocation mix, although private assets, infrastructure and real estate may edge out a ramp up in public equities (which stands at 27% currently). Even so, more frequent portfolio rebalancing should create equity flows in the opposite direction to performance; i.e. buying dips and selling rips to maintain allocation targets. Finnish pension reform, legislation for which is scheduled to be debated and voted on in parliament from March 2026, is expected to see an increase in equity allocations by upwards of 10%. While not a large amount (>€11bn) based on €272bn in pension assets (Exhibit 61), it is another data point from European pension reform that is supportive of greater risk-taking in asset allocation.



Exhibit 58: From >€500bn in '25, EU defence spending is expected to rise, with Germany proposing to spend >35% more in '26

2025 defence expenditure in €bn & % of GDP for European NATO members (and 2026 draft proposals for the 2 highest spenders: Germany & UK)

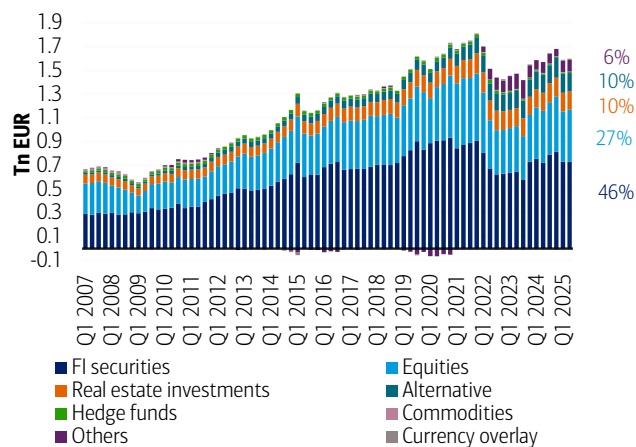


Source: BofA Global Research, NATO: [NATO - News: Defence Expenditure of NATO Countries \(2014-2025\), 28-Aug-2025](#) *These countries have laws/political agreements calling for 2% of GDP or more to be spent on defence annually; future spending may change. ** NATO 2025e data (USD) & UK's 2026 draft budget (GBP) converted to € as of 11-Nov-25. *** Using 2025's approved budget.

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Exhibit 60: Dutch pensions are undergoing transition, likely leading to greater risk-taking in the new Defined Contribution (DC) system

Total amount of Dutch PFs' investment assets, including derivatives



Source: DNB, BofA Global Research. Data from Q1'07 to Q2'25. Note: "Others" include Repo operations, Structured notes and Total return swaps. "Alternative" includes private equity, infrastructure and microfinancing.

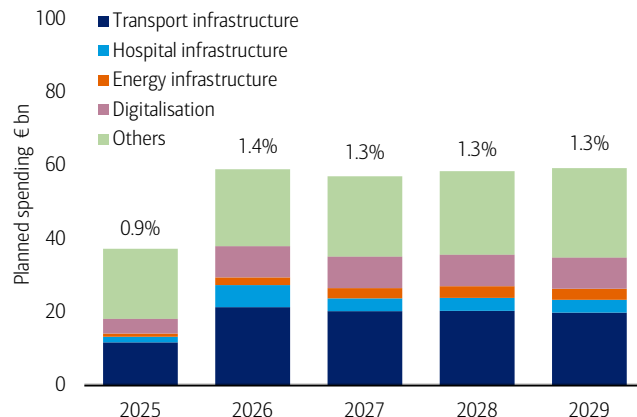
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European stock buybacks at 10yr highs another pillar of support and dip-buying

While US corporates are better known for stock buyback activity, Exhibit 62 shows that EU buybacks have been on the rise since the pandemic, with 2025 on track to be the year with the highest volumes since at least 2015. Indeed, EU corporates have been buying back just shy of 1.5% of their market cap in each of the last 4 years. While buybacks are typically a positive market signaling tool, we're also able to isolate the impact on stock price action by examining how quickly buyback stocks bounce back after a downside shock. Indeed, Exhibit 63 shows that the average return of EU buyback names after a worse-than 1 sigma down move is 7% greater than average, while non buyback stocks see a 7% decline in return after a shock.

Exhibit 59: Also, Germany's infrastructure spending in 2026 is projected to grow to 1.4% of German GDP*

Planned spending from the Special Fund for Infrastructure and Climate Neutrality (in € bn and as % of BofA's German GDP estimates)

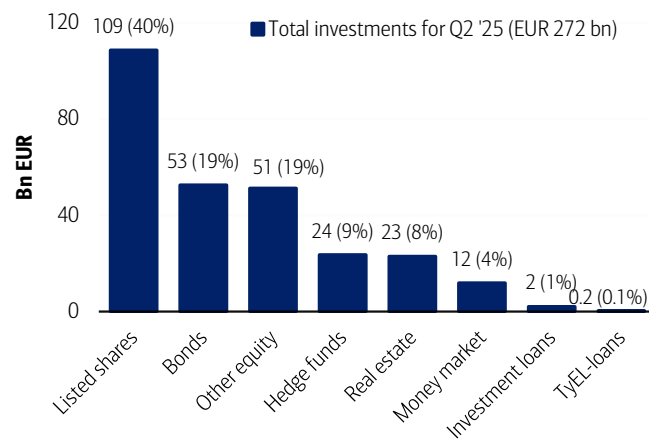


Source: BofA Global Research, bundesfinanzministerium.de (Federal Ministry of Finance).

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Exhibit 61: Finnish pensions are also undergoing reform, with reports of this leading to a 10% increase in equity allocations

Total amount of Finnish PFs' investment assets

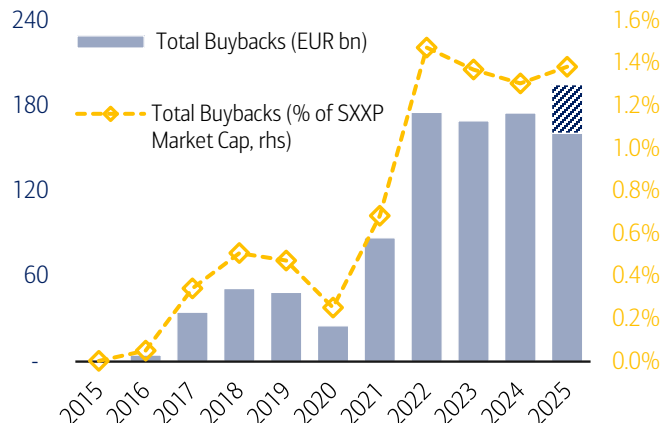


Source: TELA, BofA Global Research. Data as of Q2'25. "Listed shares" includes equity funds. "Other equity" comprises private equity investments and investments in unquoted shares.

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Exhibit 62: EU stock buybacks are on the rise, with 2025 on track to be the year with the largest buyback volumes in at least a decade

Total annual (extrapolated for 2025) buyback volumes for (current) STOXX 600 constituents, in EUR bn (lhs), and as % of the market capitalisation (rhs)



Source: BofA Global Research, Bloomberg. Data from 01-Jan-2015 till 15-Sep-2025. Data shown for current SXXP constituents (hence may introduce survivorship bias). However, >80% of the current constituents have been in the index for the last 10yrs (90% in the last 4 yrs) thus limiting survivorship bias..

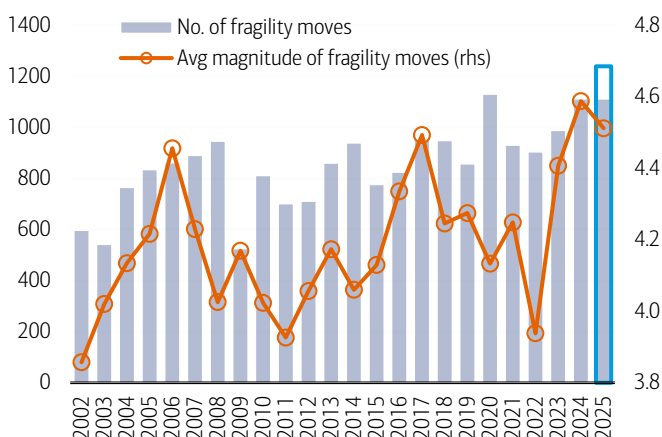
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European derivatives opportunities in 2026: Long volatility, dispersion & RV

While the structural themes above are long run positives for Europe, markets may ignore them in the near term and focus on headlines du jour. Indeed, the main risks for European risk assets are poor execution and administrative bottlenecks holding up much-needed reforms, political/fiscal stress taking center stage (e.g., in France) while US exceptionalism returns, or if geopolitical conflict in the region escalates beyond Ukraine.

Exhibit 64: EU stocks continue to show elevated fragility in 2025, with near-record frequency and magnitude

Frequency & average magnitude of fragility events (sigma move magnitude > 3)* for the historical top 200 names within STOXX 600



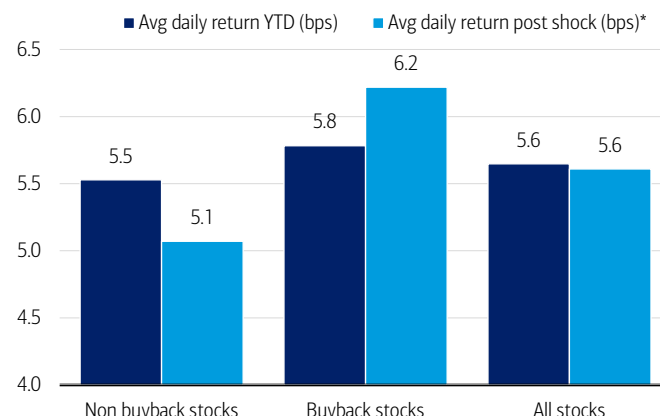
Source: BofA Global Research, Bloomberg. Data from 31-Dec-01 to 20-Nov-25. Components of index kept fixed at the start of the calendar year. *"Sigma move" is measured by dividing a stock's 1d return by its 21d trailing volatility. It is a measure of how large a shock is vs its risk projected by recent volatility and is one measure of price "fragility".

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Consistent with our view on vol being supported as the AI bubble builds further, we like owning EU equity vol, via carry-friendly SX5E forward variance (see *Long vol, light on carry: SX5E fwd var leads globally*) Not only does owning SX5E forward variance have better risk/reward vs global peers, but our carry-conscious forward var alternative has a superior cost of carry in muddle-through scenarios, where Europe is in familiar territory.

Exhibit 63: The market impact of buybacks is visible in European stocks, with buyback names bouncing back harder after a shock.

Avg daily return of SXXP stocks, Buyback stocks (281/600) & stocks with no announced buybacks YTD, unconditioned & conditioned on a down shock*

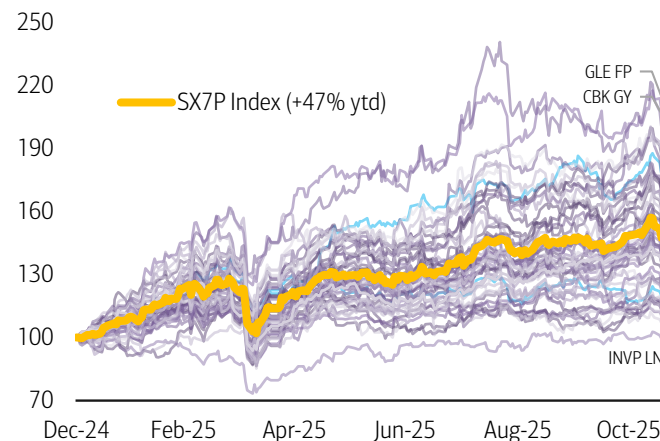


Source: Data: 30-Dec-24 to 3-Nov-25. *Shock defined as a daily decline for a stock that is worse than 1std dev, based on the stock's 1y realised vol (a measure of long term risk)

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Exhibit 65: EU banks have had a stellar year (SX7P up ~50%), but stock returns (nearly flat to >2x ytd) reflects diverse sector drivers

Rebased year-to-date spot price for SX7P and its current constituents



Source: BofA Global Research, Bloomberg. Data from 31-Dec-24 till 21-Nov-25.

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Second, we highlight dispersion opportunities in a region that is often overlooked in favour of the US, given relatively less option liquidity and less concentrated idiosyncratic risk. However, Exhibit 64 shows that EU stocks are making new fragility records, underscoring that the rise in single stock volatility (a positive for dispersion) isn't just a US or purely retail-driven phenomenon (Europe doesn't have the same extent of retail participation in markets as the US). Now, dispersion can be traded in many forms, and one we like in particular is the dispersion of returns in European banks (see *Once bitten twice shy: Hedging FX for non-USD investors*). Distinct themes like French fiscal stress, Ukraine reconstruction, German stimulus and the US banking cycle (deregulation) have all helped drive diverging performance. Exhibit 65 is a good visual depiction of this, as while the Banks sector is up nearly 50% YTD, some stocks are down while others have doubled.

Finally, while the initial euphoria (in Q125) on German infrastructure spend and defence has moderated, we maintain that Germany is where European upside risk is worth hedging. We favour funding German upside by fading strength in French equities (CAC) given French fiscal stress is unlikely to be resolved in the near term. Structures like DAX calls contingent on CAC not rising will typically offer steep discounts vs vanilla calls, as we showed in our GEVI (see [28-Oct GEVI](#)).

Asian AI stocks driving massive regional vol dislocations

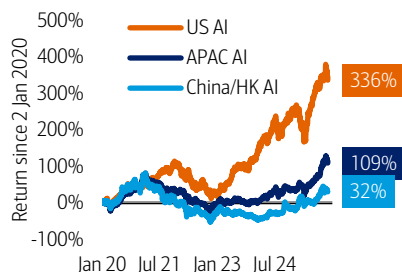
While most markets agree on pricing stocks near all-time highs, they do not agree on the risks and volatility. Kospi vols are at "distressed" levels, near the highest since 2009 (97th %-ile) while Nifty vols hover above single digits, in the 3rd %-ile (Exhibit 68). The key differentiator is the extent to which the market is driven by AI stocks. The top 10 AI-linked stocks in **Korea make up 56% of the local market** (see bubble size in Exhibit 69) while the corresponding number is **only 6% in India**. In fact, China's top 10 AI stocks only make up **8% of the China/HK market** compared to **27% in the US**.

Indeed, in contrast to Nvidia et al in the US, the AI trade in China hasn't played out within large caps. While China's tech giants Tencent and Alibaba have been "grinding" higher, the AI frenzy appeared in stocks such as Cambricon, sometimes called China's Nvidia. This US\$77 Bn market cap stock – about a tenth of Tencent – tripled in value this summer. With the top China AI stocks less dominant than in the US – at the same time as valuations are cheaper (Exhibit 69) – this could indicate more room to the upside in China while also leaving the overall market less susceptible to a hypothetical AI bust. Key facts regarding Asia's AI market:

- Size of top 10 China AI stocks = 8% of the US top 10 (\$1.4 Trn vs. \$18.5 Trn)
- China/HK top AI stocks still trade below 2021 levels (Exhibit 66)
- While Korea and Japan have signaled bubble-like price action of late, APAC AI vol has been trending sideways, lacking spot-up/vol-up bubble dynamics (Exhibit 67)
- APAC AI (22x) is 29% cheaper than US AI (32x), based on forward PE (Exhibit 69)
- Kospi Jun-26 OTM call vols trade above S&P vols. Going back 20 yrs, the Kospi has never realized that much more vol than the S&P (Exhibit 70 & Exhibit 71)
- The word clouds in Exhibit 72 help visualizing the AI specialty of each of the key tech countries in Asia. The bubble chart in Exhibit 73 illustrates the size of the key AI stocks in China/HK, Japan, Korea and Taiwan

Exhibit 66: US top 10 AI stocks have rallied strongly for 3 years and China top 10 AI stocks for 1 year. Coinciding with the Nov-22 release of ChatGPT and the Jan-25 release of DeepSeek, respectively

Performance of key AI stocks since Jan 2020

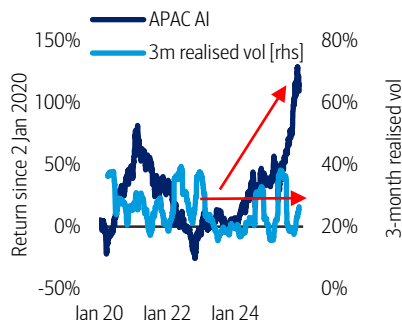


Source: BofA Global Research, Bloomberg. US AI and China/HK AI consists of the top 10 AI linked large caps stocks in each region. APAC AI is similarly comprised of the top 10 AI-linked large caps in China/HK, top 5 from Japan, top 5 from Korea and top 5 from Taiwan. Data from 2-Jan-20 through 18-Nov-25.

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Exhibit 67: APAC AI in aggregate is not yet showing typical bubble signs such as strong spot-up/vol-up dynamics

APAC AI stock performance and 3m realised vol

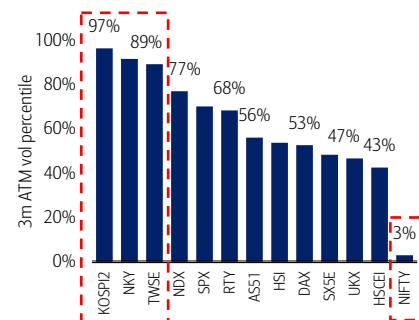


Source: BofA Global Research, Bloomberg. APAC AI is comprised of the top 10 AI-linked large caps in China/HK, top 5 from Japan, top 5 from Korea and top 5 from Taiwan. Data from 2-Jan-20 through 18-Nov-25.

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Exhibit 68: Massive vol divergence across Asia. Kospi and Nikkei vols near the highest since 2009 while Nifty vols are near the lows

3m ATM vol percentiles vs data since Jan 2009

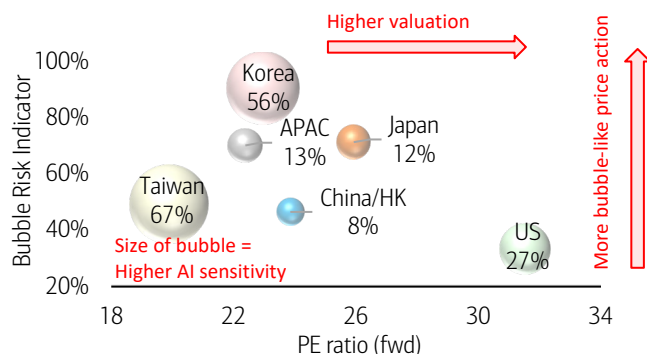


Source: BofA Global Research. The percentiles are computed based on daily vol data since 2-Jan-2009 through 19-Nov-25

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Exhibit 69: While exhibiting more bubble-like price action, Korean AI stocks are considerably cheaper than US AI stocks (23x vs 32x PE). The Taiwanese stock market has the highest relative AI sensitivity (67%) while China/HK has the lowest (8%)

Bubble Risk Indicator, valuations and AI sensitivity* of each market, measured on the top 10 largest AI-linked stocks in each country

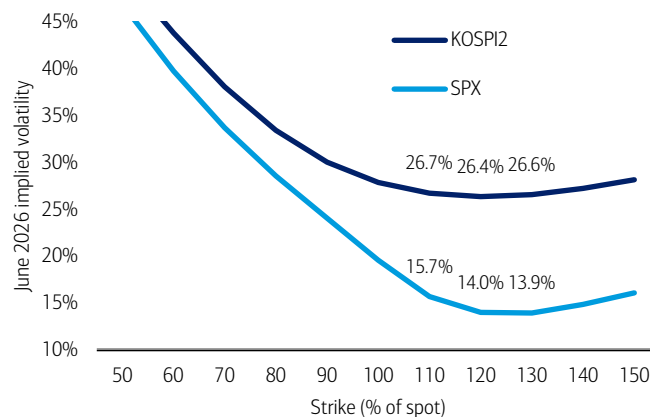


Source: BofA Global Research, Bloomberg. * AI sensitivity measured as the market cap of the top 10 AI linked stocks' market cap as a percentage of each country's total market cap. Data from 2-Jan-2020 through 18-Nov-2025.

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Exhibit 70: 7-month Kospi skew vs S&P skew illustrates how bid Kospi call vols have been lately

7-month vols as a function of strike (%) for Kospi 200 and the S&P 500



Source: BofA Global Research. Data as of 21-Nov-2025

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2025 marks the year of the Asian AI catch-up rally – but still 2 years behind

Exhibit 66 illustrates well how US AI accelerated after the ChatGPT release in Nov-22, while APAC AI accelerated after the DeepSeek moment in Jan-25. Viewed through that lens, APAC AI is just 1 year into the AI trade, 2 years behind the US. While the 2025 catch-up has been aggressive in certain areas, the performance acceleration does not look vastly different to the last 3 years in the US. And as such, may not warrant more caution with respect to Asia AI vs US AI. If anything, the valuation metrics in Exhibit 69 suggest Asia AI may still have room to run and catch up with the US.

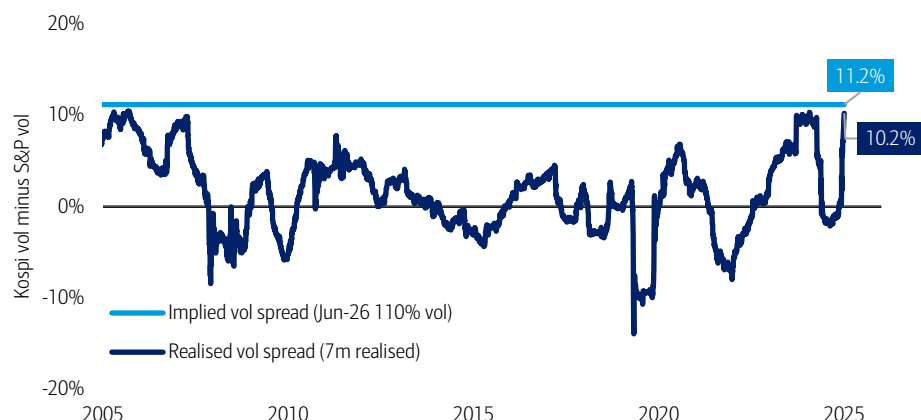
Parts of Asia have been catching up quite aggressively of late, exhibiting bubble-like price action. The primary example being the Korean Kospi, led by Samsung and Hynix. However, while the +150% 2-month Hynix rally may seem excessive, its forward PE has only re-rated from 6x to 12x, clearly nowhere near that of most other AI-linked stocks.



The top 10 large cap AI-linked stocks per country were picked based on our assessment of AI focus and comprises of the following names for each country. **US:** NVIDIA, Microsoft, Alphabet, Amazon.com, Broadcom, Meta, Oracle, Palantir, AMD, Adobe. **China/HK:** Tencent, Alibaba, SMIC, Meituan, Cambricon, iFlytek, SenseTime, Beijing Fourth Paradigm, Mininglamp, Ping An Healthcare & Tech. **Japan:** Sony, SoftBank, Hitachi, Tokyo Electron, Advantest, Keyence, Fujitsu, NEC, Murata Manufact, Denso. **Korea:** Samsung Elec, SK Hynix, LG Energy, Hyundai Motor, Doosan Enerbility, Hanwha Aerospace, Celltrion, NAVER, Kakao, LG Electronics. **Taiwan:** TSMC, Hon Hai Precision (Foxconn), Delta Electronics, MediaTek, Quanta Computer, Winyann Accton Tech, Alchip Tech, Advantech, Gigabyte Tech.

Exhibit 71: The Kospi has never (in the last 20 years) realised as wide a vol spread as implied by Jun-26 110% strike vol of Kospi vs the S&P

7-month Kospi realised vol minus S&P realised vol as well as the spread priced by Jun-26 110% strike vol

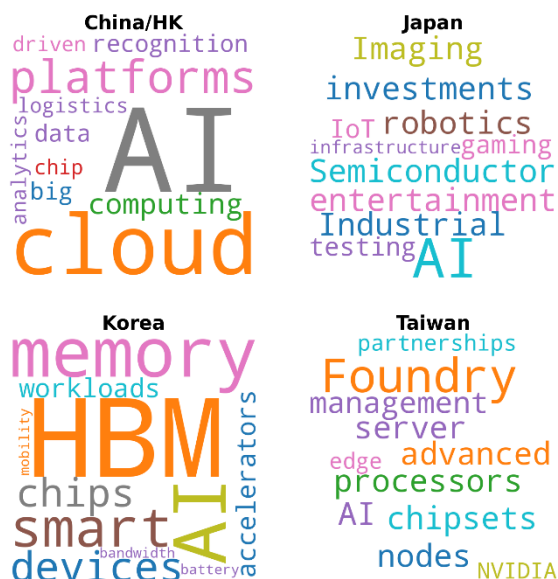


Source: BofA Global Research and Bloomberg. Data from 19-Nov-2005 through 19-Nov-2025.

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Exhibit 72: The four key “tech regions” in Asia have developed their unique edge and specialty, ranging from TSMC’s Taiwan foundry to HBM chips in Korea (not including India as AI tech is relatively small)

Word clouds aim to capture the focus of each country’s top 10 AI stocks

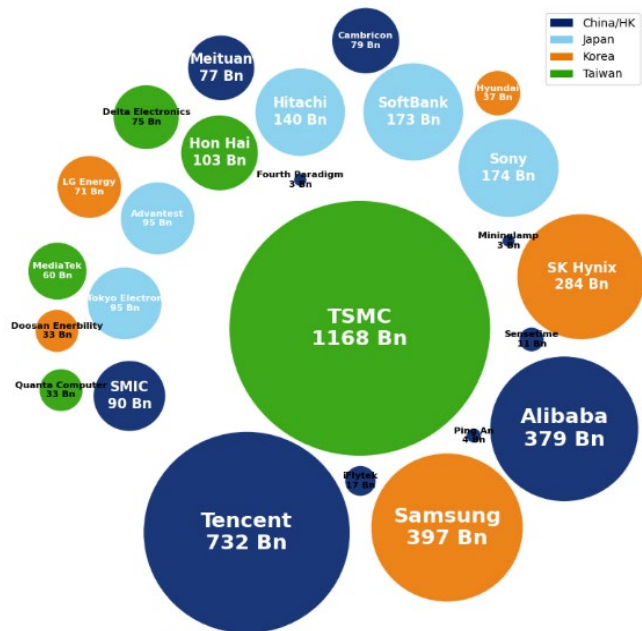


Source: BofA Global Research and Bloomberg.

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Exhibit 73: The top 25 AI-linked stocks across Asia, using 10 from China/HK, 5 from Japan, 5 from Korea and 5 from Taiwan

AI stocks with their respective US\$ market caps



Source: BofA Global Research and Bloomberg. Data as of 19-Nov-2025.

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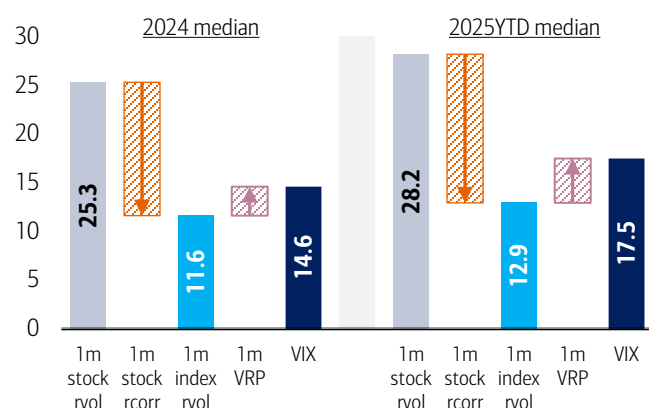
Why lower US equity vol is unlikely in 2026

2025 recap: no 2017 vol crush; both equity tails realized; VIX regime 3pts higher

In our 2025 outlook (see [The roaring 2020s?](#)), we anticipated rising risks in both tails, equity volatility holding flat to modestly higher, and dismissed the likelihood of a 2017-style (Trump 1.0) vol crush. Indeed, the VIX floor shifted higher from 12 in 2024 to 15 in 2025, the median VIX level also climbed ~3 points from 14.6 to 17.5 (Exhibit 74), and owning volatility paired with long equities continued to generate positive returns (Exhibit 75). S&P realized volatility also rose in 2025, albeit more modestly from 11.6% to 12.9% (Exhibit 74), constrained by historically low stock correlation.⁵

Exhibit 74: US equity volatility rose in 2025, with the VIX regime shifting roughly 3pts higher. The increase in S&P realized volatility was more subdued, held back by historically low stock correlation

Waterfall from SPX single stock realized vol to VIX (calendar-year median)

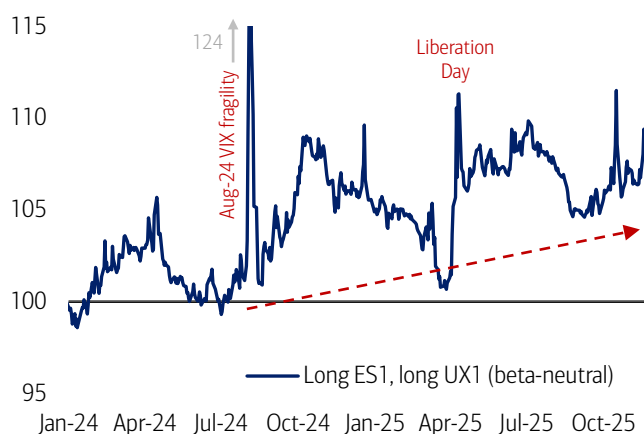


Source: BofA Global Research. Data from 1-Jan-24 to 21-Nov-25.

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Exhibit 75: Pairing long VIX futures with long equities would have continued to generate positive returns in 2025 despite the summer lull in volatility, as the VIX floor proved resilient during the equity rally

Hypothetical performance of long ES1, long UX1 with beta-neutral sizing



Source: BofA Global Research. Data from 1-Jan-25 to 21-Nov-25. Futures are rolled 1 business day before their respective expiries. Beta neutral sizing uses trailing 1y realized beta. This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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2026 base case: equity vol to stay supported...risks rise as bubbles build

With our core view that AI remains in the early stages of building towards a larger bubble, we expect further upside to equity volatility in 2026. Asset bubbles tend to grow increasingly unstable as they inflate, driven both by classic vol up/spot up dynamics as well as the tendency for volatility to reset higher following interim shocks that inject more uncertainty but fail to derail the broader bubble.

Risks also rise alongside valuations, with the S&P historically exhibiting a positive relationship between starting P/E ratios and subsequent realized volatility. In that context, today's elevated multiples should support S&P realized vol in 2026, possibly near 20% if the relationship in Exhibit 76 holds as firmly as it did in 2025.

If instead the AI boom stalls or reverses in 2026, whether due to an exogenous shock (e.g., US midterm elections) or an endogenous catalyst (e.g., overinvestment concerns grow), we suspect it's too late for it to fizzle out quietly (Exhibit 33 & Exhibit 34). The ensuing bear market would also be vol-positive, particularly given the historic concentration of AI/tech stocks in the S&P.

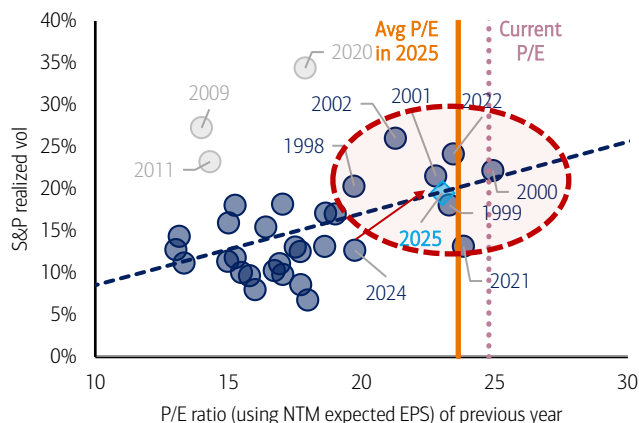
⁵ Calendar-year SPX realized vol for 2025 is closer to 19.5%, as seen from Exhibit 76, but median 1m realized vol remains much lower at ~12.9%, reflecting short-lived outlier moves around Liberation Day.



And bigger picture, the framing of AI as either utopia or existential threat (Exhibit 77) can feed “doomer” narratives that amplify fear, uncertainty, and doubt—fueling more political instability, and by extension, asset volatility.

But what could challenge this view? The VIX is ultimately anchored to S&P realized volatility, which *mechanically* depends on how much its component stocks move and how correlated those moves are. Hence, in what follows, we stress-test our outlook for equity vol by assessing *downside risks* to single stock volatility and correlation.

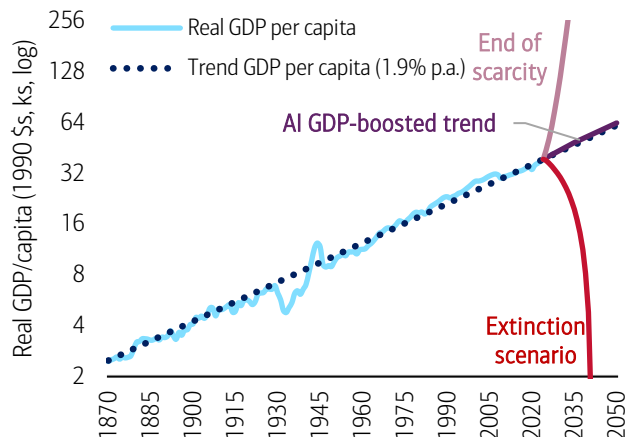
Exhibit 76: Historically, the S&P has shown a positive relationship between valuations (P/E) and subsequent vol, suggesting that today's lofty valuations should support, or drive even higher vol, in 2026
S&P 500 calendar year realized volatility vs P/E ratio of prior year



Source: BofA Global Research. Data from 1-Jan-90 to 21-Nov-25. P/E ratio based on NTM expected EPS and is calculated as calendar year average. The 2025 point reflects 2025 YTD realized of the S&P 500 vs its average P/E ratio from 2024.

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Exhibit 77: The framing of AI as either utopia or existential threat fuels “doomer” narratives that amplify fear, uncertainty, and doubt—which could drive more political instability, and in turn, asset volatility
Real GDP per capita (in 1990 dollars) under different AI scenarios



Source: BofA Global Research, Wynne & Derr, ‘Advances in AI will boost productivity, living standards over time’ (Federal Reserve Bank of Dallas, 2025). *AI-boosted path = 2.1% p.a. for 10y. Tech singularity scenarios (end of scarcity & extinction) are stylized.

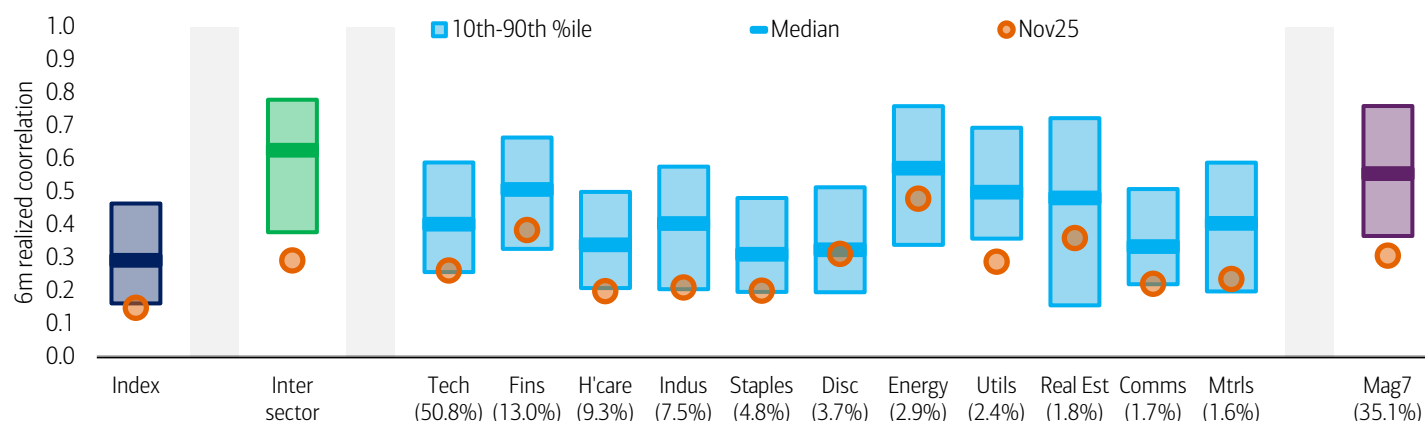
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Could S&P stock correlation fall even further?

Correlation between S&P 500 stocks—already historically depressed in 2024—remained so in 2025, outside of the Liberation Day shock (Exhibit 79). This decorrelation trend has been broad-based, with both inter-sector and intra-tech correlations (particularly between the “Magnificent 7” stocks) recently sitting near their bottom deciles since 1990 (Exhibit 78). The drivers have been diverse across stocks and sectors, including AI vs non-AI exposure, decoupling within the AI theme, and frequently shifting macro narratives around inflation, rates, and growth. While we expect correlations to remain low as long as an AI bubble remains plausible, similar to the 1990s tech bubble (Exhibit 79), we see limited statistical downside from already depressed levels. In other words, low correlation may persist, but it’s unlikely to fall meaningfully further.

Exhibit 78: Similar to 2024, S&P stock correlations remained historically low in 2025, both between sectors (e.g., tech/AI vs non-tech/non-AI) and within the tech stocks that dominate the S&P, particularly the Magnificent 7

Current levels & historical distribution of SPX 6m stock, inter-sector, and intra-sector realized correlations (sector weights in parentheses)

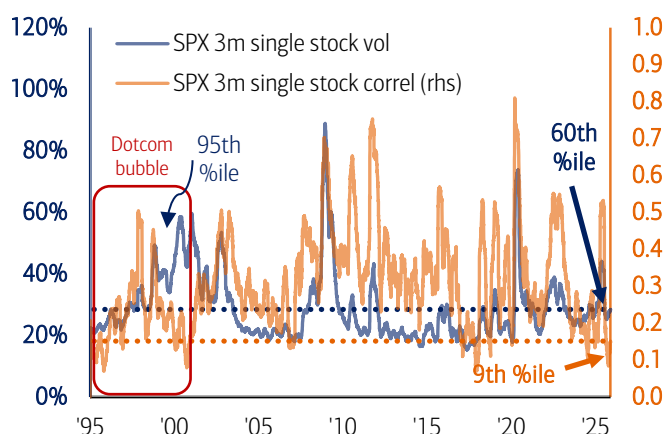


Source: BofA Global Research. Data from 1-Jan-90 to 21-Nov-25. Mag7 distribution since 2021. Sectors are defined as per the Global Industry Classification Standard, along with re-classification of certain companies (primarily those not in tech that are largely internet/tech driven, e.g., Tesla, Amazon, Uber etc.). Intra-sector correlations are ranked by decreasing current sector weight in the S&P (shown in parentheses).

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Exhibit 79: S&P 500 single stock volatility has been in a structural uptrend since 2017 yet remains below dotcom bubble levels, suggesting further upside if the AI bubble inflates

SPX 3m single stock realized volatility and realized correlation

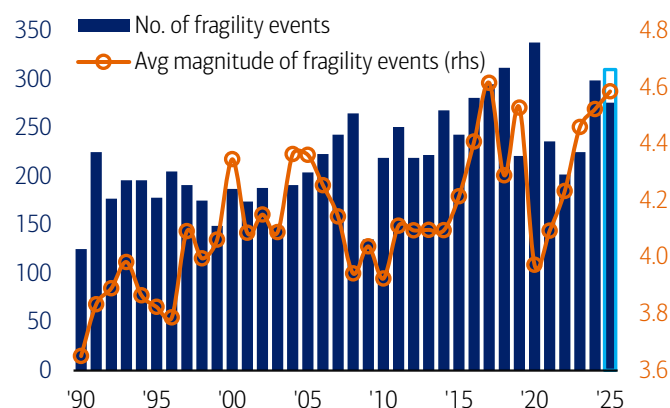


Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25.

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Exhibit 80: US large caps continued to show elevated fragility in '25, with near-record frequency & magnitude of outlier events

No. & average magnitude of fragility events (sigma move magnitude > 3) for the largest 50 stocks in the S&P



Source: BofA Global Research. Data from 1-Jan-90 to 21-Nov-25. Fragility event = absolute value of sigma move > 3, where sigma move = a stock's 1d return divided by its 21d trailing volatility. It is a measure of how large a shock is vs the risk projected by recent volatility and is one measure of price "fragility". Components of index kept fixed from the start of every calendar year.

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Could single stock volatility reverse its multi-year uptrend?

Unlike correlation, single stock volatility has far more room to decline given the elevated levels of idiosyncratic risk seen in recent years (Exhibit 79). We saw a glimpse of this earlier in the summer when the S&P's steady grind higher produced an information ratio above 6. Still, we view a *sustained* period of materially lower single stock vol unlikely for the following reasons.

First, although S&P single stock vol has been on a structural uptrend since 2017, it remains well below the extremes reached during the latter stages of the late-1990s dotcom bubble (Exhibit 79). Within that analogue, further inflation of the AI bubble is more likely to push single stock vol higher, not lower.



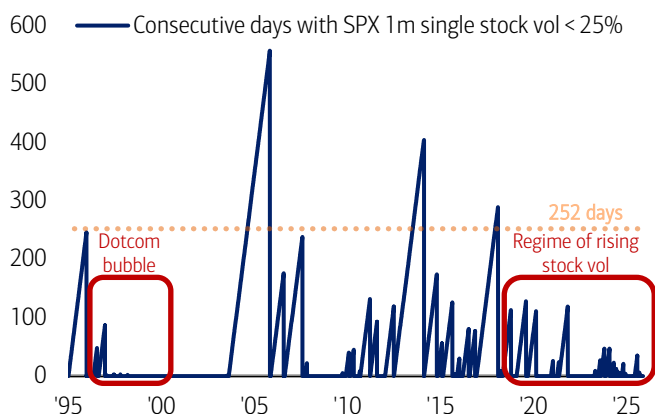
Second, single stock fragility (large, idiosyncratic price moves that support volatility) remains near record highs in both frequency and magnitude (Exhibit 80). Fragility is unlikely to dissipate as the AI bubble builds and price action becomes increasingly violent. Indeed, our new BofA Bubble Risk Indicator, which quantifies how return distributions evolve as asset bubbles form, suggests we should expect more extremes in return, volatility, momentum, and fragility before the AI bubble peaks.

Finally, recent years and the history of the dotcom bubble suggest little precedent for single stock volatility to sustain levels materially below today's (Exhibit 81). During the dotcom bubble, the largest seven tech stocks in the S&P saw a successively higher floor in realized volatility as the bubble progressed (Exhibit 82). The Mag7 stocks haven't yet exhibited the same ratchet higher in realized vol, even briefly approaching ChatGPT-era vol lows during this summer's post-Liberation Day lull (another sign that we may still be in the early innings of an AI bubble). However, aggregate S&P single stock volatility did not decline to the same extent, suggesting that other pockets of the market "took the baton" from the Mag7 in supporting broader stock vol (Exhibit 82).⁶

In short, the **bear case for single stock volatility in 2026** would seem to require the following **narrow path** to be traversed: (i) a prolonged grind higher in the core of AI as its size and strong earnings prevent it from disconnecting from fundamentals; coupled with (ii) a lack of sufficient micro-bubbles in the periphery that boom & bust.

Exhibit 81: Across both the dotcom bubble and the uptrend in S&P single stock volatility over recent years, it has been rare for stock volatility to remain materially below today's levels on a sustained basis

Consecutive days with SPX 1m single stock realized volatility less than 25%

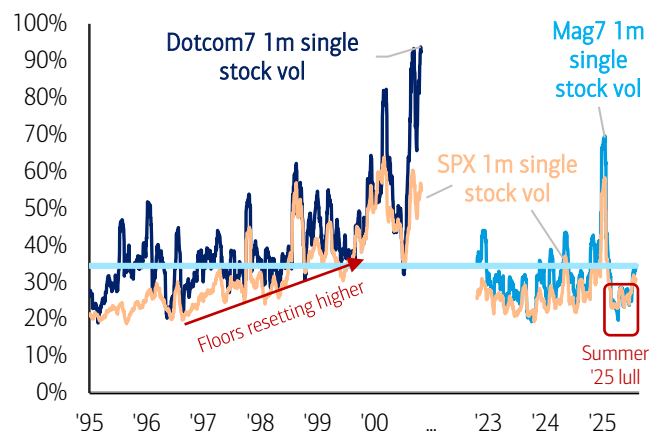


Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25.

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Exhibit 82: In the dotcom bubble, realized vol for the largest 7 S&P tech stocks rose as the bubble progressed. The Mag7 broke this trend in summer '25, though other pockets of froth helped support stock vol

Dotcom7*, Mag7** and SPX 1m single stock realized volatility



Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25. *Dotcom7 are the largest 7 tech stocks in SPX at the peak of tech bubble: Cisco, Microsoft, Intel, Oracle, IBM, Nokia, Nortel. **Mag7 or Magnificent 7 stocks are Apple, Amazon, Alphabet, Meta, Microsoft, Nvidia, Tesla.

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What about policy vol?

Policy volatility was a feature, not a bug, of the US administration in 2025. The April tariff shock around "Liberation Day" delivered (i) the largest joint drawdown in US equities, Treasuries, and the dollar since the 1987 crash; (ii) more S&P realized volatility in seven trading sessions than in all of 2024; and (iii) only the third episode since 1990 in which the VIX closed above 50 (joining the GFC & Covid).

⁶ For instance, July saw vol support stem from non-Mag7 tech stocks like AVGO, TXN and AMD as well as healthcare stocks like LLY, UNH, and ELV.

Looking ahead, the administration's incentives to take risk and experiment with radical policy likely diminish as the Nov-26 US midterm elections approach, given that losing represents a significant tail risk. This suggests the strike of the "Trump put" should rise in 2026, removing a key tailwind for equity volatility in 2025. Whether the put is struck first in US equities is debatable, but a rising Trump put strike likely means policy shocks become less frequent in the run-up to the midterms.

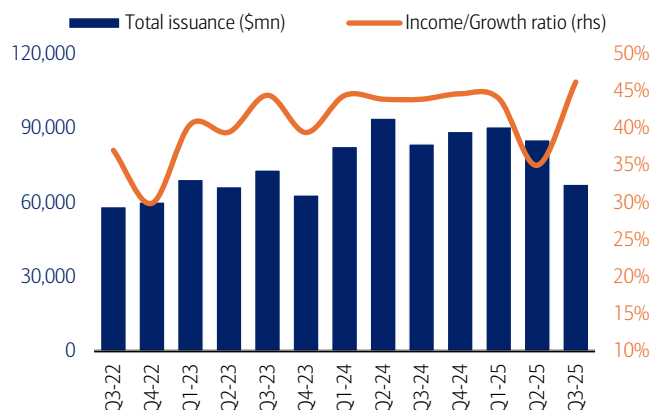
As for the outcome of the midterms, a loss of unified government could introduce gridlock—typically associated with lower policy volatility. However, it may also re-embolden Trump to wield executive powers more liberally, with little to lose. So, while policy volatility may ease heading into the midterms, we may see it re-emerge afterwards.

Booming derivatives income & its market impact

The retail investment landscape has become more focussed on income products in recent years. Evidence of this can be found in the higher proportion of income vs growth-oriented retail structured products, ~\$80bn of which is issued in EMEA & North America every quarter (Exhibit 83). Strikingly, we see a similar trend in ETFs, arguably the most successful/versatile financial instrument in history. Indeed, income ETFs have grown by 61% in the last 3yrs, significantly outpacing broader ETF growth (Exhibit 84).

Exhibit 83: There is growing focus on Income (vs participation) in equity-linked retail structured products.

Qtrly issuance of equity-linked struct. products in EMEA+ N. America (\$mn)



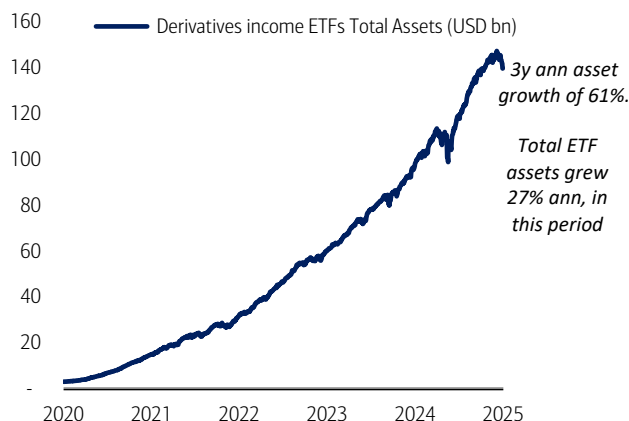
Source: SPI, part of WSD, BofA Global Research. Quarterly data from Jul22 to Oct-25.

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In 2025, we've also seen ETFs embrace the autocallable, a payoff that's historically been the preserve of structured products distributed through financial intermediaries. What's driving this thirst for yield? In our view, it is partly a generational preference for attractive 'coupons' on offer (~15%p.a., see Exhibit 85), which rival average annualised US equity returns. And partly, it is the potential for tax-efficient distributions (e.g., return of capital) from some income vehicles. In any case, (attractive) yield is only half the story in our view as investors should also consider drawdown and capital depreciation risks. Indeed, a number of the most popular income ETF strategies have not yet been tested in prolonged, difficult market conditions as many of the largest were launched after COVID.

Exhibit 84: Derivatives income ETF AuM has grown to \$140bn at a pace that is >2x faster than all ETFs.

AuM of derivatives income ETFs based on Bloomberg's FSRC function*



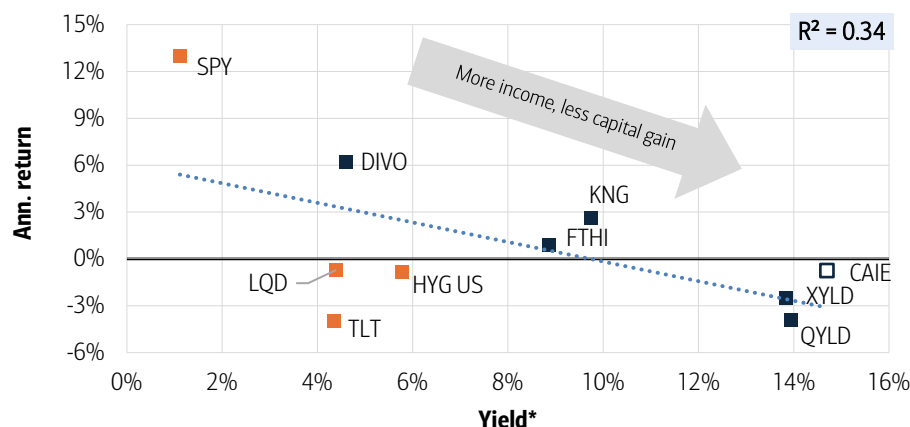
Source: BofA Global Research, Bloomberg (* & expert judgement), EPFR. 23-Nov-20 to 21-Nov-25.

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Exhibit 85: High yielding income products tend to forgo capital gains.

Ann. price return (-axis) vs div yield* (x-axis). Blue squares: Derivatives-based income ETFs, orange squares: x-asset ETFs for equities (SPY), rates (TLT) & credit (LQD, HYG) investments, unfilled square: Autocall ETF



Source: BofA Global Research, Bloomberg. Data: 27-Mar-18 to 20-Nov-25. Start date reflects when at least 5 of the top 15 derivatives income ETFs by AuM were available. *Using trailing 12m div yield, except for CAIE where this is the indicated yield.

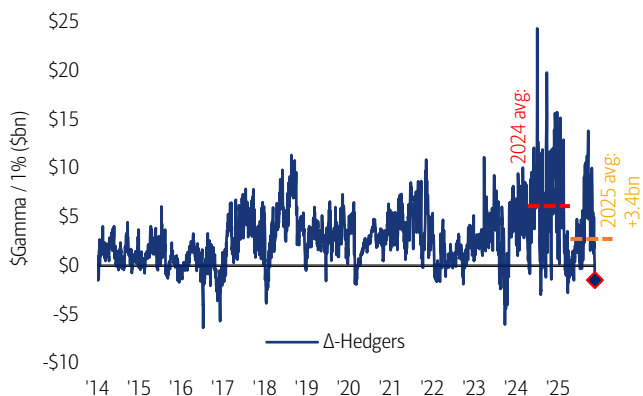
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Market impact of derivatives income: A story of (SPX) gamma & (SX5E) vega

Most questions we field about derivatives income strategies centre around their market impact. Given that most derivatives income ETFs are net sellers of short-dated equity options, this naturally leads to a discussion on dealer gamma positioning. Exhibit 86 shows this for SPX options using signed trade data from the CBOE. As expected, option dealers are generally net long SPX gamma. However, '25 saw the imbalance nearly halve vs '24. Moreover, Exhibit 87 shows that the likely market impact (vol suppression) of this gamma imbalance is only between 1-2pts for 1m SPX vol (which was recently ~13v).

Exhibit 86: 2025 saw a lower imbalance in gamma positioning vs '24; delta-hedgers were long \$3.4bn of SPX gamma vs \$6.2bn last year.

EOD SPX gamma for Δ -Hedgers

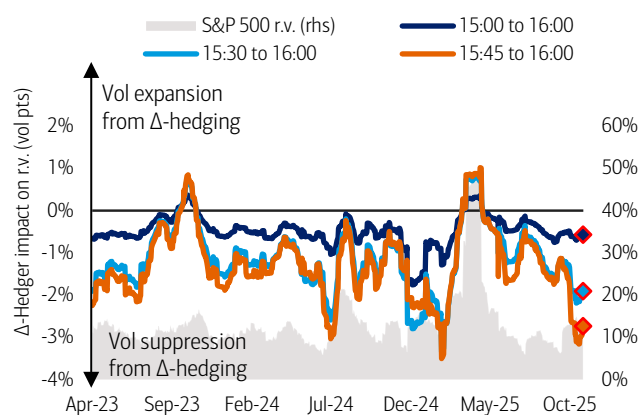


Source: BofA Global Research, CBOE. Data as of 19-Nov-25. The gamma is derived from exchange data which classifies trades by market participant type. The gamma level above combines the gamma from regular Firm trades and all (regular + flex) Market-Maker trades.

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Exhibit 87: The impact of long gamma for delta-hedgers is likely suppressing 1m SPX vol by 1-2v points (~10% of the level of vol)

Est. impact of SPX gamma on S&P fut 1m RV (different Δ -hedging windows)



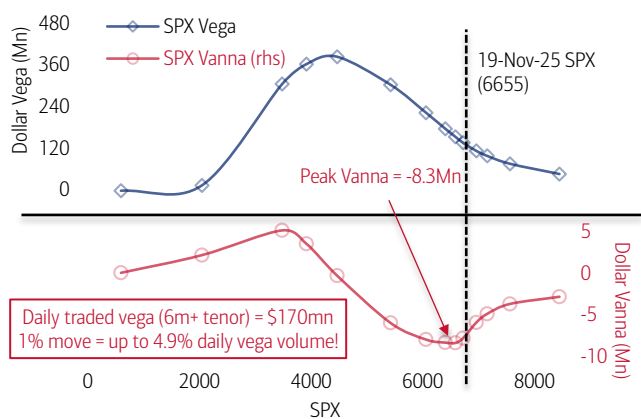
Source: BofA Global Research, CBOE. Data as of 19-Nov-25. Impact figures are quoted as market observed realized vol - estimated vol without gamma.

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What about autocallables and longer-dated income structures? Here, we look at the vega profile (from a dealer's perspective) for retail structured products linked to SPX (Exhibit 88) and SX5E (Exhibit 89). While peak vega and corresponding strikes are useful to gauge when autocallable hedging flow may change direction, the market impact on volatility is better estimated by how much vega changes with spot (vanna). Now while peak vanna is close to current spot levels for SPX and SX5E, it is a significantly higher (4x) fraction of traded vega volume for SX5E than it is for SPX; i.e., greater SX5E market impact.

Exhibit 88: Peak market impact of autocallable issuance (vanna) in the SPX is near the 6400 strike, but this is <5% of (listed vega) volume...

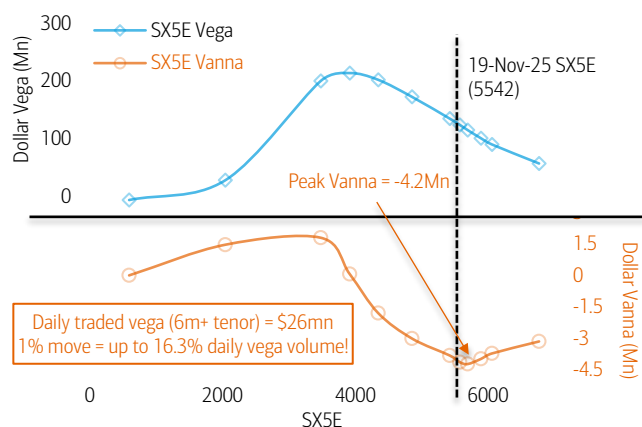
Vega/Vanna profile of live SPX-linked structured products issued in NA/EU



Source: BofA Global Research, SPi, part of WSD, Bloomberg. Vega derived from the universe of products live as of 21-Oct-25. Vanna calc using adjacent vega values. Daily traded vega = 1m avg.
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Exhibit 89: ...meanwhile, peak vanna for the SX5E is near the 5700 strike, but this is over 16% of listed daily vega traded

Vega/Vanna profile of live SX5E-linked structured products issued in NA/EU



Source: BofA Global Research, SPi, part of WSD, Bloomberg. Vega derived from the universe of products live as of 21-Oct-25. Vanna calc using adjacent vega values. Daily traded vega = 1m avg.
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How to trade our global themes

Table of Contents

Don't fight the future: A plan for the AI right tail	42
Cheapest call options on the planet includes China AI	43
How to buy-the-dip without having to catch a falling knife	44
Tail hedging the AI titan: Structured flows = opportunity	47
Hedgecraft: Systematic hedging for equity-bond portfolios	49
Hedging long-end pain with low rates vol & 10x hybrids	53
Long vol, light on carry: SX5E fwd var leads globally	54
Once bitten twice shy: Hedging FX for non-USD investors	55
Cheaply gear for EZ Banks dispersion after blistering rally	56
Nikkei up on a stronger yen for an 80% discount	57
The China repo arb – CSI 1000 calls vs ASHR calls	58

Don't fight the future: A plan for the AI right tail

What: Buy (i) NDX Dec26 ATM upvar, currently 19.1v but target an entry level closer to 17v (t & t-1 convention; var ref. 29.9v), and/or (ii) QQQ Jun26 660-750 call spreads for \$18 (~35D/10D strikes, 5x max payout ratio, ref. 605.16). Risks involve upfront premium paid for the call spread & variance strike paid for the upvar.

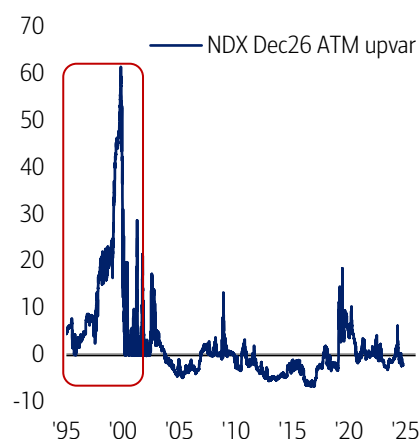
Why: As noted in our Outlook, major technological leaps fuel major asset bubbles, and an eventual AI bubble seems hard to avoid. Often driven by retail's fear of missing out, bubbles create a painful dilemma for value-oriented investors: chase the rally, or risk underperforming until the bubble pops? The key is having a plan for the right tail.

As vol rising alongside prices is a hallmark of asset bubbles, we continue to find long delta, long vol trades like owning NDX up variance ('upvar') well-suited for further bubble inflation. At current prices, upvar would have performed extremely well in the late 90s tech bubble (Exhibit 91) and has carried well this year despite periods of muted realized vol. Moreover, cutting off exposure to dislocated deep OTM put vols creates an historically attractive entry point over 10pts below vanilla var (Exhibit 90).

However, the timing of an AI bubble's potential blow-off top remains uncertain, and the risk of a slow grind higher is a key challenge for upvar. To address this, we like pairing upvar with, for example, long 6m QQQ call spreads rolled prior to expiry. These offer high max payout ratios due to relatively flat call skew (Exhibit 92) and are well-suited for a grind higher, especially when rolled early to avoid steep rolldown. Indeed, in periods when upvar struggled amidst a benign NDX rally, the call spread would have complemented it well (Exhibit 93). Alternatively, one could also pair upvar with systematic grind higher strategies that fund delta-replicated calls with short-dated upside vol.

Exhibit 91: At current pricing, NDX ATM upvar would have performed particularly well as the dotcom bubble inflated in the late 90s

Backcast of NDX Dec26 ATM upvar (at current prices)



Source: BofA Global Research. Data from 1-Jan-95 to 21-Nov-25. This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 92: Given the flat call skew, 6m QQQ call spreads offer historically high max payout ratios

QQQ 6m 35d/10d call spread payout ratio

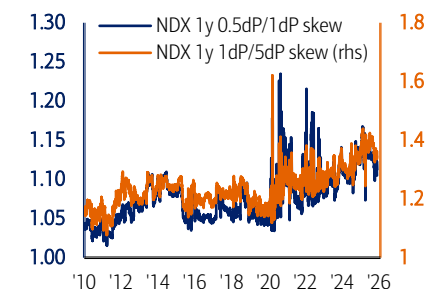


Source: BofA Global Research. Data from 1-Jan-12 to 21-Nov-25. Indicative payout ratios based on mid prices.

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Exhibit 90: By cutting off exposure to the expensive left tail, NDX upvar trades at an attractive 10v+ discount to vanilla variance

NDX 1y 0.5d/1d and 1d/5d put skew

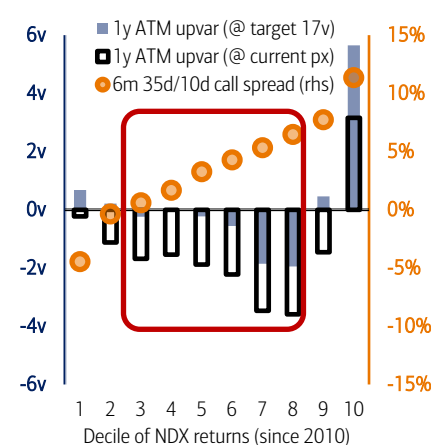


Source: BofA Global Research. Data from 1-Jan-10 to 21-Nov-25.

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Exhibit 93: Pairing NDX upvar with risk-limited long delta can help mitigate some of the upvar's challenges in a grind higher

Hypothetical backtest of (i) NDX ATM upvar (at current/target prices) and (ii) NDX call spreads*



Source: BofA Global Research. Data from 1-Jan-10 to 20-Nov-25. *6m call spreads rolled after 3m. This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Cheapest call options on the planet includes China AI

Long China small cap innovation at artificially low risk – 7.9x max payout

What: Buy 6-month 105% 125% call spreads on a CSI 1000 tracking basket† for **2.53%** (indic. 20d, 27.1v/28.7v, quanto USD)

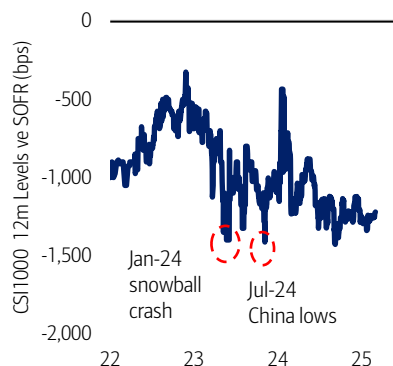
Why: The “artificially” cheap CSI 1000 forward corresponds to an 8 vol discount. Investors can buy the most exciting part of the Chinese stock market, including AI innovation, at unusually low & limited risk

Attractive asymmetry in China: A favorable flow outlook into the Chinese stock market but with uncertain macro backdrop begs for asymmetrical exposure. Large upside potential and limited downside. The uniquely expensive repo and borrow cost for Chinese small caps translates to very depressed forwards and cheap call options.

Money flowing into stocks: The household savings rate in China remains high, estimated around 32-34% (U.S. personal savings rate is about 4.6%). Savings used to go into real estate and wealth management products (WMP) earning high yields (“shadow banking”), but the property slump has all but removed that option and Beijing has clamped down on shadow banking.

TINA in China: “There Is No Alternative” to stocks. At the same time as properties and WMP are no more, China has some of the lowest interest rates globally. Chinese 10yr government bonds yield a meagre 1.8%. But for how long can stocks rally if the macro doesn’t improve? This is where attractive asymmetry via CSI 1000 calls comes in. Should markets crack by, say 20-30%, your max losses are just 2,53%.

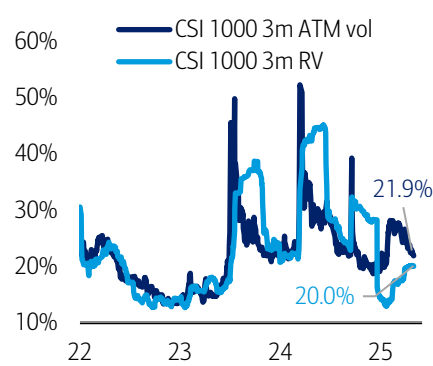
Exhibit 96: CSI 1000 swap levels remain low
12-month swap spreads over SOFR (in bps)



Source: BofA Global Research. Data from 1-Aug-22 to 24-Nov-25.

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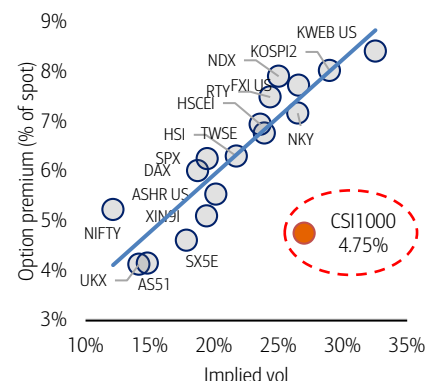
Exhibit 97: CSI 1000 vols have fallen lately
3-month ATM vol and realised vol



Source: BofA Global Research. Data from 22-Jul-22 to 19-Nov-25.

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Exhibit 95: CSI 1000 calls can be seen as the cheapest calls globally if translating the artificially low forward to a ~8 vol discount
6-month ATM call cost vs implied vol



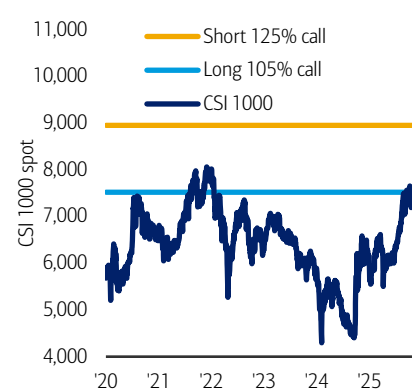
Source: BofA Global Research. Data as of 21-Nov-25. Indicative offers.

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†: several banks offer CSI tracking baskets

Risks to call spread limited to premium paid

Exhibit 98: The strikes of the call spread
A 6-month 105% 125% call spread



Source: BofA Global Research. Data from 2-Jan-20 to 24-Nov-25.

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How to buy-the-dip without having to catch a falling knife

What: Leverage buy-the-dip (BTD) with limited risk by selling next day 40d-25d SPX put spreads to fully fund calls if a (sigma shock-based) signal triggers.

Why: A Fed put, Trump put, and conditioned upside-fearing investors have driven BTD strength to near 100y highs, with the phenomenon having most asymmetric risk-reward for short-dated (1-2day) SPX returns. We show that a simple, risk-limited, zero premium option strategy that buys calls fully funded by the sale of put spreads when a BTD signal triggers, helps diversify classic upside vol selling strategies. The strategy would have generated an info ratio of 0.67 in the last 15y, with the signal improving risk-reward of the 'always on' option strategy by >60% and the structure yielding ~30% greater upside convexity than using just long calls on BTD triggers.

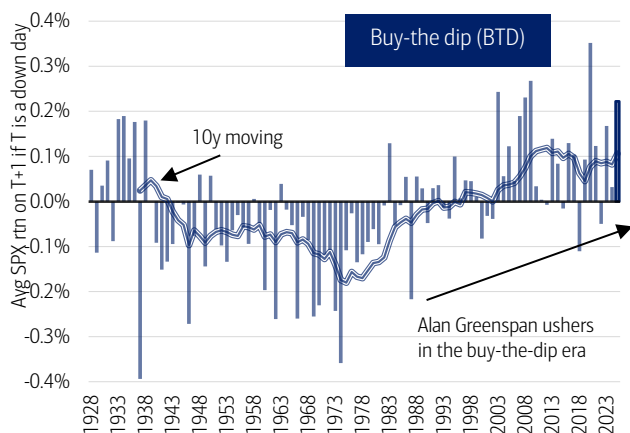
Risk: Limited at each BTD trigger to width of short put spread (60bp on avg).

2025 is shaping up to be the year with close to the second-highest dip-buying strength in nearly 100 years (after COVID in 2020, Exhibit 99). The trifecta of a Fed put (room to ease if growth fears materialise), a Trump put (the US administration's willingness to reverse course if their words/policies upset markets), and upside-fearing investors acting on a learned response to prior market dips, have all contributed to this phenomenon.

Exhibit 100 shows that the strength of buy-the-dip in SPX returns is greatest for short term (1-day) horizons, where the statistical skewness of returns conditioned on a prior downside shock is positive (unconditioned returns are negatively skewed). Moreover, history shows that SPX 1-5day returns are up to 2x greater than usual after a simple 'Buy-the-dip' (BTD) signal triggers (based on trailing daily and weekly SPX returns being worse than a fraction, e.g., -half, of trailing realised volatility, Exhibit 101).

Exhibit 99: Ever since Alan Greenspan 'invented' the Fed put in the '80s', Buy-the-dip (BTD) has become a stronger force in equities

Avg daily SPX return on the day after a down day, by year

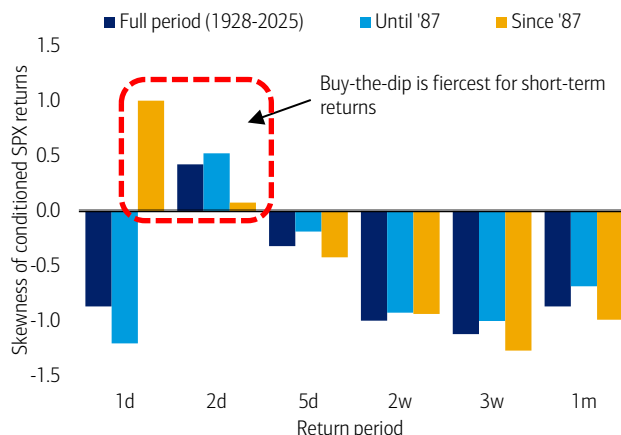


Source: BofA Global Research. Data from 4-Jan-1928 to 25-Nov-2025.

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Exhibit 100: Indeed, short-term equity returns are skewed positively if we conditioned on a prior downside shock

Skewness[†] of SPX returns conditioned on a prior downside shock*



Source: BofA Global Research. Data from 4-Jan-1928 to 30-Oct-2025. *Defined as trailing daily SPX return worse than -2sigma, where sigma is trailing EWMA vol (using 0.95 decay). †Skewness referring to third central moment calculated using historic daily SPX returns

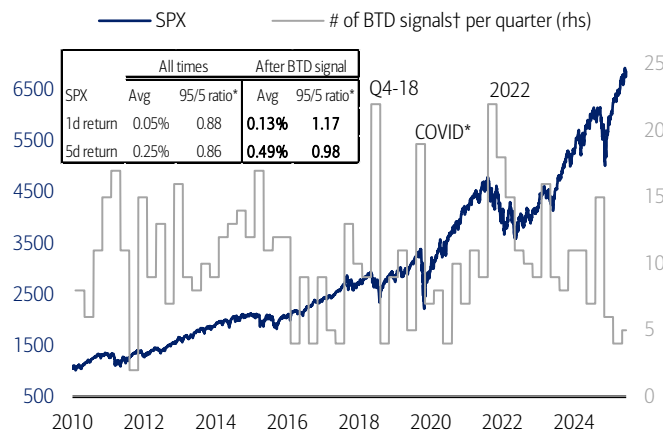
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Whenever signals are involved, parameter choices are important. While we don't wish to be prescriptive, the choice of the BTD signal we use here is guided by trying to achieve a balance between hit ratio (signal success rate), sensitivity (i.e. not so hard-to-trigger as to be too infrequent) and risk-reward (how right vs how wrong), as Exhibit 102 shows.



Exhibit 101: Moreover, history shows that SPX daily & weekly returns are ~2x than usual after a simple Buy-the-dip (BTD) signal triggers

Historical distribution of SPX BTD signals (when trailing daily & weekly returns are $< -0.5\sigma$ sigma¹). Relevant SPX return stats are shown in the box

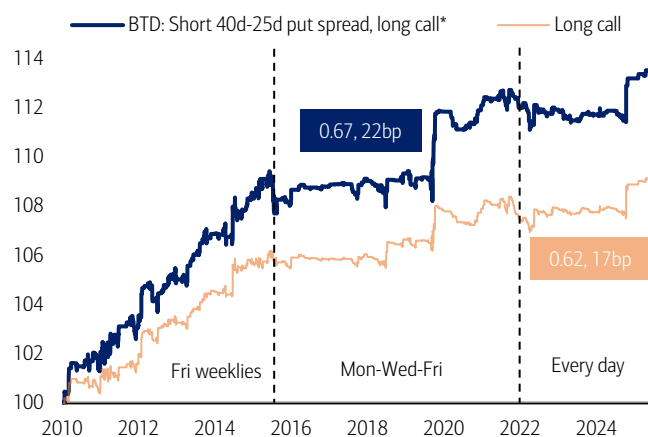


Source: BofA Global Research. Data: 1-Jun-10 to 10-Nov-25. *95/5 = Ratio of avg ES return $> 95^{\text{th}}$ %ile and abs (avg SPX return $< 5^{\text{th}}$ %ile). ¹A BTD signal is triggered based on the 6min TWAP of SPX futures centred at 1445 ET, if the daily return and weekly return are both worse than a 0.5 sigma decline, where sigma is calculated using trailing EWMA vol ($\lambda = 0.97$).

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Exhibit 103: An option strategy that sells SPX put spreads to fully fund calls on a BTD trigger would have been successful historically

Hypothetical back-test: SPX BTD strategy sells SPX 40d-25d put spreads to buy a call for nearest listed expiry on signal days (avg strikes: 99.7%, 99.1% & 100.8%). Info ratio & avg of best ($> 95^{\text{th}}$ %ile) daily returns shown in boxes



Source: BofA Global Research. Data: 1-Jul-10 to 10-Nov-25. *On BTD signal days, the strategy sells a 40d-25d put spread to buy a call for zero premium for the closest listed expiry at end of day (Until 2016, this is weeklies (Fri expiry), then Mon-Wed-Fri expiries until 2022, then daily listed expiries since). The signal itself is observed using a 6min TWAP of SPX e-minis centred at 1445 ET. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

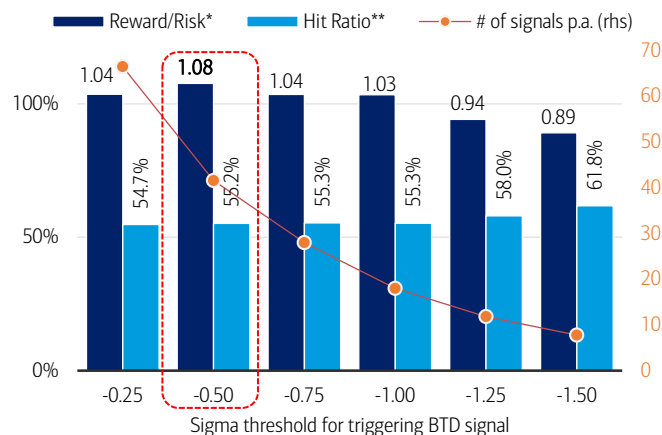
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Constructing a risk-limited SPX option-based BTD strategy

Exhibit 103 shows the back-test of a simple BTD option strategy; it purchases short-dated (1-5day) SPX calls that are fully funded by the sale of same tenor SPX 40d-25d put spreads whenever the BTD signal triggers (which is about 17% of the time). We see that monetising risk aversion by selling put spreads to fund calls would have not only increased risk-adjusted returns (0.67 vs 0.62) versus buying calls outright, but also

Exhibit 102: Our choice of a BTD signal is guided by balancing risk/reward (of subsequent SPX returns) versus signal sensitivity

Reward/risk ratio of subsequent SPX returns, Hit (success) ratio of our signal and the number of signals generated p.a. (rhs) for different signal thresholds

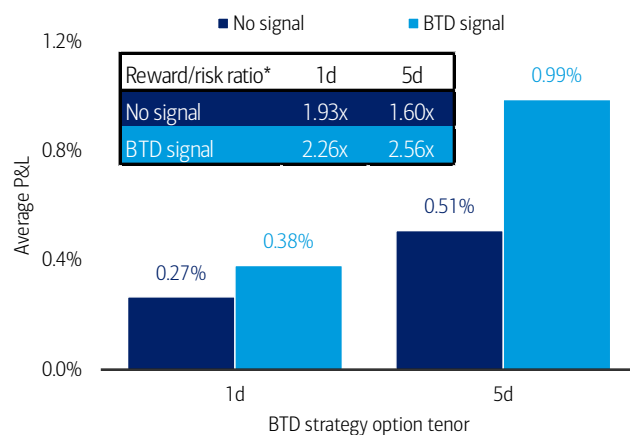


Source: BofA Global Research. Data: 1-Jun-10 to 10-Nov-25. *Reward/Risk = ratio of subsequent avg SPX return $> 95^{\text{th}}$ %ile and abs (avg SPX return $< 5^{\text{th}}$ %ile) after a BTD signal triggers, averaged for subsequent 1d and 5d returns. **Hit ratio = fraction of time the median SPX return after a BTD signal $>$ median SPX return, averaged for 1d and 5d returns.

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Exhibit 104: Crucially, the BTD signal would have greatly improved risk/reward (+60%) of selling short-dated put spreads to fund calls

Hypothetical back-test of selling 1d and 5d 40d-25d put spread to full fund same tenor calls (avg strikes: 99.7%, 99.1% & 100.8%)



Source: BofA Global Research. Data: 1-Jul-10 to 10-Nov-25. *Reward/risk ratio = Avg P&L if positive / Avg P&L if negative. A BTD signal is triggered based on the 6min TWAP of SPX futures centred at 1445 ET, if the daily return and weekly return are both worse than a 0.5 sigma decline, where sigma is calculated using trailing EWMA vol ($\lambda = 0.97$). The strategy sells a 40d-25d put spread to buy a call for zero premium for the closest listed expiry at end of day (until 2016, this is weeklies (Fri expiry), then Mon-Wed-Fri expiries until 2022, then daily listed expiries since). This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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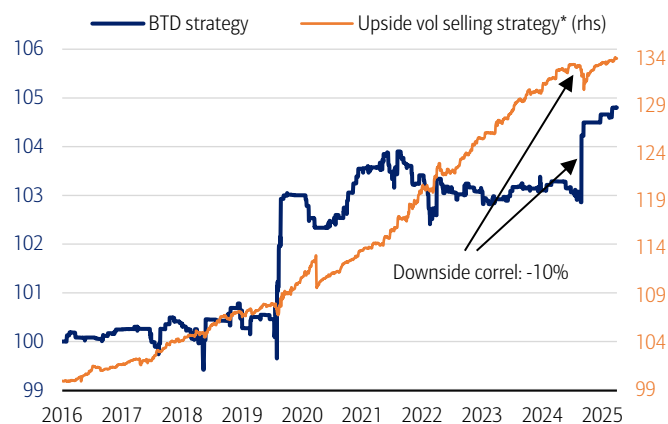
improved upside convexity (average best daily return of 22bp vs 17bp). Moreover, Exhibit 104 shows that the BTD signal would have helped improve the risk/reward of selling put spreads to fund calls systematically by up to 60%.

Where does a BTD strategy fit in your portfolio?

At its core, a BTD strategy's aim is adding equity risk at the right time, more often than not. Given the signal-driven nature of the strategy (in our example, it's active 17% of the time), it's sensible to either embed it within or overlay it on 'always-on' strategies. For example, we look at how our BTD strategy can diversify a classic (vol) carry strategy that typically sells short-dated calls. Exhibit 105 and Exhibit 106 show that not only would BTD be uncorrelated to systematic upside vol selling, it would have been profitable in the worst decile of upside vol selling returns.

Exhibit 105: Avoid selling calls into a drawdown. A BTD option strategy is complementary to vol carry strategies that sell short-dated calls...

Hypothetical back-test: SPX BTD strategy sells SPX 40d-25d put spreads to buy a call for nearest listed expiry on signal days (avg strikes: 99.7%, 99.1% & 100.8%). *BAEISMIH (intra-day hedged short-dated SPX call selling strategy)

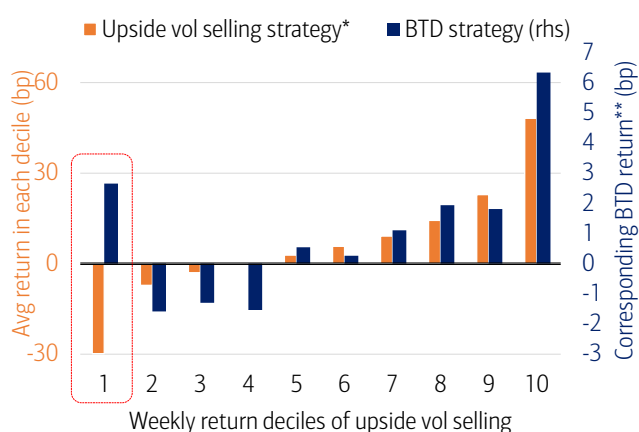


Source: BofA Global Research. Data: 1-Jul-10 to 10-Nov-25. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 106: ...moreover, BTD strategies can help offset the worst drawdowns of call selling. Embed BTD within or overlay accordingly.

Hypothetical back-test: SPX BTD strategy sells SPX 40d-25d put spreads to buy a call for nearest listed expiry on signal days (avg strikes: 99.7%, 99.1% & 100.8%). *BAEISMIH (intra-day hedged short-dated SPX call selling strategy)



Source: BofA Global Research. Data: 1-Jul-10 to 10-Nov-25. **Avg BTD strategy weekly return in each decile of upside vol selling returns. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Tail hedging the AI titan: Structured flows = opportunity

What: Buy long-dated OTM NVDA puts (e.g., 2y 10-delta), delta-hedged.

Why: As the preeminent AI stock, NVDA may seem “too big to fail”, yet it remains exposed to significant tail risks—both exogenous (e.g., China-Taiwan tensions disrupting semiconductor supply chains) and endogenous (e.g., reassessment of AI-related demand growth). While debate on how significant these risks are is high, they can be hedged cheaply by owning longer-dated NVDA put vol and skew. More structured product issuance and “autocallification” of NVDA vol mean these parameters could see a strong bid in a tail event as dealer hedging exacerbates the moves, potentially driving payouts on the order of 35-40x.

Risk: Losses in a prolonged vol-down, spot-down grind lower in NVDA.

Retail structured products making their way to US-listed tech stocks

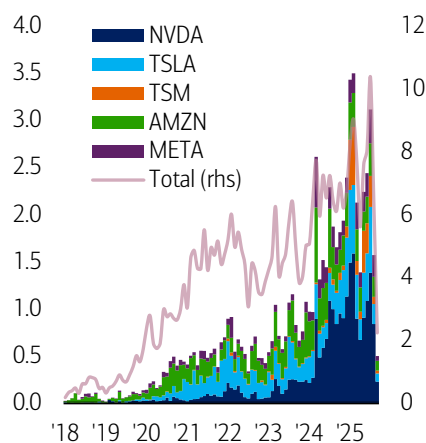
Equity-linked retail structured products have long been a major driver of European and Asian equity derivatives markets. North American issuance has grown substantially in recent years, however. And while the S&P and Russell 2000 still dominate, single stock products have become more relevant. Issuance on US-listed single stocks, which was negligible pre-Covid, surged in 2020 and has nearly tripled since, averaging ~\$7bn per month in 2025 (Exhibit 107). Tech stocks have also become increasingly prominent, with NVDA, TSLA, TSM, AMZN & META the top 5 single-name underlyings in 2025 (Exhibit 107).

Signs of “autocallification” in NVDA longer-dated put volatility?

Dealer hedging of autocallable structured products, for example, tends to leave a mark on underlying vol markets—driving longer-dated put vol to be positively correlated with spot (as seen for the Russell 2000 in Exhibit 109) and unusually flat long-term deep left tail skew. Interestingly, we see evidence of both dynamics in 2y NVDA implied vol (Exhibit 108 & Exhibit 109), though this was also true at times before the surge in NVDA issuance.

Exhibit 107: Retail structured products are making their way to US-listed tech stocks

Structured product issuance on US-listed single stocks (\$bn), top 5 by 2025 YTD issuance + total

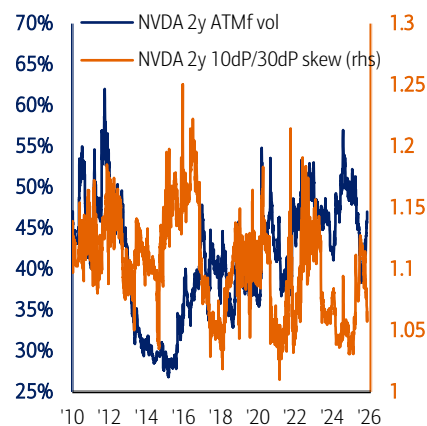


Source: BofA Global Research, SPI, part of WSD. Monthly data as of Sep-2025.

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Exhibit 108: Signs of “autocallification” in NVDA? Flat long-term deep left tail skew...

NVDA 2y ATMf vol and 10d/30d put skew

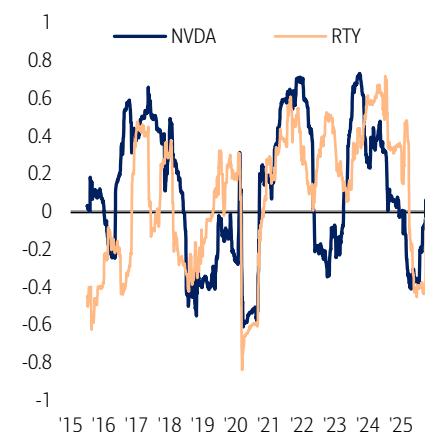


Source: BofA Global Research. Data from 1-Jan-10 to 20-Nov-25.

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Exhibit 109: ...and +ve spot-vol correl à la RTY, a good proxy for structured product influence

NVDA & RTY 6m correlation between spot and 2y 80% fixed strike vol



Source: BofA Global Research. Data from 1-Jan-15 to 10-Nov-25.

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Leveraging derivatives dislocations to construct an efficient NVDA tail hedge

We have previously exploited these supply-demand imbalances in vol to construct low-cost tail hedges that benefit from a convexity kicker when dealer vega supply flips to demand (see our NKY 18m 1x3 put ratio from our [2020 Outlook](#)). Here, we propose a similar structure on NVDA—a stock that is perhaps “too big to fail”, yet still exposed to both exogenous (e.g., China-Taiwan tensions disrupting semiconductor supply chains) and endogenous tail risks (e.g., a reassessment of demand growth for AI-related tokens).

Exhibit 110 shows a hypothetical backtest of a simple long NVDA 2y 10-delta put that is entered monthly, held to expiry, and delta-hedged daily. At mid-prices, the strategy would have delivered the much-coveted tail profile of positive carry and positive convexity. Gains during 2025's DeepSeek and Liberation Day shocks were muted as 2y NVDA vol largely shrugged off these fragility events, but the Covid shock produced the prototypical convexity. Even the 2022 selloff, which unfolded over 11 months and challenged many convexity strategies at the index level, generated positive performance.

Estimating a max payout ratio is challenging, as (i) the backtest period never saw the intersection of NVDA's current theoretical downside with structured product hedging flows that could amplify the reaction of vol in a tail event; and (ii) NVDA option volumes have exploded over time, making static transaction cost assumptions unreliable.

That said, recent bid-mid costs for NVDA Jan28 10-delta puts are around 7-8bps of NVDA notional exposure, implying a carry cost of 35-40bps in Exhibit 110 given the 5x leverage. Haircutting the Covid spike of ~25% to 15%, to avoid top-ticking the unwind and to reflect likely higher historical transaction costs, implies a max payout ratio of around 35-40x were that event to repeat.

Exhibit 110: While tail risks for NVDA are hotly debated, they can be hedged cheaply via longer-dated put vol and skew. More structured product issuance and “autocallification” of NVDA vol mean these parameters could see a strong bid in a tail event as dealer hedging exacerbates the moves

Hypothetical backtest of long NVDA 2y 10d put; monthly entry, held to expiry, daily delta-hedge



Source: BofA Global Research. Data from 1-Jan-19 to 19-Nov-25. No transaction costs included. Monthly entry involves entering into 1/24th notional of puts every month (portfolio of 24 positions). This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Hedgecraft: Systematic hedging for equity-bond portfolios

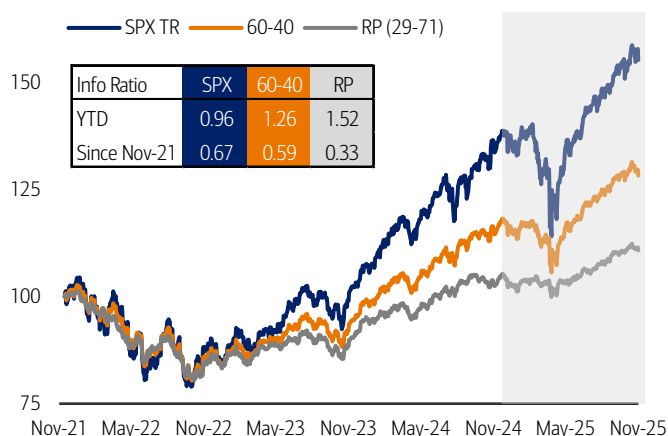
What: We propose a systematic equity-rates defensive overlay for 60-40 portfolios that combines our SkeU-trade (carry-neutral convexity in negative and positive equity shocks) and long 1y5y USD swaption straddles given recently lower rates volatility. We pair each leg with 'grind lower (equity) / higher (rates)' strategies to defend against either sharp or slower-moving selloffs in each asset.

Why: (i) A [SkeU-trade](#) & 'grind-lower' protection combo is a robust equity hedge solution in our view. The former is designed for convex shocks and sharp market rebounds, while the latter is better at hedging slow-evolving selloffs. (ii) Given duration's unreliable hedging credentials, we like leveraging recently lower rates vol to buy 1y5y USD swaption straddles for bullish or bearish rate shocks. With the risk that persistent inflation stays the Fed's hand, weighing on duration but without a spike in rates vol, we pair long 1y5y rates vol with a [rates 'grind higher'](#) strategy. The equity-rates combo overlaid onto a US 60-40 portfolio would've achieved a more balanced hedging outcome than equity or rates hedges alone in the last 10y, boosting return by 2.3%p.a., Sharpe by 40% & lowering tail risk 25%.

Risk: Lack of equity or bond sell-off, causing the overlay to carry negatively.

With US equities near record highs and the Fed expected to continue easing, investors are right to take comfort from the likelihood of a duration rally if the AI bubble bursts or growth falters. But what if inflation proves sticky and the Fed cannot ease as expected, or even considers hiking? The rise of inflation post-pandemic blindsided markets, triggering a crisis of diversification for multi-asset portfolios (Exhibit 111). And while bonds have 'worked' YTD, Exhibit 112 shows that their hedging role remains uncertain, more like the pre-2000s era than most of this century. Given this and the uncertain timing of shocks or when bond vigilantes return, we show how investors may systematically hedge both equity and duration risk in multi-asset portfolios.

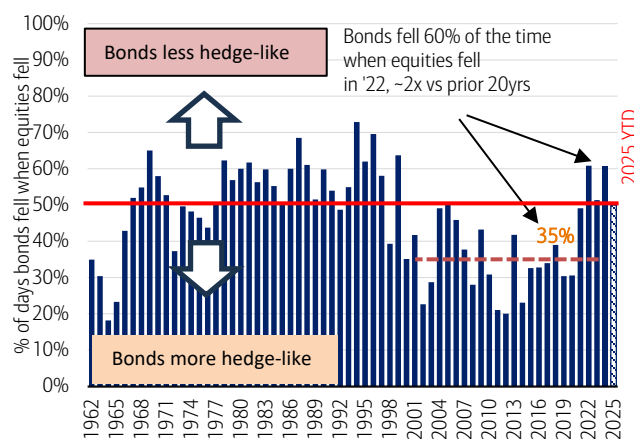
Exhibit 111: While 60-40 & Risk parity portfolios outperformed equities in risk-adjusted terms YTD, they struggled in recent years
SPX TR, 60-40 (Equity-Bond) & Risk parity (29-71, Equity-Bond) portfolios*



Source: BofA Global Research. Data: 26-Nov-21 to 13-Nov-25. *Using SPX TR & 7-10y UST returns.
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The tale of tail & grind-lower: Building a robust equity hedging overlay: The sharp early Apr-25 equity sell-off was followed by a violent rebound just days later, underscoring the need for hedges to 'lock-in' gains and for positively convex strategies to capture both downside and upside fragility shocks. The [SkeU-trade](#), introduced in our [Year Ahead 2025: The roaring 2020s?](#), remains a strong candidate for both these objectives, thanks to its symmetric gearing to up and down shocks. As a reminder, the

Exhibit 112: Indeed, even in '25, it's been a coin toss for bonds' hedging credentials - they fell 50% of the days stocks were down YTD
Fraction of S&P down days per year that 7-10y UST TR is negative



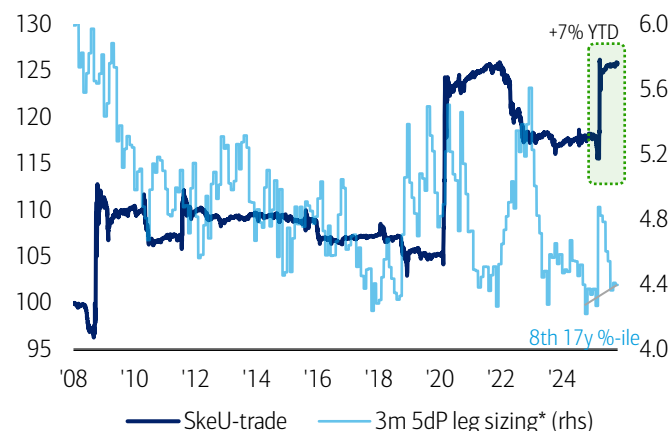
Source: BofA Global Research. Data: 2-Jan-1962 to 13-Nov-2025.

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strategy systematically trades 3m -25d/+5d SPX delta-hedged put ratios, applying a skew-adjustment to the long 5d leg and sizing for zero-premium. The skew-adjustment modifies put option delta in a way that helps in equity rebounds and offers a superior long-run cost of carry. YTD, the SPX SkeU-trade would have been up ~7% (Exhibit 113).

Exhibit 113: The SkeU-trade is well suited for sharp selloffs & strong rebounds. Even a low (4.4x) sizing of tail puts like now can be powerful

Hypothetical back-test: SkeU-trade: 3m -25d/+5d SPX delta-hedged put ratio with skew-adjustment to 5d leg, sizing 5d put for zero premium at inception



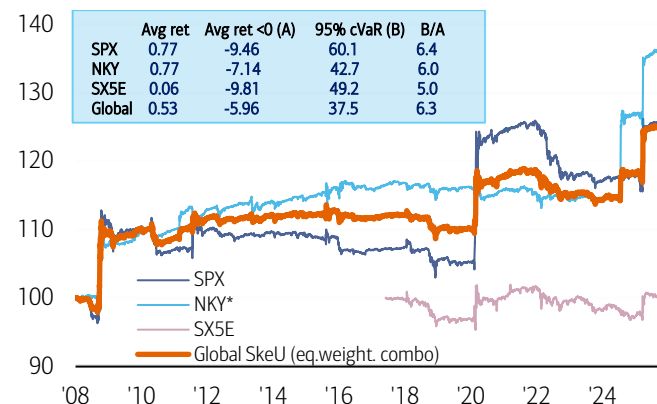
Source: BofA Global Research. Data from 18-Jan-08 to 10-Nov-25. *Number of 5d puts bought for each 25d put sold. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 114 shows the benefits of extending the SkeU trade to other indices; the NKY version would have had a superior carry profile and strong gains in both 2024 (Yenmageddon) and 2025 (Liberation day). A combination of SPX, NKY and SX5E SkeU trades can help strategy scalability and manage carry cost without sacrificing convexity.

Exhibit 114: Diversifying SkeU across regions can offer scalability and help reduce the cost of carry without sacrificing convex gains

Hypothetical backtest: SkeU-trade across major global markets. Stats in box since Jun'17 (SX5E start date) are computed on daily returns (shown in bps)

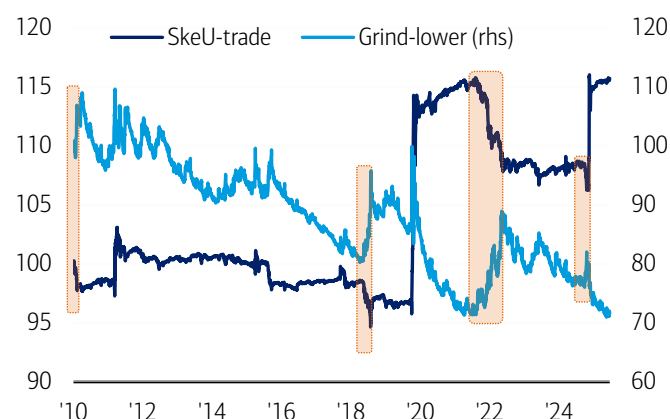


Source: BofA Global Research. Data from 18-Jan-08 to 10-Nov-25. *NKY: Dynamic sizing targets long-term avg cost (using a 5y rolling window) of a 1x3 put ratio. This performance is backtested and doesn't represent actual performance of any account or fund. Back-tested performance depicts theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 115: SkeU struggles when equity declines are more measured (shaded regions). This is where 'grind-lower' hedges can help

Hypothetical back-test: SkeU-trade & Grind-lower strategy ([Mind the grind...](#)). Latter is long SPX 1wk 40d put, short 3x 1wk 10d puts delta-hedged at close

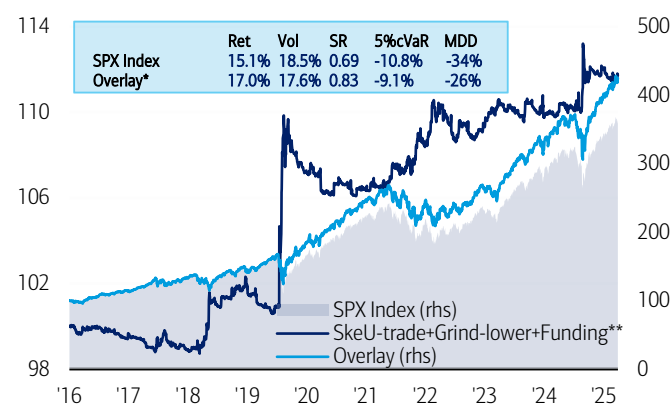


Source: BofA Global Research. Data from 28-May-10 to 10-Nov-25. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 116: A SkeU+Grind-lower overlay would have boosted SPX returns by ~2%p.a., with 20% higher Sharpe and lower tail risk

Hypothetical back-test: Combination of SPX SkeU-trade, Grind-lower strategy & a funding leg**, & the overlay of this combo onto the SPX



Source: BofA Global Research. Data from 15-Aug-16 to 10-Nov-25. *Overlay notional obtained via 5% cVaR minimization given a max tracking error of 3% p.a. Weights: SkeU trade (42%), Grind-lower (22%), funding leg (28%), expressed in 10% vol target. **BAEISMIH (delta-hedged upside call selling, contact us for more info). This performance is backtested & does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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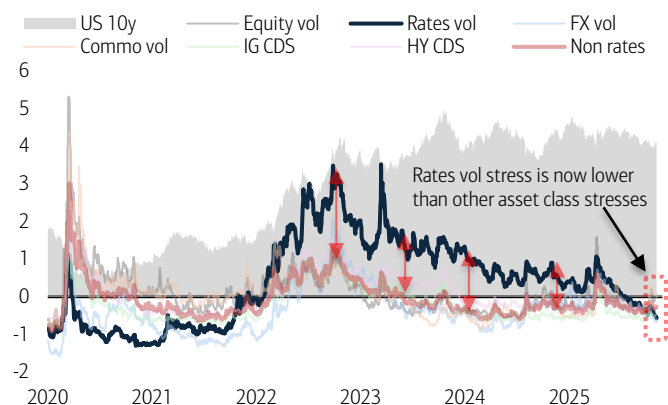


Like other tail hedges, however, the SkeU-trade can struggle in slow-moving equity declines, even if they are substantial (e.g., 2022). This is when 'grind-lower' hedges shine ([Mind the grind, not just the gap](#), Exhibit 115). Indeed, Exhibit 116 shows that combining SkeU with a 'grind lower' strategy would have hedged SPX effectively, enhancing returns by 2% p.a., risk-adjusted returns by 20% and materially reducing tail risk.

Hedging your hedge: Building a defensive overlay for duration: One of the most noticeable developments in cross-asset risk in 2025 has been the repricing of rates vol (lower) versus other asset classes, after several years of it trading at a substantial premium (Exhibit 117). While this can be rationalised by assuming that rates markets deem risks to duration are tilted towards mildly bearish moves (rather than a full-blown repricing of the Fed's trajectory), it also leaves rates volatility an attractive own for those concerned about Fed independence, fiscal largesse or a resurgence of inflation.

Exhibit 117: After years of being persistently elevated across asset classes, the rates vs cross-asset vol gap has closed in recent months

GFSI stress (modified z-score*) in equity, rates, fx & commodity vol, IG & HY CDS, and US 10y rates (x100bps)

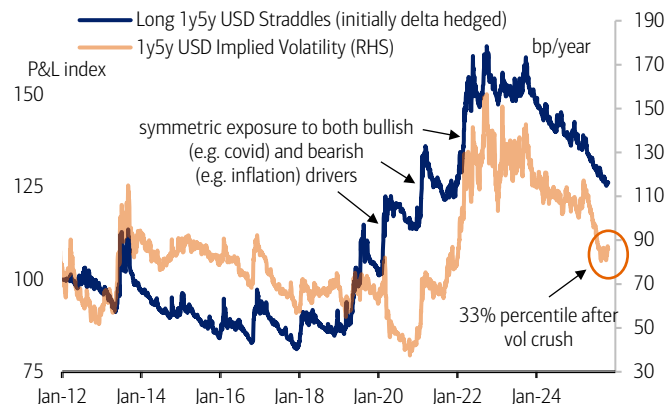


Source: BofA Global Research. Data: 3-Jan-20 to 21-Nov-25. *See our latest primer: [BofA's Global Financial Stress Indicator](#). Non rates = stress in x-asset vol & CDS spreads ex rates vol.

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Exhibit 119: While long 1y5y USD swaption straddles offer positive convexity in bullish or bearish rate shocks...

Hypothetical back-test of long 1y5y ATM USD swaption straddles, rebalanced weekly, held for 6m and rolled, vol-targeted to 10%/year



Source: BofA Global Research, Bloomberg. Data from 1-Jan-12 to 21-Nov-25. This performance is back-tested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 118: 1y5y USD rates vol is a sweet spot in the rates vol grid with low vol vs history, flat vega carry & a negative vol risk premium*

Normal implied vol percentile (%) / Average vega carry (% of vol / yr) / Average Vol Risk Premium* (bp/year), all since May 2020

tail	2y	[5y]	10y	30y
expiry	Positive VRP and/or Negative vega Carry			
3m	33 / -19 / -1	34 / -4 / 0	33 / -2 / 2	21 / -1 / 6
6m	33 / -18 / -9	33 / -1 / -4	33 / 2 / 0	27 / 4 / 3
[1y]	32 / -12 / -10	33 / 0 / -6	33 / 2 / -2	32 / 4 / 1
5y	33 / 1 / -9	34 / 2 / -10	36 / 2 / -8	54 / 3 / -9
10y	47 / 2 / -9	57 / 2 / -10	65 / 2 / -12	71 / 2 / -15
20y	86 / 1 / -13	86 / 1 / -15	87 / 1 / -17	86 / 1 / -16
	Rich Implied Volatility			

Source: BofA Global Research, Bloomberg. Data from May-20 to 21-Nov-25. * The Vol Risk Premium (VRP) is defined across tails as the ATM Implied vol minus subsequent delivered vol over a holding period of the minimum between either option expiry or 6 months. Positive values mean implied vol was on average higher than delivered vol, and vice versa negative values mean average implied vol was lower than realised vol over the period.

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Exhibit 120:... our rates Grind Higher strategy can benefit from bearish moves in yields that don't necessarily lead to higher implied vol

Hypothetical back-test of our 'Grind Higher' TY strategy* and breakdown between its long-and short-vol components, vol-targeted to 10%/year



Source: BofA Global Research, Bloomberg. Data from 1-Jan-12 to 10-Nov-25. This performance is back-tested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein. *See [Hedging USTs on the long & grinding road](#)

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However, there is a (for some, a more likely) scenario where persistent inflation doesn't allow the Fed to cut as much as markets anticipate, weighing on duration but without causing a spike in rates vol. This is where a [Grind Higher Strategy](#) in rates can be useful.

Long rates vol: The case for long 1y5y USD straddles... In our view, volatility in the belly (5y point) of the curve will be impacted in the majority of bearish duration scenarios. Indeed, the 1y5y point on the vol surface is a sweet spot for long vol strategies with desirably flat vega carry, low vol vs history and a negative vol risk premium (Exhibit 118). Our hypothetical backtest (see Exhibit 119) confirms value in owning long 1y5y ATM straddles systematically, with convex gains in both bullish or bearish rate shocks.

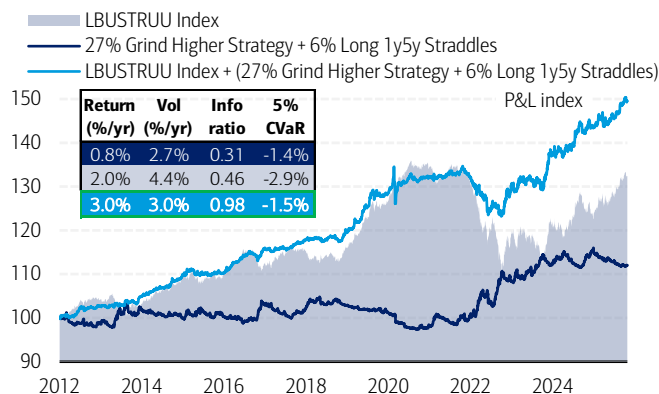
And if rates grind higher without a rise in rates vol... Our (rates) grind higher strategy buys (unhedged) 1m 45d puts on TY futures and sells (premium-sized) 3x 20d puts (daily delta-hedged). Our hypothetical backtest in Exhibit 120 demonstrates that the strategy would have performed when yields moved higher while implied volatility was largely rangebound (in H2-22 & 2023). Moreover, resilience in times of declining implied vol (2025) is a feature that helps diversify outright long rates vol strategies. Indeed, aggregating long 1y5y USD straddle and grind higher strategies into a defensive overlay for a bond benchmark would have increased returns by 100bp and doubled risk-adjusted returns (Exhibit 121).

Doubling down on defence: Hedging 60-40 with a defensive equity-rates combo

We now blend our equity and rates hedges (Exhibit 116 & Exhibit 121) into a diversified overlay applied to a US 60-40 portfolio (BMA6040). Our hypothetical back-test (Exhibit 122) suggests this would have boosted return by 2.3%p.a. with a nearly 40% jump in Sharpe, and a ~25% reduction in tail risk. When viewed through the lens of our Hedging framework (link here: [note](#)), this equity-rates defensive aggregation would have offered a more balanced hedging profile vs its constituents vis-à-vis the metrics in Exhibit 123.

Exhibit 121: Long rates vol and 'grind higher' would have helped increase bond returns by 100bp p.a. and doubled risk-adjusted returns

Hypothetical back-test of a defensive Rates strategy overlaid on LBUSTRUU

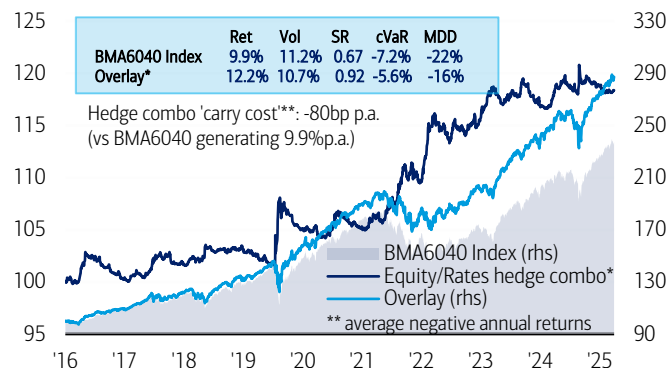


Source: BofA Global Research, Bloomberg. Data from 1-Jan-12 to 10-Nov-25. LBUSTRUU Index is the FI component of BMA6040 Index. *Notional obtained for min 5% cVaR and 3% max tracking error. This performance is back-tested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 122: An overlay of defensive equity & rates alternatives would have boosted 60/40 returns (+23%), Sharpe (+37%) & reduced tail risk

Hypothetical backtest of the equity and rates overlays (Exhibit 116 and Exhibit 121) and their impact to protect a 60-40 benchmark (BMA6040 Index)

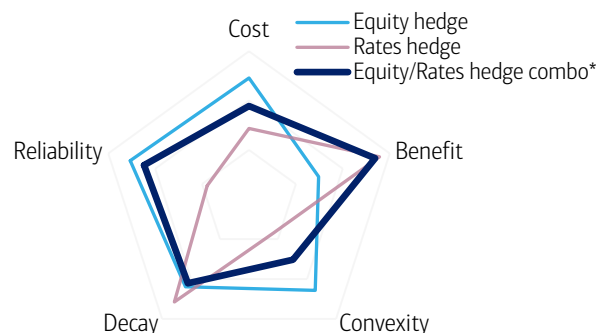


Source: BofA Global Research. Data from 15-Aug-16 to 10-Nov-25. *Notional obtained for min 5% cVaR and 3% max tracking error. Weights: Equity hedge (77%), Rates hedge (63%) for a combo that has a 10% vol target. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 123: This equity rates hedge combo would have offered a more balanced hedging profile than its individual constituents

Hypothetical backtest of our overlays (Exhibit 116, Exhibit 121, Exhibit 122) evaluated against hedging framework**. We use BMA6040 as the benchmark



Source: BofA Global Research. Data from 15-Aug-16 to 10-Nov-25. *See Exhibit 122. **For explanation of these hedging metrics, read here: [The Leading Edge of your Hedge](#). This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Hedging long-end pain with low rates vol & 10x hybrids

What: Buy (i) TLT (20+Yr Treasury Bond ETF) Mar26 86 puts for \$1.03 (~25D strike, ref. 90.01), or (ii) 30y SOFR > ATMf + 20bps / XAU > 106% Mar26 dual digitals for 7.95% (**12.5x max payout ratio**, -18 correl offer, ref. 3.955%, 4078.70).

Why: In our view, the biggest downside risk for markets in 2026 is tightening financial conditions on the back of either inflation, or fiscal deficit concerns, both of which would be marked by long-end yields rising.

Indeed, we have seen this dynamic repeatedly since 2023 (Exhibit 124), with either inflation concerns (e.g., 3Q-23 and 1Q-24) or fiscal concerns (e.g., Q4-24) pushing long-end rates higher. Additionally, a stronger-than-expected economy in the near-term could also push yields up, as markets are forced to unwind expectations of easy policy that were predicated on recent labour market cracks.

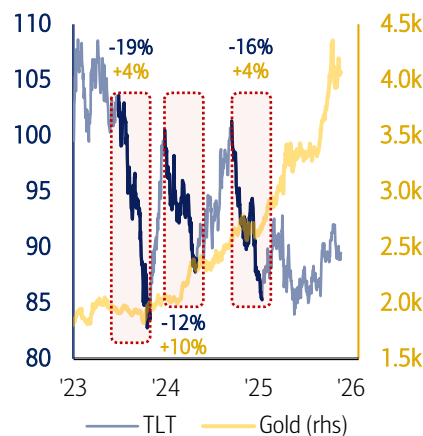
With TLT implied volatility currently near post-Covid lows and the put wing pricing in little premium vs recent history (Exhibit 125), we like tactically hedging the risk of another long-end rates selloff using TLT puts.

Alternatively, one could leverage the historical strength of gold in response to inflation and fiscal fears by hedging with 30y rates up, gold up dual digitals. The realized correlation between 30y rates and gold has been rising in 2025 while the implied correlation remains more negative (Exhibit 126), creating an attractive entry point for long correlation hybrids.

Risks: Losses on both trades would involve upfront premium paid.

Exhibit 124: The past two years highlight TLT's vulnerability (and gold's strength) amid a resurgence of inflation & fiscal concerns

TLT (20+Yr Treasury Bond ETF) & gold price since 2023

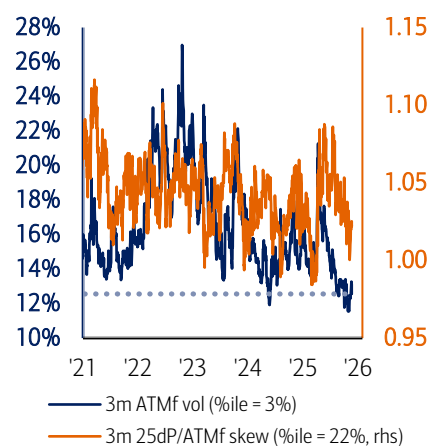


Source: BofA Global Research, Bloomberg. Data from 1-Jan-23 to 21-Nov-25.

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Exhibit 125: TLT vol, especially the put wing, is near post-Covid lows, creating an attractive entry point to hedge higher long-end yields with TLT puts

TLT 3m ATMf vol & 25dP/ATMf skew

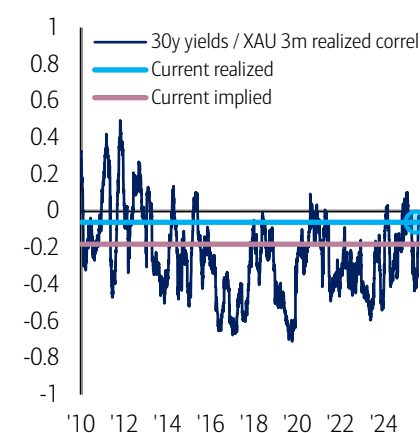


Source: BofA Global Research. Data from 1-Jan-21 to 21-Nov-25.

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Exhibit 126: 30y yields and gold have seen their realized correlation rise, creating an attractive entry point for hybrids given the more negative implied correlation

30y SOFR vs gold 3m correlation



Source: BofA Global Research. Data from 1-Jan-10 to 21-Nov-25. Current implied correl is indicative only.

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Long vol, light on carry: SX5E fwd var leads globally

What: Long SX5E 1.7x Jun27 var (23.7v) & short 1x Jun26 var (22.8v), sized for best carry-conscious risk/reward. *Risk: Jun27 var in Jun26 fails to rise relative to an increase in realised vol.*

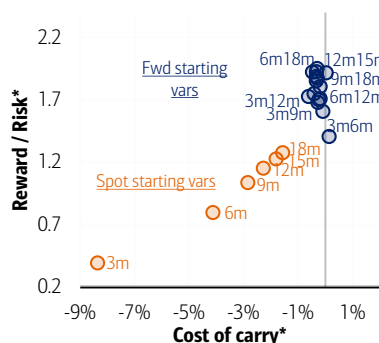
Why: Long fwd var is most attractive for SX5E vs global peers. Resize legs to cut carry cost by 4v (45% better vs regular sizing) while preserving risk/reward.

The case for SX5E fwd var: We expect growth/tariff risks, geopolitical uncertainty, potential hedging flows from EU pension reform and risk aversion ahead of US mid-term elections to keep EU equity vol supported in 2026. Long spot-starting var is unattractive, especially on SX5E given rich VRP (Exhibit 127). In contrast, long SX5E forward-starting var is compelling: it has no RV exposure, has superior carry and a better risk/reward profile vs spot-starting var (Exhibit 129). Moreover, SX5E fwd var is most attractive vs global peers on risk/reward (Exhibit 128). We favor owning SX5E 6m18m (~Jun26/Jun27 equiv.) to avoid shorting the Dec26 tenor that may be impacted by US mid-terms.

Cheapen via resizing: Rich var convexity and elevated tail-liquidity risk premium make cheapening fwd var key (6m18m fwd var ~48th 5y %-ile). A simple way (vs var replication) to cut carry cost while preserving convexity is to buy less Jun27 var than implied by a Jun26/Jun27 fwd var swap. This improves carry at the expense of introducing residual short RV exposure. The trade still retains a hedge bias, preserving the reward/risk profile while cutting carry costs by 4v (~45% better vs regular sizing; Exhibit 130-Exhibit 131).

Exhibit 129: Long fwd var offers superior carry and better reward/risk than spot var

Reward / risk ratio vs cost of carry for SX5E spot-starting and fwd-starting vars, for 3m holding

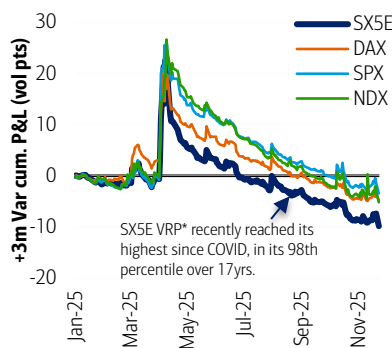


Source: BofA Global Research. Data from 24-Nov-20 to 24-Nov-25. *Please see Exhibit 128 for reward/risk ratio; cost of carry is computed similarly (e.g. 6m18m vs 3m15m current fwd var levels). For spot-starting vars, we use trailing 3m RV.

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Exhibit 127: Outright long var positions would have posted losses YTD given rich VRP

Cum P&L (in vol pts) for +3m Varswaps across major global indices (1 unit vega notional)

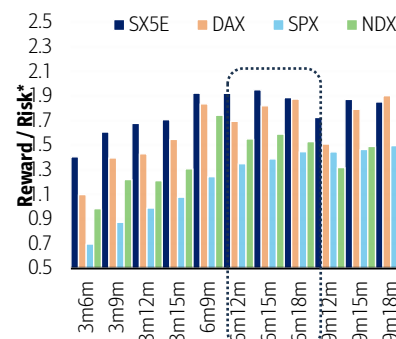


Source: BofA Global Research. Data from 02-Jan-25 to 24-Nov-25. We enter monthly with a notional of 1/3. *6m Variance Risk Premium (VRP) as: $(\text{Var } K^2 - \text{subsequent RV}^2) / (2 * \text{Var } K)$. †: See backtesting disclaimer.

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Exhibit 128: SX5E fwd var is most attractive vs global peers on a reward/risk basis

Reward / risk ratio for fwd vars across combinations of tenors and major global indices



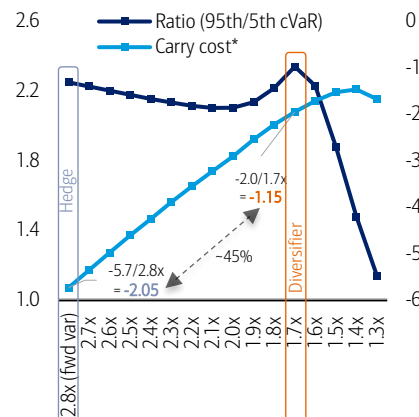
Source: BofA Global Research. Data from 24-Nov-20 to 24-Nov-25. *Reward/risk ratio assumes trade held for 3m. 6m18m reward = $(95^{\text{th}} \text{ %-ile of } 3\text{m}15\text{m}^2 - \text{current } 6\text{m}18\text{m}^2) / (2 * \text{current } 6\text{m}18\text{m})$. Similarly for risk with 5th %-ile (5y history).

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†: This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein

Exhibit 130: Buying <2.8x Jun27 var cuts carry cost without compromising reward/risk

MtM at expiry for (-1x) 141d var vs (+Nx) 395d var (equivalent to Jun26/Jun27 fwd var)

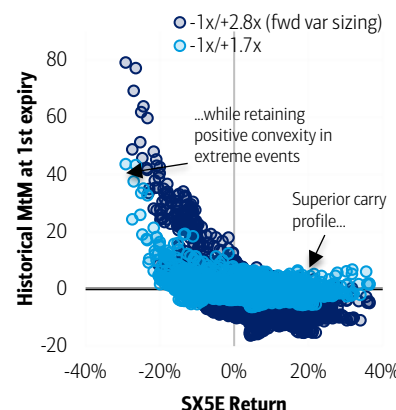


Source: BofA Global Research. Data from 27-Nov-15 to 24-Nov-25. *Avg P&L when negative. †: See backtesting disclaimer.

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Exhibit 131: The trade's carry profile is improved, yet its defensive tilt is retained

MtM at expiry with different sizing for short 141d var vs long 395d var (equiv. Jun26/Jun27)



Source: BofA Global Research. Data from 27-Nov-15 to 24-Nov-25. †: See backtesting disclaimer.

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Once bitten twice shy: Hedging FX for non-USD investors

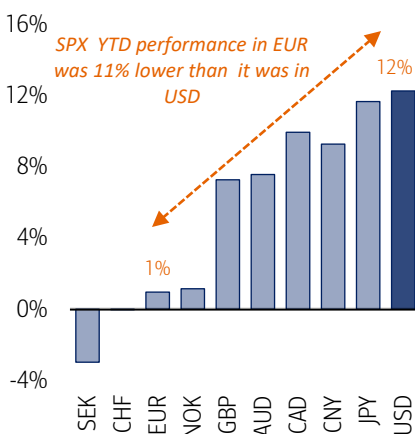
What: Long 110% Dec26 SPX calls qEUR for 4.33% (17 bps over vanilla); or long 90% Dec26 SPX puts qAUD for 3.66% (16bps less than vanilla)

Why: Unhedged FX risk in stocks exposes foreign investors to USD swings. SPX qEUR calls hedge this for just 17bps more (YTD this would've saved 1100 bps)

2025 underscored the FX translation risk in foreign allocations to US assets. While the SPX rose 12% in USD terms, foreign investors saw lower returns in local FX (Exhibit 132).

FX hedge ratios are low (one estimate is 24% for non-USD investors¹), leaving portfolios exposed to FX volatility even as structural allocations to US assets rise (Exhibit 133). Foreign equity holders in the UK, Canada, Japan and Europe have large US stakes (Exhibit 134), making FX risk a key concern. With AI optimism likely to sustain US equity demand into 2026, overlaying FX hedges is compelling. Quanto* calls offer clean SPX upside in local currency for only a modest premium over vanilla calls, enabling participation in the AI theme without FX drag. With vol subdued in both FX and equities, quanto forward adjustments favor SPX qAUD puts given strong negative equity-FX correlation (Exhibit 135), while SPX qEUR calls provide FX-safe upside (Exhibit 136).

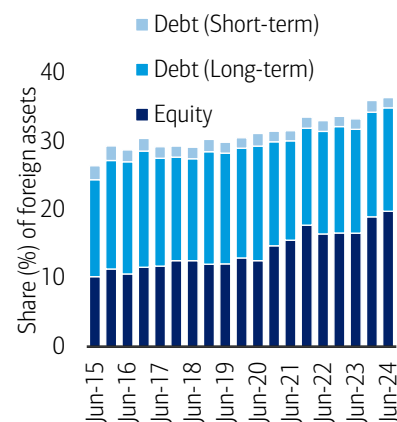
Exhibit 132: US equities are up ~12% YTD, but not for FX-unhedged foreign investors
YTD SPX Returns across Currencies



Source: BofA Global Research, Bloomberg. Data from 01-Jan-25 to 21-Nov-25.

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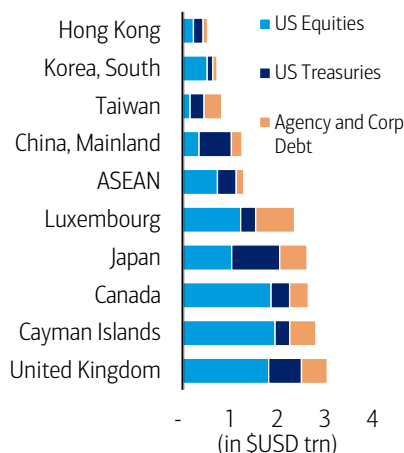
Exhibit 133: US equities' share in foreign portfolios has nearly doubled in the last 10y
Share of foreign portfolios held in US securities



Source: Data from BIS Bulletin: US dollar's slide in April 2025: the role of FX hedging. Chart reproduced by BofA Global Research. Data from June-2015 to June-2024

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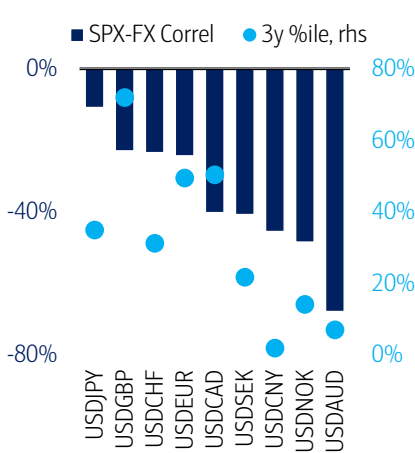
Exhibit 134: Most foreign equity holders of US equities are in UK, Canada, Europe, Japan
Foreign holders of US securities link



Source: Data from BIS Bulletin: US dollar's slide in April 2025: the role of FX hedging. Chart reproduced by BofA Global Research. Data from March 2025

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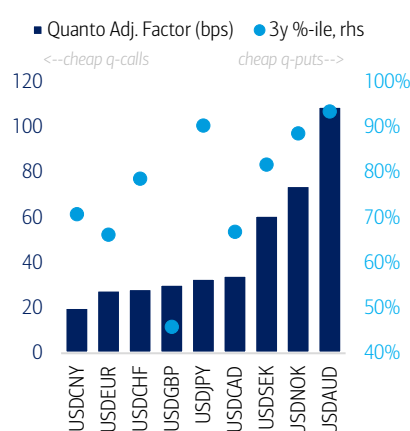
Exhibit 135: SPX has decorrelated with USD vs all FX pairs, most so vs AUD
3m correlation of SPX and FX, and its 3y %ile



Source: BofA Global Research, Bloomberg. Data from 18-Nov-22 to 21-Nov-25. Quanto Forward Adjustment = $\text{Correlation}(\text{Eq-FX}) * \text{Vol}(\text{Eq}) * \text{Vol}(\text{FX})$, where FX has USD as base currency, e.g. USDEUR. 3-month daily return realised correl

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Exhibit 136: SPX qAUD puts are cheap, while qEUR calls aren't very expensive
SPX-FX Quanto Forward adjustments



Source: BofA Global Research, Bloomberg. Data from 21-Nov-22 to 21-Nov-25. Correlations computed over 3-months of 3-day returns

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Cheaply gear for EZ Banks dispersion after blistering rally

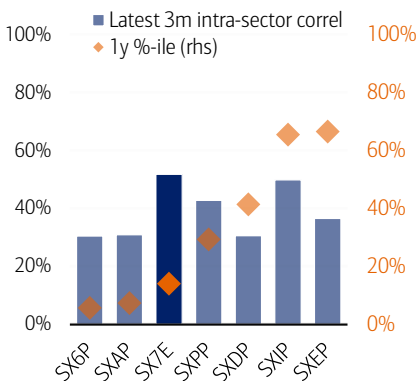
What: (i) Long Mar26 straddle on liquid EZ banks* vs short SX7E straddle for 2.36%; (ii) Long Dec-26 ATMf call on return dispersion on bank names for 3.50% and a strike of 17.3% (SX7E spot ref: 234.93). **Risk:** (i) short straddle losses occur in strongly correlated moves; (ii) limited to premium paid.

Why: EU banks' thematic tilts support ongoing dispersion. Call on dispersion trade would have gained 12% recently, offering 2.2x reward/risk historically.

Despite a >60% sector (SX7E) rally, French fiscal stress, Ukraine reconstruction, German stimulus and US bank deregulation themes have driven significant performance divergence within European banks (Exhibit 65). Indeed, SX7E is among the sector indices with the lowest intra-sector correlation historically (Exhibit 137). Dispersion of YTD returns is also near 5y highs (Exhibit 138). We like positioning for continued dispersion in the sector given the continued relevance of these themes. Our basket includes 12 stocks* (Exhibit 140; see our [04-Nov GEVI](#) for a mapping of Banks to themes). Notably, the basket vs SX7E vol spread has carried well recently, underscoring the appeal of owning sector dispersion via optionality (Exhibit 139).

Exhibit 137: Correlation among EZ Banks is among the lowest (historically) in Europe...

Latest 3M (realized) intra-sector correlation and its 1y %-ile rank for major European sectors

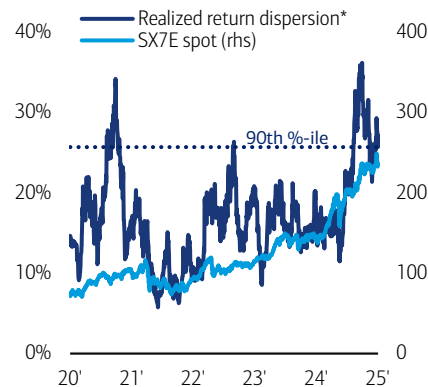


Source: BofA Global Research. Data from 20-Nov-24 to 20-Nov-25. We use current membership and weights. Intra-sector correlation is computed using 3M stock and index RV.

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Exhibit 138:... while return dispersion between liquid SX7E stocks is near 5y highs

*Avg absolute deviation of 11m (~YTD) returns for SX7E Liquid names** vs Index spot



Source: BofA Global Research, Bloomberg. Data 20-Nov-20 to 21-Nov-25. **We select SX7E's names that are within the top 100 SXXP liquid names (see Exhibit 140).

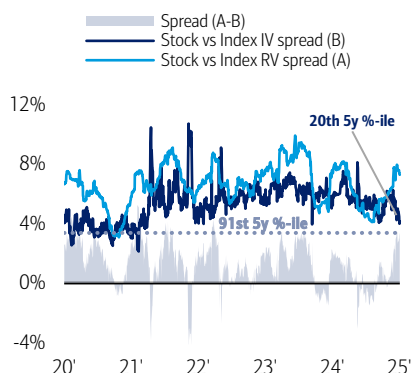
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†: This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein

Dispersion attractive, vanilla or OTC: We favor trading banks' return dispersion via vanilla straddles or a call on return dispersion (for limiting risk). At current pricing, long 3M straddles on liquid SX7E names vs short an SX7E straddle would have gained 1.1% recently, being profitable on average historically. Similarly, a call on return dispersion would have gained as much as 12.3% recently with 2.2x historical risk-reward (last 3y).

Exhibit 139: Long liquid EZ Bank stocks vol vs SX7E would have carried well recently

Wgt* stock vs index vol spread (using 3M atm IV, 3M RV spread and RV minus IV spreads)

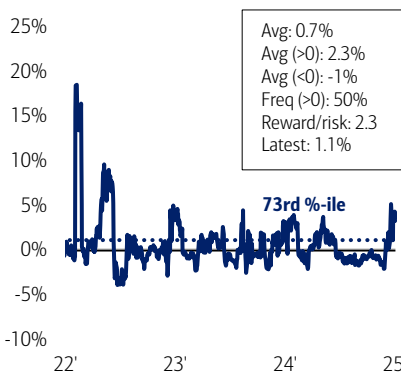


Source: BofA Global Research, Bloomberg. Data 20-Nov-20 to 21-Nov-25. *We use historical SX7E index weights for the names in Exhibit 140.

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Exhibit 140: Straddle dispersion would have being profitable on average historically

Back-cast (†) at current pricing (2.36%) for 3M ATMf wgt* stock vs SX7E straddle (unhedged)

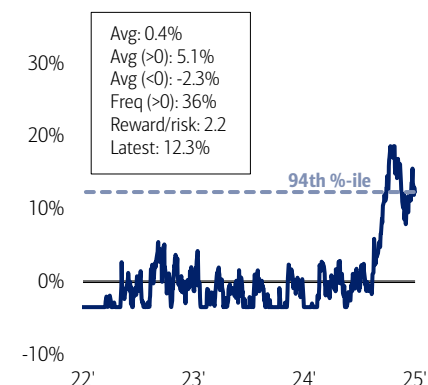


Source: BofA Global Research, Bloomberg. Data: 20-Nov-22 to 21-Nov-25. †: See backtesting disclaimer. *Names /weights (25-Nov-25) : BNP FP, ABN NA, GLE FP, BBVA SQ, INGA NA, UCG IM, BAM IM, ACA FP, SAN SQ, ISP IM, CBK GY, DBK GY.

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Exhibit 141: Call on return dispersion on the selected names would've paid 12.3% recently

Back-cast (†) at current pricing (3.5%) of Dec-26 ATMf call on return dispersion for liquid names*



Source: BofA Global Research, Bloomberg. Data from 21-Nov-22 to 21-Nov-25. †: See backtesting disclaimer. *Equally-weighted basket (see Exhibit 140 for names).

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Nikkei up on a stronger yen for an 80% discount

Higher JP rates & lower US rates – a setup to exploit correlation

What: Buy Jun26 NKY>104% & USDJPY<96% dual digital for 9.7% (+48 correl offer, ref: 48,626, 157.00) 10.3x max payout (Exhibit 145)

Why: Moderate Nikkei upside trades can be cheapened by 80% if willing to lean optimistic on the yen. Narrowing US-JP rate differentials could weigh on USD/JPY in '26 but doesn't have to be Nikkei negative

Taking advantage of conventionally priced correlation:

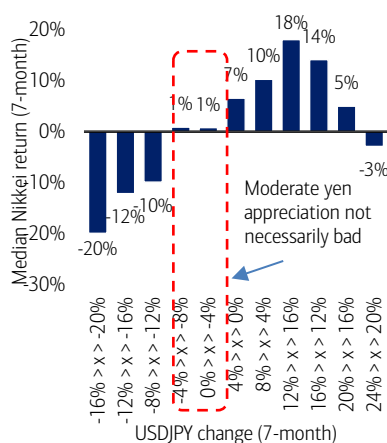
The Nikkei tends to thrive on a weaker yen and vice versa. However, the more exciting gains can be made when identifying the right periods to take the other side of that correlation. While our base case is a weaker yen in the near term – especially if the BoJ doesn't hike in Dec – the 80% discount one gets by betting on yen returning to last month's levels looks attractive.

Why in 2026? What if interest rates start to matter again?

As US rates are falling, Japanese rates keep climbing. Rate differentials have not been key USD/JPY drivers of late, but one has to wonder for how long this increasing disconnect can last (Exhibit 143). A moderate yen rally back to Oct levels may not pull Nikkei lower (Exhibit 142).

Exhibit 142: Moderate yen appreciation isn't necessarily Nikkei negative

Typical Nikkei returns vs USD/JPY changes

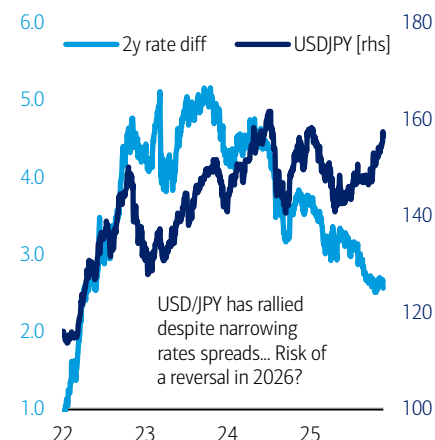


Source: BofA Global Research and Bloomberg. Data from 2-Jan-15 to 20-Nov-25. Time from now to June expiry is 7 months

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Exhibit 143: What if US vs Japan rate differentials start to matter in 2026

US & Japan 2yr gov bond yield spread & USD/JPY



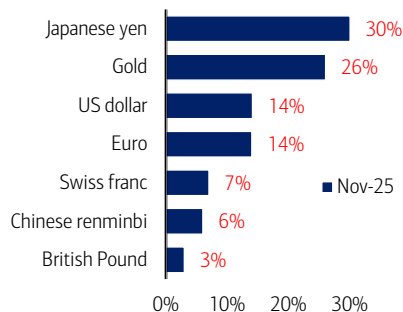
Source: BofA Global Research and Bloomberg. Data from 2-Jan-22 to 21-Nov-25.

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Talks of yen intervention may heat up near 160: USD/JP may rise towards 160 in case the BoJ doesn't hike in Dec. At which point, the talk of intervention will likely heat up. While BofA leans bearish the yen, the currency is deeply undervalued on a PPP basis (by 40-50%) and many investors believe the yen might outperform next year (Exhibit 144).

Exhibit 144: BofA's Fund Manager Survey: expect the yen to outperform in '26

Which of the following currencies do you expect to outperform in 2026?

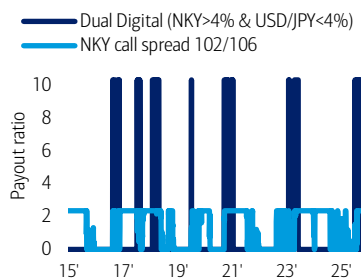


Source: BofA Global Fund Manager Survey, 18-Nov-2025.

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Exhibit 145: This dual digital is best used tactically – as stocks rally on a weaker USD – and is not meant to work systematically

Historical payout ratios if using current cost

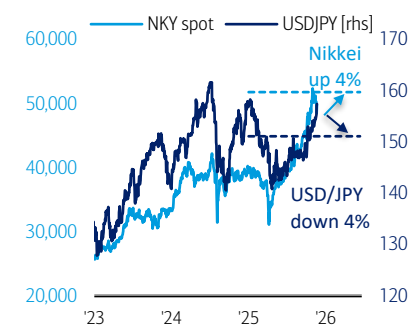


Source: BofA Global Research. Data from 22-Jul-22 to 19-Nov-25. The 102/106 call spread is used as a reference point for a digital 104% Nikkei call (104 is the mid-point between strikes)

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Exhibit 146: A 4% Nikkei rally is priced as if it is 80% less likely if the yen rallies 4%

Nikkei and USD/JPY spot and the barrier levels



Source: BofA Global Research and Bloomberg. Data from 2-Jan-23 to 21-Nov-25.

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The China repo arb – CSI 1000 calls vs ASHR calls

1.0 information ratio and small up when small caps underperformed

What: Buy 1x 20-Mar-2026 106% call on a CSI 1000 tracking basket* for 2.32% (+27d, 26v). Sell 1.05x ASHR US 20-Mar-2026 34 calls for \$0.55 (~1.73% of spot; the 34.00 strike is ~107% of spot, ref 31.76)

Why: The long/short trade extracts value from the rich repo for Chinese small caps while limiting directional risk. Trading calls vs calls manages the risk of small caps underperforming (Exhibit 148). Beta-neutral sizing yielded 8% p.a. on gross notional at 8% vol (Exhibit 147, Exhibit 149)

Target cheapest forward vs near

richest: The trade benefits from being long the most persistently discounted forward (CSI 1000) while selling one of 2nd richest forwards (ASHR) among popular China indices (Exhibit 151).

Calls instead of delta-1: While both are on China A-shares, ASHR contains large caps and CSI 1000 small caps, making delta-1 implementations exposed to broad selloffs. As illustrated in Exhibit 148, this risk is limited if doing calls vs calls instead (both legs expire worthless in broad market declines).

China A50 instead of ASHR? Investors wanting to avoid mark-to-market risks arising due to asynchronous opening hours and holiday calendars, can consider selling China A50 (XIN9I) calls instead. While the forwards are not as elevated as on ASHR (USA listed CSI 300 ETF) (Exhibit 151), risk and return characteristics are still attractive (Exhibit 150).

*= Long 1x CSI 1m 103% call vs. short 1x ASHR 1M 103% call. **= Long 1x CSI 103% call vs. short 1x ASHR 103/110% call spread. ***= Long 1x CSI 103% call vs. short ASHR 103% call, sized to be beta neutral. ASHR carries currency risk (CSI 300 × FX), while CSI 1000 options are typically quanto USD. The China A50 backtest in Exhibit 150 was done on the 2823 ETF due to availability of listed option prices but calls can be sold directly on XIN9I (OTC). ASHR is the CSI 300 ETF listed in the US. For details on various China indices, see [14-Oct GEVI](#).

Exhibit 149: A beta-neutral CSI 1000 call vs ASHR call strategy would have generated a 1.0 info ratio and 0.86 Calmar ratio

Hypothetical backtested performance

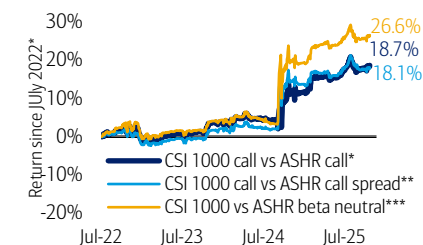
	Ann. Return	Ann. Vol	Info ratio	Calmar ratio
Call vs Call*	5.7%	6.2%	-5.8%	0.93
Call vs Call beta neutral***	8.0%	8.0%	-9.3%	1.00
Call vs Call spread**	5.6%	7.0%	-8.3%	0.79

Source: BofA Data from 22-Jul-2022 to 13-Nov-25. †: See backtesting disclaimer.

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Exhibit 147: CSI 1000 1M 103% calls vs. ASHR 1M 103% (beta-neutral sizing) calls delivered 8.0% return per annum at 8.0% vol

Hypothetical backtest of CSI 1000 vs ASHR US

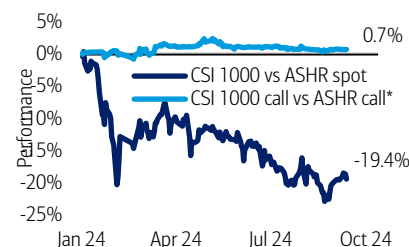


Source: BofA Global Research and Bloomberg. Data from 22-Jul-2022 to 13-Nov-25. †: See backtesting disclaimer. Backtest only includes days when both markets were open.

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Exhibit 148: Calls vs calls limits the risk of small caps underperforming during selloffs

Performance during the deep 2024 selloff



Source: BofA Global Research, Bloomberg. Data from 2-Jan-24 to 13-Sep-24. †: See backtesting disclaimer.

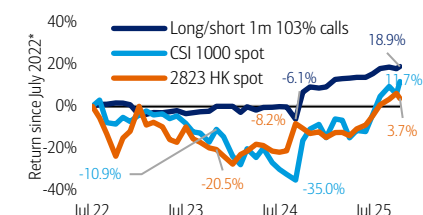
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†: This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein. *= several banks offer CSI tracking baskets; indic. in quanto USD).

Risks: That the gain on the CSI 1000 calls are not sufficiently large to compensate for hypothetical losses on the short ASHR leg

Exhibit 150: Mitigate mark-to-market risks by selling calls on the FTSE China A50 index instead (XIN9I or the HK listed ETF 2823 HK)

Long/short CSI 1000/2823 HK 1m 103% calls

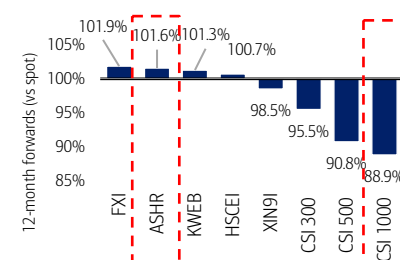


Source: BofA Global Research and Bloomberg. Data from 22-Jul-2022 to 13-Nov-25. Note that 2823 has currency risk as the underlying index is the FTSE China A50 * FX while CSI 1k options are typically quanto USD. †: See backtesting disclaimer.

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Exhibit 151: CSI forwards are the most compressed, particularly CSI 1000, while ASHR ranks among the most elevated

Estimated mid forward pricing of popular China indices



Source: BofA Global Research and Bloomberg. Data as of 20-Nov-2025. All except for the CSI indices are theoretical mid

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More ideas on our global radar

Lock in long-dated AI optionality today with convertibles

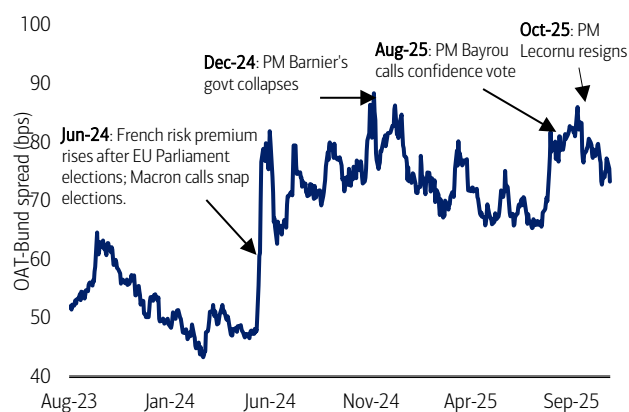
Global convertible bonds (corporate bond + equity warrant) have been a beneficiary of the vol up/spot up dynamic as issuers, many of which are exposed to the AI theme, have been able to bring new deals to market on favorable terms. In fact, the market has priced over \$150bn globally year-to-date, a record pace, 20% of which is from AI-linked data centers, energy providers, and semiconductors. The surge in volumes allows investors new opportunities to get long-dated optionality on AI—a key benefit given it may take years before we know who are the winners and losers. For outright investors, we like long positioning in “balanced” mid-delta names, which allow for asymmetry versus common shares (in fact, AI names that meet this criteria have delivered strong convexity year-to-date, capturing 1.3x of their underlying stocks’ upside relative to downside). For hedged investors, synthetic put trades (long deep ITM CB, short common stock) allow managers to harvest gamma on equity moves while maintaining structural protection via the CB’s bond component. In our [Global Convertibles 2026 Year Ahead](#), we offer single-name ideas that show investors how they could potentially benefit from each of these trades.

Cheapen German upside by >60% by fading CAC strength

French political risk continues to simmer as budget negotiations drag on into year-end and beyond. Despite the enormity of fiscal challenges that lie ahead for France (as reflected in the still-elevated OAT-Bund spread, Exhibit 152), the CAC and more domestically-focused French stocks have recently outperformed the DAX (Exhibit 153). We like risk-limited trades that fade recent strength in CAC given the French fiscal overhang to fund German equity upside in case optimism returns for German growth.

Indeed, Jun26 DAX 105% calls contingent on the CAC < 105% at expiry cost 1.24%, indicatively, offering a 63% discount vs DAX calls outright.

Exhibit 152: Persistent political uncertainty and fiscal stress have weighed on the OAT-Bund spread, a key risk barometer for France...
(10yr) OAT-Bund spread over the past 2 years.

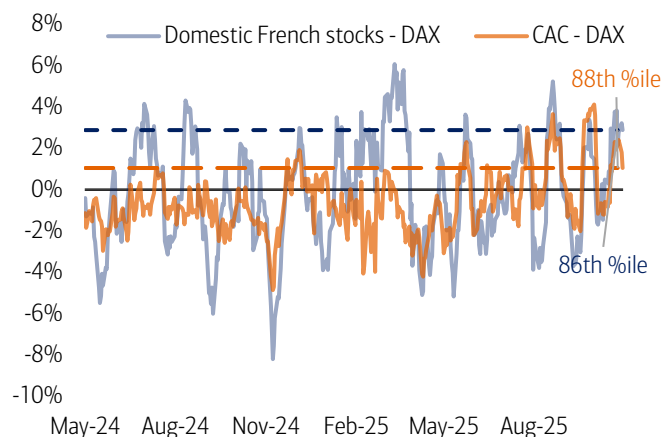


Source: BofA Global Research, Bloomberg. Data: 28-Aug-23 to 25-Nov-25.

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Exhibit 153: ... which is inconsistent with continued French vs German stock outperformance, if history vs OAT-Bund spreads is any guide.

Regression residual of (i) MLD1FRAN (Domestic French stocks) vs DAX, & (ii) CAC vs DAX 10d returns vs 10d OAT-Bund spread changes, since 2024.



Source: BofA Global Research, Bloomberg. Data from 26-Apr-24 to 25-Nov-25. We show 2-week excess returns vs 18m regression betas.

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Appendix

Introducing the BofA Bubble Risk Indicator

Since ChatGPT's release, the AI trade has quickly become central to global markets and driven tremendous growth in equities, particularly in the US. Similar technological developments in the past like railways, radios, and the internet ultimately drove the formation of large asset bubbles, prompting fears of a similar outcome for AI. In our view, it is increasingly likely for the AI trade to eventually result in an asset bubble, which we first outlined in 2024 (see 2026 Outlook & [Volatility suggests AI isn't a bubble...yet](#)).

Monitoring bubble progression can be challenging. However, beyond traditional lenses of valuation and positioning, the hallmark historically of bubbles has been volatility rising with prices. The **BofA Bubble Risk Indicator ("BRI")** aims to expand on this observation and apply a broader distributional approach to asset price movements to detect potentially frothy price action today and the potential for asymmetric risks ahead.

The BRI is intended to be an indicative metric only and may not be used for reference purposes or as a measure of performance for any financial instrument or contract, or otherwise relied upon by third parties for any other purpose, without the prior written consent of BofA Global Research. The BRI was not created to act as a benchmark.

Detecting frothy behavior in financial assets with a distributional lens

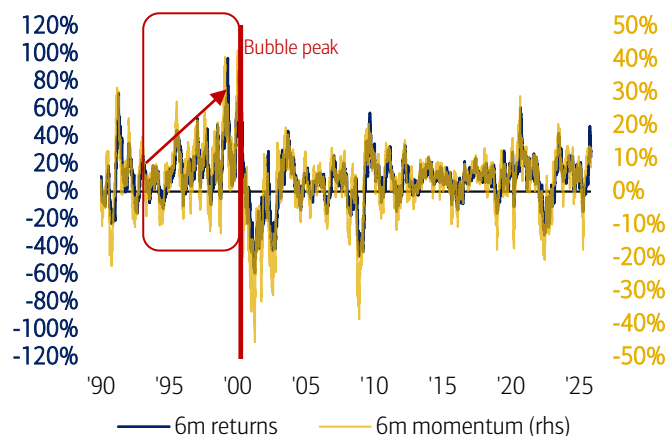
The BRI aims to quantify the degree of bubble-like behaviour exhibited by a financial asset's price action. It uses inspiration from the way the four statistical moments describe a distribution, and centres around the moments of an asset's price action: (i) returns, (ii) realized vol, (iii) price momentum⁷ (proxy for skew), & (iv) fragility/convexity⁸ (proxy for kurtosis). We first introduced this framework in our [16-Sep-25 GEVI: Market's moments still say no bubble yet...](#) Why centre around such a framework?

Inferring positioning and sentiment from price action

Asset bubbles historically have typically exhibited elevated levels of these four moments at their frothiest points, with high returns, high volatility, extreme upside momentum, and a high degree of fragility. For example, the dotcom bubble in the late 90s exhibited historically elevated levels of these four metrics in the Nasdaq (Exhibit 154, Exhibit 155).

Exhibit 154: Both price returns and price momentum of the Nasdaq rose as the dotcom bubble inflated

NDX 6m price returns and 6m price momentum⁷

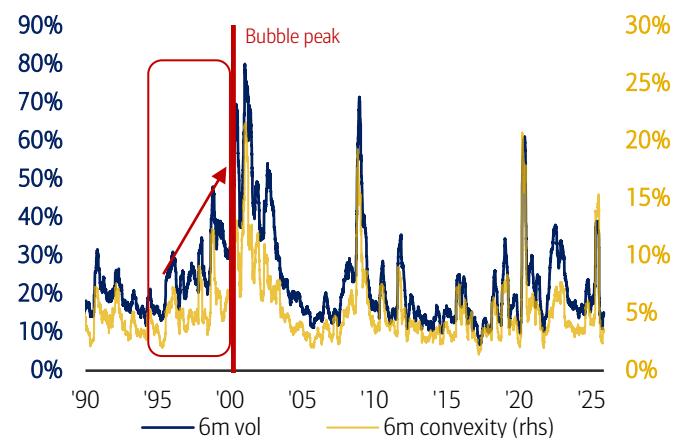


Source: BofA Global Research. Data from 1-Jan-90 to 7-Nov-25.

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Exhibit 155: Nasdaq's vol and fragility also rose into the peak of the dotcom bubble as tech stock price rose

NDX 6m realized vol and 6m realized convexity⁸



Source: BofA Global Research. Data from 1-Jan-90 to 7-Nov-25.

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⁷ X-month price momentum = % distance of the current price from the X-month moving average price.

⁸ X-month fragility/convexity = X-month realized vol – X-month realized mean absolute deviation.



The intuition behind why this might be the case lies in the potential follow-through of positioning and investor sentiment into price action. The fear of the upside and exuberant sentiment typically present in bubbles often results in one-sided positioning and imbalanced demand vs supply dynamics, all of which can translate into price action with high volatility, upside momentum and fragility, particularly at peak exuberance.

Parsimonious & purely price-based measure

The BRI provides a parsimonious estimate of potential froth by conveying how elevated the four moments of price action are vs the asset's historical tendencies. Historically elevated levels of the moments, and by construction the indicator, suggests more violent, bubble-like behaviour and potentially increased risks in the near term.

One of the primary reasons we find the BRI a useful tool to include along with more traditional bubble measures including valuation is its reliance purely on daily price data (no additional fundamental & positioning data). Indeed, it aims to *infer* the extreme one-sided positioning and exuberant sentiment that develops as asset bubbles form and pop by detecting elevated levels in the four moments of the price action's distribution. As a result, the BRI can be applied to assets without explicit fundamental valuation methodologies (e.g., commodities) or those with opaque positioning visibility.

Note that the indicator's exclusive use of price data means that it would primarily provide utility under the assumption that the asset in question might be in (or turn into) a bubble.

Construction of the indicator

The BRI is constructed as an average of 3 sub-indicators, which each range between 0 and 1: a short-term, medium-term, and long-term sub-indicator. The aggregation of these different indicators helps to capture the asset's dynamics on numerous time horizons, giving a more complete assessment of the price action's bubble-like dynamics. Higher values of the BRI would suggest more bubble-like price behaviour.

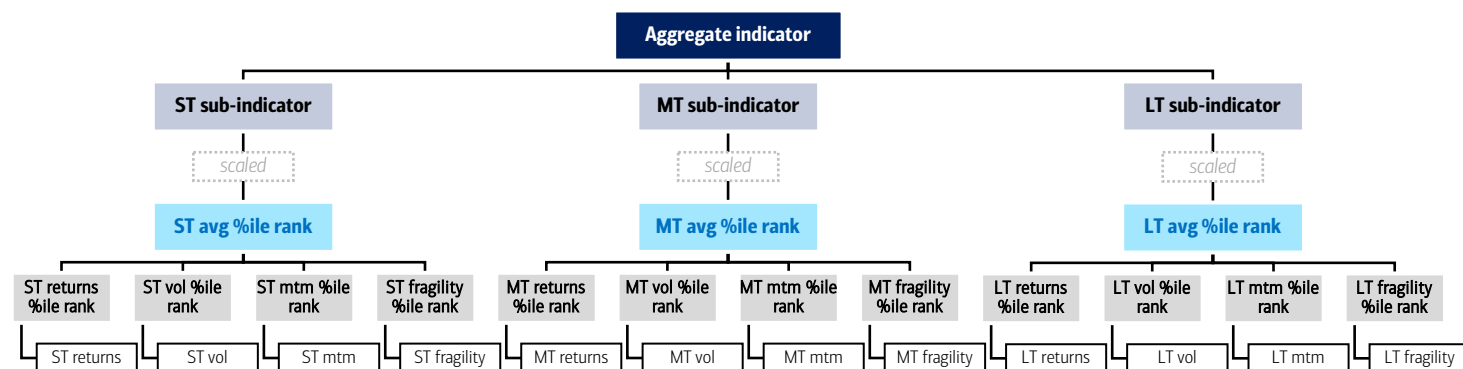
Each sub-indicator aims to capture how stretched the asset's four moments (returns, vol, momentum, and fragility) are relative to its history, and is computed as the average of each of these four moments' percentile rank vs history.⁹ However, given detection of bubble-like behaviour is the primary aim of the indicator, we condition this average percentile rank on periods when the asset is rallying by scaling it up or down depending on how positive or negative returns are.¹⁰ Exhibit 156 summarizes the construction of the aggregated indicator and its microstructure.

⁹ This report uses the following parameters for the 3 sub-indicators: (i) short-term = 3m moments vs 1y percentile rank lookback, (ii) medium-term = 6m moments vs 3y percentile rank lookback, (iii) long-term = 1y moments vs 5y percentile rank lookback. These parameter choices are for illustrative purposes and may differ in their efficacy depending on the dynamics of the asset being analyzed.

¹⁰ Scaling factor ranges between 0 & 1. It is computed using a logistic function transformation of the trailing return (tenor based on sub-indicator). Highly positive returns give a factor of ~1 and highly negative returns give a factor of ~0.

Exhibit 156: The aggregated indicator is an average of 3 sub-indicators which capture short-, medium- and long-term price dynamics. Each sub-indicator is computed as a scaled average percentile rank of the four metrics (returns, vol, momentum & fragility)⁹

Construction summary of BofA Bubble Risk Indicator



Source: BofA Global Research. ST, MT & LT refer to short-, medium- & long-term; 'mtm' refers to momentum component; '%ile rank' refers to percentile rank.

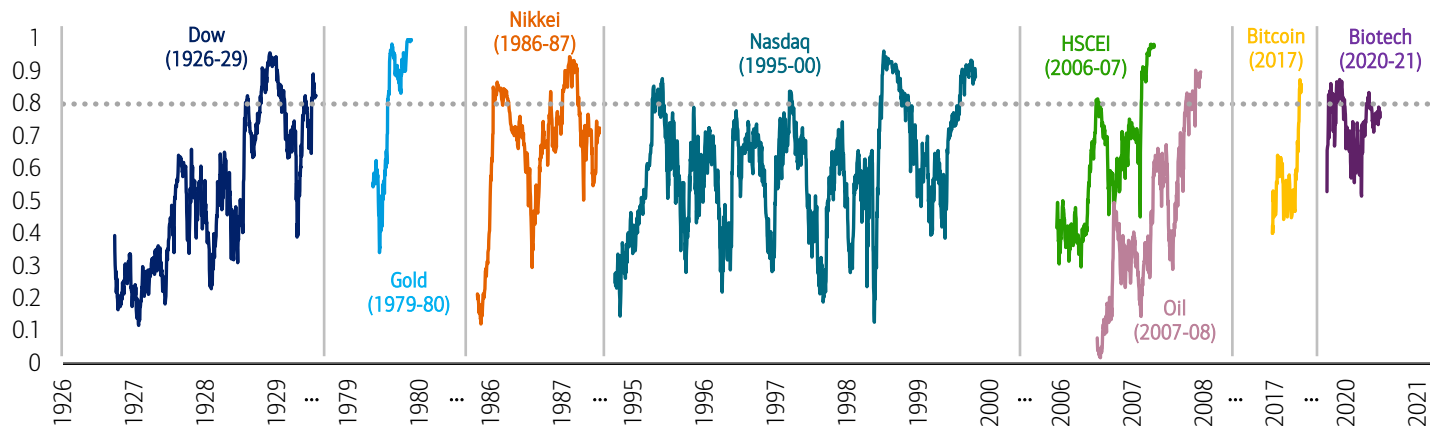
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Retrospective view of historical bubbles from the indicator's lens

For illustrative purposes, Exhibit 157 shows the aggregated indicator applied to 8 historical widely recognized asset bubbles. As these bubbles formed, the indicator rose towards 1 driven by elevated levels of returns, vol, momentum, and fragility.

Exhibit 157: Looking at eight historical asset bubbles from a distributional lens using the BRI methodology reveals increasingly elevated returns, vol, momentum and fragility in the asset bubble formation process, as seen by rising levels of the indicator towards 1

BofA Bubble Risk Indicator during historical asset bubbles



Source: BofA Global Research. Data from Jan-1926 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow (Oct-1926 to Sep-1929), XAU for Gold (Jul-1979 to Jan-1980), NKY for Nikkei (Jan-1986 to Oct-1987), NDX for Nasdaq (Jan-1995 to Mar-2000), HSCEI for HSCEI (Jun-2006 to Oct-2007), CO1 for Oil (Jan-2007 to Jul-2008), XBTUSD for Bitcoin (Jul-2017 to Dec-2017), SPSIBI for Biotech (May-2020 to Feb-2021).

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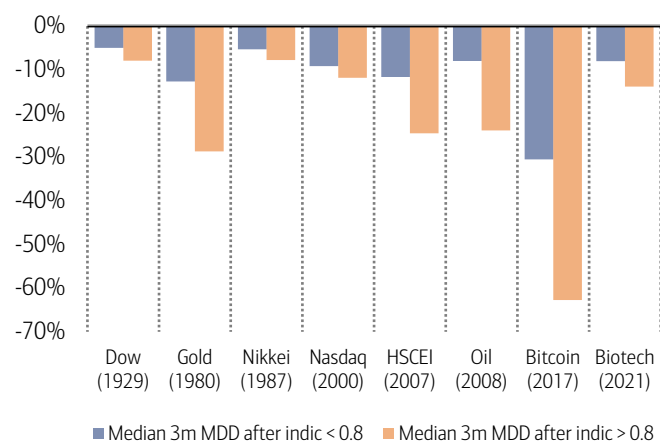
“Watch out” (not “get out”) when indicator at nosebleed levels

Interestingly, within this universe, the peaks of these bubbles would have typically corresponded to the indicator breaching 0.8 (top quintile), territory rarely breached otherwise. We note the forward distribution of returns during these bubbles conditional on the indicator's level suggests increased downside risks at high levels of the indicator. The median max draw-down over the 3m period following an indicator reading above 0.8 would have been greater than in periods following readings below 0.8 in each of the eight asset bubbles (Exhibit 158). In this sense, one can use the top quintile as a rule of thumb after which one should remain wary of (i) a potential pullback or (ii) the ultimate pop of the bubble.

However, the median max draw-up levels would have been similar (or at minimum materially as large) both in the indicator's top quintile or otherwise (Exhibit 159), suggesting that while high indicator levels may suggest increased downside risk in the future, completely de-risking may risk major underperformance as well. As a result, we believe the indicator is best positioned as a cautionary tool to add asymmetric protection/equity replacement ("watch out") rather than an explicit sell signal ("get out").

Exhibit 158: The median 3m max draw-down following indicator readings above 0.8 would have been more adverse than those below 0.8 during historical asset bubbles

Median 3m max draw-down during historical asset bubbles (i) after indicator < 0.8, (ii) after indicator ≥ 0.8

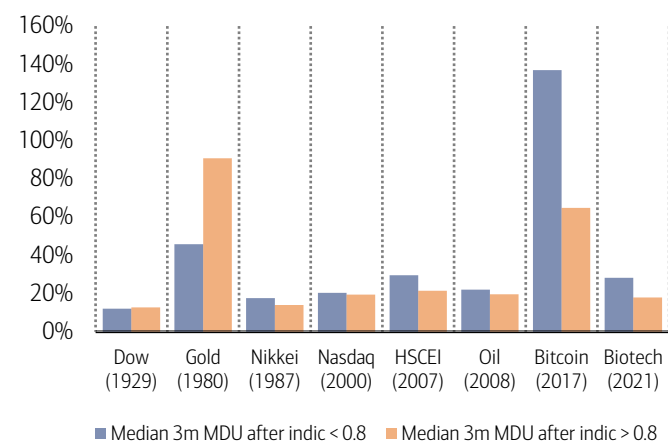


Source: BofA Global Research. Data from Jan-1926 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow (Oct-1926 to Sep-1929), XAU for Gold (Jul-1979 to Jan-1980), NIKY for Nikkei (Jan-1986 to Oct-1987), NDX for Nasdaq (Jan-1995 to Mar-2000), HSCEI for HSCEI (Jun-2006 to Oct-2007), CO1 for Oil (Jan-2007 to Jul-2008), XBTUSD for Bitcoin (Jul-2017 to Dec-2017), SPSIBI for Biotech (May-2020 to Feb-2021).

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Exhibit 159: The median 3m max draw-up would have been similar at both high and low indicator readings, suggesting that de-risking completely following high indicator levels can risk underperformance

Median 3m max draw-up during historical asset bubbles (i) after indicator < 0.8, (ii) after indicator ≥ 0.8



Source: BofA Global Research. Data from Jan-1926 to Feb-2021. Underlying Bloomberg tickers: INDU for Dow (Oct-1926 to Sep-1929), XAU for Gold (Jul-1979 to Jan-1980), NIKY for Nikkei (Jan-1986 to Oct-1987), NDX for Nasdaq (Jan-1995 to Mar-2000), HSCEI for HSCEI (Jun-2006 to Oct-2007), CO1 for Oil (Jan-2007 to Jul-2008), XBTUSD for Bitcoin (Jul-2017 to Dec-2017), SPSIBI for Biotech (May-2020 to Feb-2021).

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Potential risks

- Fundamental transformation of asset:** There could be instances of 'false positives' when the indicator appears stressed but the asset doesn't appear to be in a bubble based on other metrics (e.g., fundamentals also inflate with asset prices and leave valuations unchanged, like in semiconductor equities in the immediate aftermath of ChatGPT's release in Nov-22). Indeed, as previously mentioned, we believe the BRI could be most useful as an *added* signal when a bubble in the asset is suspected, rather as a primary indicator to detect a bubble in the absence of other information. It only aims to provide a view of how bubble-like an asset's behavior has been.
- Assumption of bubble resulting in instability:** By construction, the indicator's lack of direct visibility into the positioning and valuation makes its froth detection capabilities reliant on these non-price dynamics bleeding into an asset's price action. While the typical evolution of a bubble sees this happen as stretched positioning and exuberance translate into volatile & fragile rallies with high momentum into the peak, outlier cases where the bubble peak sees the bull run end in a whimper can make it challenging for the indicator to detect froth and flash warning signs of future downside risk.
 - The Nikkei bubble in the 1980s is an example of such an outlier, where the most violent upside period was in 1987, which was followed by a relatively tame rally until the price peak in 1990. However, the Nikkei's case very much appears to be the exception, given 7 out of 8 asset

bubbles in our universe saw their equity market peaks when the indicator was in its top quartile. Moreover, even in the case of the Nikkei, stress in the indicator was visible during the bubble build (in 1987).

- **Vulnerability to exogenous shocks:** Another consequence of the indicator's isolated focus on price dynamics is its susceptibility to exogenous shocks (like the 1987 or Covid crash). For one, assets can be susceptible to drawdowns driven by exogenous factors even when the indicator may not be at high levels. More critically, the indicator's level can be harder to interpret in the immediate aftermath of such exogenous shocks given the indicator's attempt to benchmark prevailing moments to a lookback containing an exogenous left tail event.

Lastly, while we present the construction and analysis using the aggregate indicator (i.e., average of the short-, medium-, and long-term sub-indicators, see footnote 10 for details), we also find it useful to analyze the sub-indicators to better gauge the disaggregation of shorter vs longer-term dynamics. For instance, shorter-term dynamics can often be more useful to assess froth when the underlying asset has had a fundamental change in risk profile or suffered an exogenous shock during a longer-term lookback period.

Summary of Trades Opened (26-Nov-25)

Price data for open level reflects the price on open date and does not necessarily reflect the price at which the trade could be executed at the date of this report. Our trades are structured to be executed on the open date and are not necessarily appropriate to execute as formulated beyond that date.

Exhibit 160: Summary of trades opened in this report

For a full list of our open trades and trades closed over the last 12 months, please see [Global Equity Volatility Insights: Vol to go higher as AI boom continues...](#)

Trade Description	Open Date	Open Level	Expected Trade Term	Rationale	Risks
Buy Dec26 80% put on basket of HYG (38%), SPX (29%), XLF (29%) & EEM (4%) for 1.46% (27% cheaper vs avg of individual legs)	26-Nov-25	1.46%	18-Dec-26	The convergence of private and public market risk creates opportunities for hedging private risk with liquid, public market hedges. Crucially, stress in public assets closely linked to private markets may ultimately impact private valuations, while potential hedge gains will give investors the flexibility to add risk at better terms.	Limited to premium paid.
Buy a basket of Dec26 90-70 put spreads on individual underlyings: SPX (53%), XLF (32%), TLT (9%) & EEM (6%) for 2.66% (weighted max payout ratio of put spreads is 7.7x)	26-Nov-25	2.66%	18-Dec-26	Given the higher volatility of the PE hedge constituents and significant decorrelation potential, we prefer buying put spreads on individual constituents rather than on the basket	Limited to premium paid.
Buy NDX Dec26 ATM upvar, currently 19.1v but target an entry level closer to 17v (t & t-1 convention, var ref. 29.9v)	26-Nov-25	19.1v / target 17v	18-Dec-26	Trade long-delta, long vol via upvar to capture the spot-up, vol-up typically seen in asset bubbles, with a deep discount to vanilla var	Limited to variance strike paid
Buy QQQ Jun26 660-750 call spread for \$18 (~35D/10D strikes, 5x max payout ratio, ref. 605.16)	26-Nov-25	\$18	19-Jun-26	Asymmetrically capture equity upside while hedging a grind higher, which upvar can be vulnerable to	Limited to premium paid
Buy 6-month 105% 125% call spreads on a CSI 1000 tracking basket for 2.53% (indic. 20d, 27.1v/28.7v, quanto USD)	26-Nov-25	2.53%	15-May-26	The "artificially" cheap CSI 1000 forward corresponds to an 8 vol discount. Investors can buy the most exciting part of the Chinese stock market, including AI innovation, at unusually low & limited risk	Limited to premium paid.
Buy long-dated OTM puts on NVDA, delta-hedged (e.g., NVDA 2y 10-delta puts)	26-Nov-25	\$8.30 / 49v	21-Jan-28	Tail hedge the AI trade with long-dated NVDA vol, which historically would have shown >35x payout vs carry cost.	Prolonged spot down, vol down in NVDA
Buy TLT (20+Yr Treasury Bond ETF) Mar26 86 puts for \$1.03 (~25D strike, ref. 90.01)	26-Nov-25	\$1.03	20-Mar-26	With TLT implied volatility currently near post-Covid lows and the put wing pricing in little premium vs recent history, we like tactically hedging the risk of another long-end rates selloff using TLT puts.	Limited to premium paid
Buy 30y SOFR > ATMf + 20bps / XAU > 106% Mar26 dual digitals for 7.95% (12.5x max payout ratio, -18 corrol offer, ref. 3.955%, 4078.70)	26-Nov-25	7.95%	20-Mar-26	Leverage the historical strength of gold in response to inflation and fiscal fears by hedging with 30y rates up, gold up dual digitals	Limited to premium paid
Sell SXSE Jun26 var (22.8v) & buy 1.7x Jun27 var (23.7v)	26-Nov-25	22.8v/23.7v	19-Jun-26	Long fwd var is most attractive for SXSE vs global peers. Resize legs to cut carry cost by 4v (45% better vs regular sizing) while preserving risk/reward	Jun27 var in Jun26 fails to rise relative to an increase in realised vol.
Buy 110% Dec26 SPX calls qEUR for 4.33% (17 bps over vanilla)	26-Nov-25	4.33%	18-Dec-26	Foreign investors with unhedged FX risk in their equity investments can be subject to large swings on USD volatility, which quanto options can protect against.	Limited to premium paid
Buy 90% Dec26 SPX puts qAUD for 3.66% (16bps less than vanilla)	26-Nov-25	3.66%	18-Dec-26	Foreign investors with unhedged FX risk in their equity investments can be subject to large swings on USD volatility, which quanto options can protect against.	Limited to premium paid
Buy Mar26 straddle on liquid EZ banks vs short SX7E straddle for 2.36%	26-Nov-25	2.36%	20-Mar-26	EU banks' broad thematic tilt supports ongoing return dispersion. Trade gains 50% of time (including recently) and offers a 2.3x reward/risk.	Short straddle losses occur in strongly correlated moves.
Buy Dec26 ATMf call on return dispersion on bank names for 3.50% and strike of 17.28% (SX7E spot ref: 234.93).	26-Nov-25	3.50%	18-Dec-26	EU banks' broad thematic tilt supports ongoing return dispersion. Trade gains 50% of time (including recently) and offers a 2.3x reward/risk.	Limited to premium paid
Buy Jun26 NKY > 104% & USDJPY < 96% dual digital for 9.7% (+48 corrol offer, 10.3x max payout, ref. 48626, 157).	26-Nov-25	9.7%	12-Jun-26	Moderate Nikkei upside trades can be cheapened by 80% if willing to lean optimistic on the yen. Narrowing US-JP rate differentials could weigh on USD/JPY in '26 but doesn't have to be Nikkei negative.	Limited to premium paid
Buy 1x 20-Mar-2026 106% call on a CSI 1000 tracking basket* for 2.32% (+27d, 26v). Sell 1.05x ASHR US 20-Mar-2026 34 calls for \$0.55 (~1.73% of spot; the 34.00 strike is ~107% of spot, ref 31.76)	26-Nov-25	2.32%/-1.73%	20-Mar-26	The long/short trade extracts value from the rich repo for Chinese small caps while limiting directional risk. Trading calls vs calls manages the risk of small caps underperforming. Beta-neutral sizing yielded 8% p.a. on gross notional at 8% vol	That the gain on the CSI 1000 calls is not sufficiently large to compensate for hypothetical losses on the short ASHR leg
Buy Jun26 105% DAX call contingent on CAC < 105% at expiry	26-Nov-25	1.24%	19-Jun-26	We like risk-limited trades that fade recent strength in CAC given the French fiscal overhang to fund German equity upside in case optimism returns for German growth.	Limited to premium paid

Source: Bloomberg, BofA Global Research. Prices reflective of most recently available data which may be delayed in some cases. "Trade Value" represents current valuation of trades initiated on the "Open Date".

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List of abbreviations

Exhibit 161: List of abbreviations

Abbreviations used in this report

ODTE: Zero-days-to-expiry	JPY: Japanese Yen
ABC: Anything But China	LDI: Liability-Driven Investments
Adj: Adjusted	LTCM: Long-Term Capital Management
AI: Artificial Intelligence	MA: Moving Average
ATM: At-the-money	Mag7: Magnificent 7 stocks are Microsoft, Apple, Amazon, Nvidia, Alphabet, Meta and Tesla
ATMf: At-the-money-forward	MHI: Mitsubishi Heavy Industries Ltd
AUM: Assets Under Management	MtM: Mark-to-market
Avg: Average	NDX: Nasdaq 100
BoE: Bank of England	NFP: Non-farm payrolls
BoJ: Bank of Japan	NIRP: Negative interest rate policy
BTD: Buy-the-dip	NKY: Nikkei 225 Index
CAPE: Cyclically-adjusted price-to-earnings ratio	NTM: next twelve months
CAGR: Compound Annual Growth Rate	OCC: Options Clearing Corporation
CB: Central Bank	OI: Open Interest
Comm. Svcs / Comms: Communication Services	OIS: Overnight Indexed Swap
Correl / corr: Correlation	OTM: Out-of-the-money
Covid: Covid-19 virus	Outperf: Outperformance
CPI: Consumer Price Index	OW: Overwriting
CPU: Central Processing Unit	P.a.: Per Annum
CS: Call spread	P/B ratio: Price-to-book ratio
CSS: Critical Stress Signal	P/E ratio: Price-to-earnings ratio
CTA: Commodity Trading Advisors (trend followers)	PDH: partial delta-hedge
cVar: Conditional Value-at-Risk	PnL / P&L: Profit and loss
DH: Delta hedged	QoQ: Quarter-over-quarter
Disc: Discount	R.V.: Realized Volatility
Discr / Cons Disc: Consumer Discretionary	Refi: Refinancing
Div: Dividend	RTY: Russell 2000
DR: Diversification ratio	S&P: S&P 500
DXY: US Dollar Index Spot Rate	SMR: Small Modular Reactor
EM: Emerging Markets	SOFR: Secured overnight financing rate
EOD: End-of-day	SPW: Equal-weight S&P 500
EPS: Earnings-per-share	SPX: S&P 500 Index
ESTX50: Euro STOXX 50 index	SVB: Silicon Valley Bank
ETF: Exchange Traded Fund	SX5E: Euro Stoxx 50 Index
ETN: Exchange Traded Note	SX6P: STOXX Europe 600 Utilities index
EU: European Union	SX7E: STOXX Europe 600 Banks index
EZ: Eurozone	SXEP: STOXX Europe 600 Oil & Gas index
FCI-G: Fed's index measuring financial conditions' impact on GDP growth	SXXP: STOXX Europe 600 index
Fed: Federal Reserve Bank	Tech: Technology
FFR: Federal Funds Rate	TPU: Tensor Processing Unit
Fins: Financials	TR: Total Return
FOMO: Fear of Missing Out	TRP: Tail liquidity risk premium
Fwd: Forward	TSMC: Taiwan Semiconductor Manufacturing Company
FX: Foreign Exchange	Tsy: Treasury
G-SIB: Global Systemically Important Banks	UKX: FTSE 100 Index
GDP: Gross Domestic Product	UNSDG: United Nations Sustainable Development Goal
GFSI: Global Financial Stress Indicator	UST: US Treasury bonds
GICS: Global Industry Classification Standard	Var: Variance
GPU: Graphic Processing Unit	VG1: Generic 1st Euro Stoxx 50 Future
HF: Hedge fund	VIX: CBOE Volatility Index
HH Index: Herfindahl-Hirschman Index	Vol: Volatility
HSI: Hang Seng Index	VRP: Vol Risk Premium
HY: High yield	WIRP: Bloomberg World Interest Rate Probability analysis
I.V.: Implied Volatility	YCC: Yield curve control
IEA: International Energy Agency	YoY: Year-over-year
IG: Investment grade	YTD: Year-to-date
Indust/Indus: Industrials	L&I: Levered and Inverse



Options Risk Statement

Potential Risk at Expiry & Options Limited Duration Risk

Unlike owning or shorting a stock, employing any listed options strategy is by definition governed by a finite duration. The most severe risks associated with general options trading are total loss of capital invested and delivery/assignment risk... all of which can occur in a short period.

Investor suitability

The use of standardized options and other related derivatives instruments are considered unsuitable for many investors. Investors considering such strategies are encouraged to become familiar with the "Characteristics and Risks of Standardized Options" (an OCC authored white paper on options risks). U.S. investors should consult with a FINRA Registered Options Principal.

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<http://www.theocc.com/about/publications/character-risks.jsp>

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Disclosures

Important Disclosures

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